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Learning theoretic reformulation and double descent

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Abstract

The analysis of learning action, machine learning and related practices in the field theoretically has been made by utilizing Computational Learning Theory (CoLT) Valiant 1984 and Statistical Learning Theory (SLT) Vapnik 1999. These two theories, while overlapped, provides a general framework in analysing and justifying learning actions and learning model constructions, aiding in the formation of modern practices. One of the famous insight using such framework is the *bias-variance tradeoff* Geman, Bienenstock, and Doursat 1992, which states that model complexity and generality inversely affect each other, thus guarantee the need for a safe bracket between them. However, recent literatures, Belkin et al. 2019a has indicated the fallout of such dilemma by the new phenomenon called *double descent*, where there exists an interpolation threshold that renders the current statistical justification for bias-variance inconsequential to a given region. Various 'anomaly' has also been detected similarly, with varying degree of sophistication and potential.In this paper, we would like to review through progress made in this particular field of research, and the phenomena potential explanations up until now.

Keywords: Statistical learning theory, double descent, deep learning, computational complexity theory, complexity theory, grokking

The modern machine learning landscape is dominated and guaranteed by the foundation of theoretical machine learning, specifically computational learning theory (CoLT) and statistical learning theory (SLT). The theory of machine learning and its modern practice has been developed and researched, of a substantial portion by empirical and heuristic approach, either by advancing new practices, architectures or by try-and-test modification. From its early onset of a regression estimator $\theta(x, y)$ on the linear regression problem, machine learning has developed substantially. Certain model concept with great successes includes regression-classification model, Bayesian modelling, generative learning model, support vector machine, Gaussian processes, and more. Of all such, the more formal and complex model architecture created, is the concept of a *neural network*. With increasingly sophisticated architecture, heuristic approach become popular, the fast-paced advancement of the field comes with new method, new results, new observations, and its far-reaching application which led to even bigger and larger scale deployment, there has been questions about the formation and status of a theoretical ground, a rigorous matter on the side of **theoretical machine learning**.

While rigorous and well-formulated in a sense, classical and theoretical machine learning was dwarfed by the modern advancement of machine learning as a whole, leading to several anecdotal problems regarding the interpretation of phenomena, the re-evaluation of the theory to fit the more updated analysis, and explanation to more sophisticatedly designed system. This and many more, plus as present, many of such advancements and improvements are heuristic, and the general theory and conceptual understanding remain limited, led to the choice to often opt for analogies and empirical workaround. Ultimately, this resulted in multiple 'failures' in explaining, interpreting, and predicting behaviours of new phenomena appeared in the modern landscape of machine learning. While there exists novelty of analysis, and different theoretical schemes to analyse such, particular problems remain in the form of irregularity and conflict of interpretations between frameworks.

Our main focus is to review on the topic, analyse existing analysis and observations, and understanding of the umbrella term classical learning theory, for the recent phenomena observed named 'double descent'. Statistical learning theory (SLT) and computational learning theory (CLT) Vapnik 1999; Mohri, Rostamizadeh, and Talwalkar 2012; Shalev-Shwartz and Ben-David 2014; Hajek and Raginsky 2021; Bousquet et al. 2020 has been prominent in constructing a well-rounded formal theory surrounding learning problems, models, and machine learners analysis. Valuable insights have been dissected from treatment of statistical theory and mathematical modelling on models, including the *bias-variance tradeoff* Geman, Bienenstock, and Doursat 1992; P. M. Domingos 2000b, which serves as a bound for efficient learning and model configuration (or complexity). However, recently there has been observations of *double descent* Belkin et al. 2019a; Schaeffer et al. 2023; Nakkiran et al. 2019; Lafon and Thomas 2024 which refute the famous tradeoff assumption, and hence brings question to the establishment of the theory, as well as several assumptions and insight in the framework. Further events and phenomena observed also includes grokking and triple, to n-descent Davies, Langosco, and Krueger 2023; Ascoli, Sagun, and Biroli 2020. Furthermore, many problems of defining and formalizing notions used in designing and implementing machine learning models are inconclusive, such as, for example, *model complexity* and others.

1. Outline

The main contributions of the paper lie in the heart of the problem of *machine learning theoretical*, and specifically *double descent*. As such, we focus on the following:

1. Theoretical analysis of the machine learning framework, and reinterpretation of the empirical-generalization framework. We explicitly redetermine perspectives and formulation, about the problem of generalization learning, and model selection principles. Additionally, we also formalize the algorithmic objective of the objective system.
2. Observations and insight reviews and critical analysis. We analyse current knowledge, different analysis of the phenomena, and provide critiques, critical analysis on limitation, constraints, different assumptions and fallacy observed in those results, and verification or extension of experiments. This means analysing existing papers specific of details, and intrinsic of certain functional class.
3. Providing our own insight to the phenomena of double descent, and hence the need for new analysis or different theory scheme to be utilized in solving the problem.

While this is non-conclusive of the problem of double descent, we want to emphasize the importance of such issue to be further than simply the dilemma between model complexity and risk error. It hints at the mismatch of knowledge, fuzzy nature of certain concepts, failure modes being apparent that indicates the intricacy of the sensitivity of the theory when encountering general structures, and the immaturity in which the theoretical landscape contains many noises.

2. Related works

On itself, the phenomena *double descent* has been investigated somewhat comprehensively, firstly introduced by Belkin et al. 2019a, which gives it distinctive saddle escape illustration. Further analysis was made by several literatures, particularly attributed the existence of double descent to the concept of model complexity and inductive bias, and potential explanations of such. Nakkiran et al. 2019 expanded the phenomena into deep neural network models. Their conclusion is reached by considering the perturbation of a learning procedure $\mathcal{T}$ on the effective model complexity $\text{EMC}_{\mathcal{D},\epsilon}(\mathcal{T})$ defined in the paper, separating the eventual phenomena into regions of observations, thereby in one way or another, predicting the tendency of double descent. This is also the first, arguably, "concise" definition of double descent, even though its nature is an empirical definition, including the notion of model complexity. Recently, there is also the Neural Tangent Kernel (Jacot, Gabriel, and Hongler 2018) which can be used to explain double descent, however further researches are still required. Preliminary works are done by Lafon and Thomas 2024, Schaeffer et al. 2023, and Liu and Flanigan 2023 on the role of optimization in double descent. Davies, Langosco, and Krueger 2023 attempted to unifying grokking with double descent, theorized a possibility of similarity during the generalization phase transition of the inference period, and Olmin and Lindsten 2024 attempted to explain epoch-wise double descent within the model of two-layer linear network. However, it is mentioned that it can also be expanded into deep nonlinear networks.

Further literatures on double descent varies, but fairly new, ever since their discovery articles in Belkin et al. 2019a and Nakkiran et al. 2019. Since then, we have Shi et al. 2024; Schaeffer et al. 2023; Quétu and Tartaglione 2023a; Lafon and Thomas 2024; Neal 2019; Davies, Langosco, and Krueger 2023; Quétu and Tartaglione 2023b; Liu and Flanigan 2023; Mei and Montanari 2020, 2019b; Adlam and Pennington 2020; Transtrum et al. 2025; Liu, Liao, and Suykens 2021; Allerbo 2025; Pezeshki et al. 2021; Brellmann et al. 2024; Nakkiran 2019a; Mei and Montanari 2019a; Heckel and Yilmaz 2020; Nakkiran et al. 2020; Belkin, Hsu, and Xu 2020; Yang and Suzuki 2024; Spiess, Imbens, and Venugopal 2023; Nakkiran et al. 2021; Y. S. Zhang 2024; Cherkassky and Lee 2024; Nakkiran 2019b. An even more exotic version of double descent, dubbed *triple descent* (or more), is discovered in Ascoli, Sagun, and Biroli 2020.

3. Theoretical foundation

For the need of analysing the phenomenon of double descent, we begin by listing and reviewing different foundational notions that describe the typical theory on the learning dynamics, and the learnability problem. Such ranges from statistical learning theory, as seen of Vapnik 1999; Hajek and Raginsky 2021; Mohri, Rostamizadeh, and Talwalkar 2012, and the more practical learning theory, as seen in James et al. 2009.

The setting of typical machine learning system comes in two-fold, either from the outer objective of the *observation space* $\{x_i, y_i\}_{i\leq n}^{n}$, denoted $\mathcal{S}$ for any 2-tuple of that form, or the general, oracle-induced view in which there exists the main form, or the oracle $c \in \mathcal{C}$, of which form the functional spaces. We would like to review these two outlooks.

Because we are to introduce certain formalism, or at least interpretation of the learning process, as well as the different dynamical components associated, the preliminary would be first reserved to clarify such notions, alongside fixing usual notations that would be used accordingly. Historically speaking, the learning theory started from the early 1970s. By the sense of classical, we do not mean by, as currently called, of non-deep learning theory, but rather the old treatment of theoretical boundaries and rigorous formalization of the concept itself. In such development, lies the formal setup of learning, and then, the task of which developing it, until we can even reach the *Probably Approximately Correct* learning (PAC) and Vapnik-Chervonenkis theory of maximal dimension, Neural Tangent Kernels (Jacot, Gabriel, and Hongler 2018), and so on.

The nature of the learning theory can also be brought back from certain proposition or argument, most probably, the fact that it is mostly from the black-box interpretation - hence making almost all model being **phenomenological** by default. An interesting aspect of such, though, is the learning process itself, or we call it so.

3.1 Classical theory of learning

Classically, we can see that the problem of double descent is not inherently a problem that can be explained only by considering model complexity and the error rate. As such, it requires a framework that consider multiple factors together, and interpreting different hypothesis. Nevertheless, classical theory does indeed point to certain factors or aspect that should be of concern. For example, we have seen plenty of analysis that indicates that *sample complexity* might be the biggest factor, considering the behaviour is intrinsically operable and perhaps observable in almost any model, except for certain model that is specialized of the data structure itself. Even by then, the error still can be obtained given large enough model disparity to the concept class. There are attempts to, indeed, analyze double descent from a VC perspective (Lee and Cherkassky 2022), however, the paper falls into the same pitfall as others in their interpretation and analysis.

Nevertheless, even though we have said plenty of the disadvantage of the classical theory in terms of statistical learning, their insight is valuable in determining the possibility of dynamic in question. There is a reason why they are versatile and structurally valuable enough that our majority of understanding are formed from such framework itself. In simplicity, within the principle of working, our model is called the **hypothesis** $h \in \mathcal{H}$ of certain **hypothesis class** $\mathcal{H}$. The target objective is typically concerned of, denoted by $c \in \mathcal{C}$, as to be of a structural class $\mathcal{C}$, in which observations creates information in the form of the observable set $\mathcal{S}$. The set of observation $\mathcal{S}$ is presentable in the space of 2-tuple $(\mathcal{X}, \mathcal{Y}) \subset (\mathbb{R}^d, \mathbb{R}^p)$. Here, $\mathcal{X}$ is called the *ground* or input space, and $\mathcal{Y}$ is called the *target* or output space. By default, in general, the input space $\mathcal{X}$ is supposed to follow some special distribution $\mathcal{D}$. This treatment branches from the ambiguity of the origin of $x \in \mathcal{X}$ for elements of the input space, that is, $x \sim \mathcal{D}$ by a random sampling process. This assumption also ensures relative population representability[a]. By this, we mean that the observation captured fully describe what was happening with the entire concept - even including the pathological and outlier. We also assume that this sample of observation is **identically and independently sampled** of a given sampling process.

The task for the hypothesis, and the designer (us) is to figure out how to best approximate c, assuming arbitrary notion of knowledge, prior information, such that h correctly *mimic* the behaviour and phenomenon observed by c. We call this loosely as the **learning problem**. Naturally, learning problem is a statistical problem, since their setting is fully observational.

The **learning problem** divides the act, with respect to configuration the chain of component into two parts. The first one is for any measurable space of the model, after defining the system, the dependency, and the question, we have to gain the *accuracy measure*, or *phenomenological aberration* of a given model, and for the respective problem. Depends on the specific requirement of the problem, the measure can change, but, they have the same landscape. The second one requires the development of a *scheme* to dynamically move the given hypothesis, under fixed description, into a supposed optimal point in which under specific threshold, our model is 'perfect' for the concept. That is, *mimic c by h in a perfect way*. It is perceived, though, that the objective of learning the concept perfectly is impossible. Hence, we will often opt for the latter half of approximating it to a given threshold θ

Setting in the second part, most of our problem is concerned with the issue of predicting the unseen states of being, or rather, the *generalization problem*. This then perhaps, classify the problem of statistical learning theory to *mostly* predictive, or rather, we are concerned of prediction model. This is subsidized by the inner *estimation problem* for an estimation model. What is the difference between the two? First, the distinction comes from the trivial on-sight fact that estimation problems and their models, comes of as a closed-form working space. That is, they work on a closed space of what is observed, and extract properties, determine statistically the behaviours, characteristic of that system alone and only. In other word, an estimator *uses data* to guess and learn about the true state, and properties of the setting (usually also said to be the *true state of nature*). However, we notice a discrepancy - not always will we get the entirety of the problem that we are interested at. For some reason, for example, our data is only quite a portion, lacking a bit valley, a bit of patterns here and there by the blank spaces where observations left behind. Then, what if we can generalize pass that? What if we also want to somewhat estimate correctly for unseen portion of our concept class? This is called the predicting problem we have said earlier, in which we use the estimation already established, and add some flavours into what can possibly constitute a larger view - a more entire view of the learning concept. From a statistical perspective, the model itself in statistical learning theory, would have to both estimate the intrinsic empirical distribution and concept $(c_{\mathcal{S}}, \mathcal{D}_{\mathcal{S}})$, and then somehow, leaves room for generalization to the actual distribution and concept $(c, \mathcal{D})$, which is particularly of more interest than not in practical settings. Here, the notation simply means two thing: the facade of which c is the true concept, and $\mathcal{D}$ is the uncertain, probabilistic presumption of the structure in which c seems to exhibit, thus there then also exists $c_{\mathcal{C} \equiv c'}$ of the proxy concept c' that 'seems right' in comparison to the true concept, and also its observation-based distribution.

If we recall the previous discussion on the informal section on machine learning, you will see that this is typically what's called the **supervised learning setting**. This is perhaps true for the classical learning theory, and the establishment of the theory as a whole. Consider unsupervised learning. Under the scope of unsupervised learning, the model essentially goes into a free-mode, where there exists no metric that is non-context enough for evaluating the theoretical form of the model. For a

a. By population representability, we mean that the density is well-illustrated, such that the data fully captures the entire operational space that is relevant and meaningful of the underlying concept.

given density estimation model ψ, there can be many ways to 'think of evaluating it' on how well it groups objects together. This means there are not so much meaningful information can be extracted from examining the behaviour of the model, including its performance. Furthermore, since it is random as much as a random process, no conclusive bounds or theoretical theorem can be made on the maximal-minimal problem of a given model. Hence, most of the learning theory is concerned of supervised learning setting, because it is a controlled, closed, well-defined system of interest.

3.2 Phenomenological aberration

Formally, the learning problem is then formulated in details as followed. Given the machine learning model expressed a hypothesis h of the hypothesis class $\mathcal{H}$, the learning theory aims for creating a procedure to learn either elements of the concept class $\mathcal{C}$ of all concepts $c : \mathcal{X} \to \mathcal{X}$, or the function class $\mathcal{F}$ of all functions $f : \mathcal{X} \to \mathcal{Y}$, which is usually set to $\{0, 1\}$ or $[0, 1]$. This distinction is trivial, hence, if it is clear, we will talk about the concept class $\mathcal{C}$ only as representative form. The learner $\mathcal{L}(h)$ consider the set of possible hypothesis $\mathcal{H}$, in which might not coincide with $\mathcal{C}$. It receives a partial image of sample $S = (x_1, \ldots, x_2, \ldots, x_n)$ drawn i.i.d. according to $\mathcal{D}$ as well as the label $(c(x_1), \ldots, c(x_n))$. This constitutes the dataset $\mathcal{S}$, which are based on specific concept $c \in \mathcal{C}$ for the hypothesis to learn. The task is then to use (or *extract*) meaningful information to select a hypothesis $h_S \in \mathcal{H}$ that accurately mimic c, with marginal error Θ. The notation h_S stands for all hypothesis that can be inferred from the range of the dataset (the first argument, $\mathcal{X}$).

The marginal error Θ is considered of two parameters, the empirical error $\hat{R}(h)$ and the generalization error $R(h)$. We give the following definition of them.

Definition 3.1 (Empirical risk). *Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and a sample $S = (x_1, \ldots, x_m)$. For some particular $\epsilon > 0$, the* ***empirical error*** *or empirical risk of h is defined by*

$$\hat{R}_S(h) = \mathop{\mathbb{P}}_{x \in S \sim \mathcal{D}}[\ell\{h(x), c(x)\} \geq \epsilon] = \frac{1}{m}\sum_{i=1}^{m} \ell\{h(x_i), c(x_i)\} \quad (1)$$

The generalization risk is then defined for the entirety of the sample set, assuming to be densely populated on the interval of specification. Hence, $x \sim \mathcal{D}$ returns infinitely many samples.

Definition 3.2 (Generalization risk). *Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and an underlying distribution $\mathcal{D}$ on $\mathcal{X}$. For some particular $\epsilon > 0$, the generalization error or risk of h is defined by*

$$\begin{aligned} R(h) &= \mathop{\mathbb{P}}_{x \sim \mathcal{D}}[\ell\{h(x), c(x)\} \geq \epsilon] \\ &= \mathop{\mathbb{E}}_{x \sim \mathcal{D}}[\ell\{h(x), c(x)\}] \\ &= \int_{x \in \mathcal{D}} \ell\{h(x), c(x)\}\, dP(x) \end{aligned} \quad (2)$$

In essence, the empirical risk measure the hypothesis to observation error, and the generalization risk captures the difference of the hypothesis to the actual concept. Notice that by doing this, we have implicitly assumed that the observations and the concept is not the same thing, under the majority of scenarios.

Using the above notion of generalization we can separate the learning problem into two goals - one to minimize $\hat{R}(h)$, and one to minimize $R(h)$. As we have been saying, and for Theorem 10.3, one of our main assumption, as well as a guarantee under uniform convergence, is the fact that we must have $\hat{R}(h) \approx R(h)$ for larger and larger dataset size, or the empirical space.

The hypothesis h^* is often called the *empirical best*, for it being the minimal, finite hypothesis of the lowest loss evaluation on the entire observation space $\mathcal{S}$. There exists no certified assumption regarding whether h^* aligns with the minimal generalization error. We can show that normally, the problem of empirical risk minimizatio and generalization risk minimization is not the same. That is, if one tries to solve the empirical learning problem to certain measure, then they will fail the generalization problem. This problem between the disparity of the empirical and generalization target is what we often called as the *Allegory of the cave*, as seen in previous chapter. Here, we risk being dependent ourselves on false hypothesis obtained by empirical data acquisition, but also have to aim for the general form of the target by itself, which is often assumed unobtainable or impossible to know of prior belief. Hence, either that we take a skewed representation of reality through empirical data, or we diverge from such in a manner that gets closer to the ground truth's concept. At least in this setting, it is then observed as above that empirical learning problem is a subproblem toward generalization problem - one have to solve the empirical learning problem before attempting the generalization one. We also want to introduce the term **data shape** and **concept shape** for data $\mathcal{S} \in c$ and the shape of c by itself, to reflect this difference.

For the generalization problem's best model in a fixed $\mathcal{H}$, we have the notion of a **Bayes model**.

Definition 3.3 (Bayes risk). *Given a distribution $\mathcal{D}$, the Bayes error R^* is defined as the infimum of the errors achieved by measurable functions $h : \mathcal{X} \to \mathcal{Y}$:*

$$R^* = \inf_{h \in \mathcal{H}} R(h) \quad (3)$$

Definition 3.4 (Bayes model). *A hypothesis h with $R(h) = R^*$ is called a Bayes hypothesis or Bayes classifier, and is taken by*

$$\forall x \in \mathcal{X}, \quad h_{\mathrm{B}} = \operatorname*{argmax}_{y \in \mathcal{Y}} \mathbb{P}[y \mid x] \quad (4)$$

The Bayes model is understood to be the minimal of $R(h)$. Overall, in the classical learning setting, we can dissect the problem into what constitute the following cleaner version of the setting, includes both the generalization learning and empirical learning problem. Again, under ideal condition, and ideal setup, the learning setting will have empirical learning and generalization learning to be on the same track. Later on, we would formalise what exactly on the same track means.

Setting 3.1. *Given an op-space $\mathcal{X} \times \mathcal{Y}$, define the concept class $\mathcal{C}$ of all $c : \mathcal{P}(\mathcal{X}) \to \mathcal{P}(\mathcal{Y})$ of all power set for both, where $\mathcal{X} \sim \mathcal{D}$ for some distribution $\mathcal{D}$ of the probabilistic assumption. The learning problem consists of defining a hypothesis class $\mathcal{H}$, for h being an arbitrary hypothesis, either with specific architecture or not, and a learning algorithm $\mathcal{A}$, of a* ***learner*** *such that the following is acquired:*

- *Set $\epsilon, \delta > 0$ be the error bound and the confidence measure (that is - the probability of failure) of h, for ℓ the evaluative measure of $\{h(x), c(x)\}$, define $\hat{R}(h)$ being the empirical risk of h on the observation set $\mathcal{S}$ available. Then, the empirical learning problem is tasked to find the hypothesis such that $\hat{R}(h) = \min_{h_0 \in \mathcal{H}} \hat{R}(h_0) \leq \epsilon$ for some arbitrary h_0, with confidence $1 - \delta$. This h is called the empirical best.*
- *Similarly, define $R(h)$ the generalization error, assumed outside $\mathcal{S}$. Then, the generalization learning problem is tasked to find the hypothesis such that $\hat{R}(h) = \min_{h_0 \in \mathcal{H}} R(h_0) \leq \epsilon$ for some arbitrary h_0, with confidence $1 - \delta$. This h is called the Bayes model.*
- *The generalization setting requires choosing the optimal point for both h^* of the empirical best, and h_B for Bayes hypothesis.*

To choose and to compare between the empirical best, and the Bayes hypothesis (generalization best), we need certain notions to compare the two, algorithm to choose and evaluate, and metric to determine how comparison should be performed, with both measurable. At best, we need to find the correlation between the sub-concept and the actual target concept itself.

3.3 PAC-learning

The PAC-learning process (Shalev-Shwartz and Ben-David 2014; Mohri, Rostamizadeh, and Talwalkar 2012; Hajek and Raginsky 2021), or rather, *Probably Approximately Correct* learning, connects three complexity measures - the sample space complexity, representation complexity, and optimization complexity, to the evaluation of the statistical learning process. It refers to the categorization of various learning procedure that satisfies the Probably Approximately Correct criteria in its name, as such being followed.

We assume no structure of h and c. They can be functions, partial functions, relations, complex algorithms, or others. In a typically learning setting, we also have the argument of *preliminary knowledge*, presented in literatures of the term *inductive bias*. For example, if the concept class $\mathcal{C}$ is assumed, then we say the setting is **model-specific**. If there exists no hard assumption on $\mathcal{C}$, then the learning setting is said to be **model-free**. Then, the PAC-learning set for model-specific setting is defined as followed, for the observable set $\mathcal{S}$ of m samples altogether, in which the major assumption is $x \sim \mathcal{D}$ of an arbitrary distribution $\mathcal{D}(\theta_d)$:

Definition 3.5 (PAC-learning)**.** *A concept class $\mathcal{C}$ is said to be* ***PAC-learnable*** *if there exists an algorithm $\mathcal{A}$ and a polynomial function $poly(\cdot,\cdot,\cdot,\cdot)$ of 4-argument such that for any $\epsilon > 0$, $\delta > 0$, for all distribution $\mathcal{D}$ on $\mathcal{X}$ and for any target concept $c \in \mathcal{C}$, the following holds for any sample size $m \geq poly(1/\epsilon, 1/\delta, n, size(c))$:*

$$\Pr_{S \sim \mathcal{D}^m} \left[R(h(S)) \leq \epsilon \right] \geq 1 - \delta$$

for a given error measure $R(h_S)$. If $\mathcal{A}$ further runs in

$$poly(1/\epsilon, 1/\delta, n, size(c))$$

, then $\mathcal{C}$ is said to be efficiently PAC-learnable. When such algorithm exists, we call $\mathcal{A}$ a PAC-learning algorithm.

This can be easily extended into the model-free case, and further on, which is left in the appendix section. The reason for the argument *size*(c) is as followed. Consider a class of concepts defined by the satisfying assignments of certain formulae. A concept from this class that satisfies such formulae, can be represented by a formula f, a truth table, or any given formulae that is tautologically equivalent formulae f' to f. PAC-learning bound argues in the sense of *computational complexities*, and *space complexities*. Specifically, the term $poly(1/\epsilon, 1/\delta, n, size(c))$ hopes to bound the required learning process to polynomial time, specified by 4 parameters. Here, n is the *input dimension*, *size*(c) is the encoding complexity of the target concept, $\epsilon > 0$ is the *accuracy bound*, and δ is the confidence bound - the probability that h fails.

The case of model-free is simple, we forgo any assumptions on the behaviours or structures of $\mathcal{C}$ and its associated c-model, and thus the observable pairs is simply $(x, y) \sim \mathcal{D}_{mf}$. Here, we denote $\mathcal{D}_{mf}$ to separate it from $\mathcal{D}$ of which is a singular x-distribution; however, we can also write $\mathcal{D}_{ms}$ and $\mathcal{D}_{mf}$ in certain specific cases. Per Hajek and Raginsky 2021, the treatment of model-agnostic (no model c assumption), targets the ingredient sets of,

1. Sets $\mathcal{X}, \mathcal{Y}$ and $\mathcal{U}$. Here, $\mathcal{U}$ is the correlated h-functional result, i.e. the hypothesis model, covering the case of **probabilistic models.**
2. A class $\mathcal{P}$ of probability distribution on $Z := \mathcal{X} \times \mathcal{Y}$. Here, the distribution covers both.
3. A class $\mathcal{H}$ of functions $h : \mathcal{X} \to \mathcal{U}$.
4. A loss function as objective, $\ell : \mathcal{Y} \times \mathcal{U} \to [a, b]$, preferably $[0, 1]$. Additional overhead measures are also required.

Under such lens, PAC-on-agnostic can be realized as the following definition instead:

Definition 3.6 (agnostic-PAC learning, Hajek and Raginsky 2021)**.** *We say that a learning algorithm for a problem $(\mathcal{X}, \mathcal{Y}, \mathcal{U})$ and interpretations $(\mathcal{P}, \mathcal{H}, \ell)$ is PAC-learnable to accuracy $\epsilon > 0$, if, we have*

$$\lim_{n \to \infty} \sup_{P \in \mathcal{P}} P^n \left(Z^n : \hat{R}_P(h_n) > \inf_{h \in \mathcal{H}} R_P(h) + \epsilon \right) = 0 \qquad (5)$$

An algorithm that is PAC-learnable to accuracy ϵ for every $\epsilon > 0$ is said to be a PAC-learning algorithm. A learning problem is said to be model-free learnable if there exists such algorithm, for every $A \in \mathcal{A}$ of such class to be ***PAC****-learning algorithm class.*

PAC-learning procedure gives, by assumption of a **consistent learner**, two different generalization bounds for two different cases. We first define the notion of a consistent learning algorithm, or consistent learner, for a concept class C and hypothesis h.

Definition 3.7 (Consistent Learner). *We say that an algorithm is a consistent learner for a concept class C using hypothesis class $\mathcal{H}$, if for all n, for all $c \in C_n$ and for all m, given*

$$S = \{(x_1, c(x_1)), (x_2, c(x_2)), \dots, (x_m, c(x_m))\}$$

as input, $x_i \in X_n$ outputs, then the algorithm $\mathcal{A}$ outputs a hypothesis $h \in \mathcal{H}_n$ such that

$$h(x_i) = c(x_i), i = 1, \dots, m$$

A consistent hypothesis is the hypothesis that is consistent with all the training data provided to it from the concept c. The two bounds then contain the debate between consistent hypothesis (i.e., for example, admitting no error on training data, almost perfect - maybe even so), and inconsistent hypothesis, of which have general defects from the actual concept, or even different by a margin. This equates to observational noise σ, typically used in simulation of noisy feature and spread, and also for randomization. Inconsistent then means the further this particular noise is increase, the higher the lower bound of the error can be for any particular measure, in comparison to the true concept c in consistent case. We need to note that this inherently pushes the consistency as for both the algorithm L, and the hypothesis $\mathcal{H}$. Decomposing such definition, we can clearly see two assumptions:

Assumption 3.1 (Consistent learner, condition 1). *The algorithm is capable of extrapolate the entire coverage structure of class $\mathcal{H}$. Thus, $\mathcal{A}(R, \ell)$ of the objective condition is guaranteed upon all class $\mathcal{H}$.*

Assumption 3.2 (Consistent learner, condition 2). *There exists $h \in \mathcal{H}$ that absolutely equals to c. That is, for the concept c of interest, on the arbitrary expression space of which contains $\mathcal{H}, \mathcal{C}$, then for a subset $\mathcal{C} \supset \mathcal{C}' \supset c$ as the size of $|\mathcal{C}'| \to 0, \mathcal{C}' \subset \mathcal{H}$.*

On such guarantee of the formalization and assumption of coverage, PAC-learning famously leaves two results on the learning bound of the learning setup. The first one concerns of the form as to be consistent, and the second case is for where the second assumption fails. However, we have no correct formalization of this failure mode. Specifically, the learning bound intricacy states that of the **global correctness**, that is, absolutely fitting. Usually this is not true, and the inconsistent case is much more possible, but yet again loose as to only examine the case where consistency fails, not *how much* or how many.

Theorem 3.3 (Learning bound - finite $\mathcal{H}$, consistent case). *Let H be a finite set of functions mapping from $\mathcal{X} \to \mathcal{Y}$. Let $\mathcal{A}$ be an algorithm that for any target concept $c \in H$ and i.i.d. samples S returns a consistent hypothesis H_S, such that $\hat{R}(h_S) = 0$. Then for any $\epsilon, \delta > 0$, the inequality $\Pr_{S \sim D^m}[R(h_S) \leq \epsilon] \geq 1 - \delta$ holds if*

$$m \geq \frac{1}{\epsilon}\left(\log|H| + \log\frac{1}{\delta}\right)$$

This sample complexity result admits the following equivalent statement as a generation bound: for any $\epsilon, \delta > 0$, with probability at least $1 - \delta$,

$$R(h_S) \leq \frac{1}{m}\left(\log|\mathcal{H}| + \log\frac{1}{\delta}\right) \tag{6}$$

Proof. See Mohri, Rostamizadeh, and Talwalkar 2012 for the original proof. □

The theorem is illustrated as followed in Figure 1 and Figure 2:

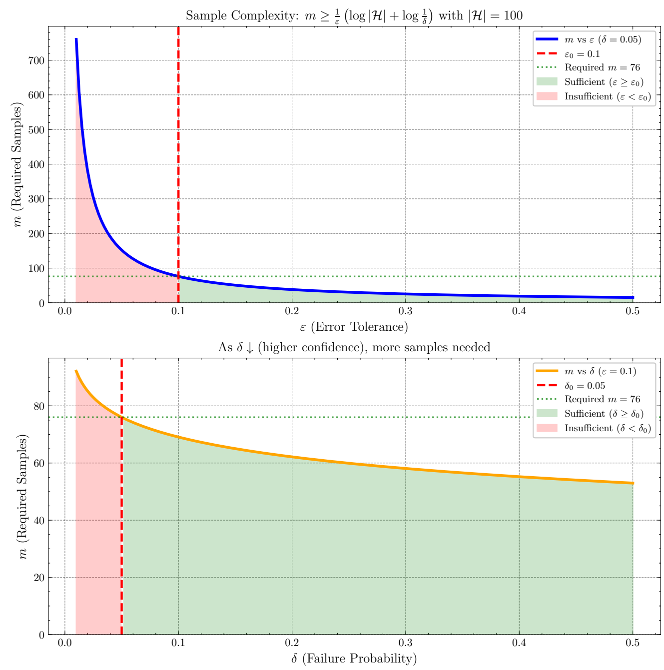


Figure 1. Illustration of of sample complexity in terms of cardinality $m = |\mathcal{S}|$, with respect to ϵ or δ measure.

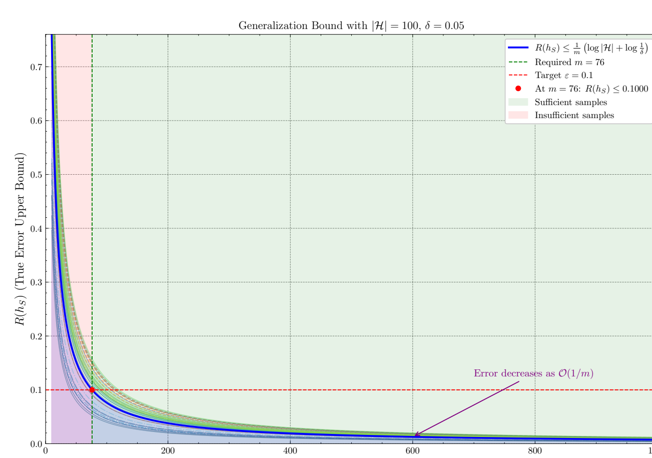


Figure 2. Illustration of of generalizational error/risk in terms of cardinality $m = |\mathcal{S}|$, with respect to changing $\mathcal{H}$ cardinality and δ-value measure. Blurred line indicates different cases or values, the deviation is rather small, thus the bound is very tight.

As for inconsistent case, the margin of error exists, thus $\mathcal{H}$ is bounded like the following.

Theorem 3.4 (Learning bound - finite $\mathcal{H}$, inconsistent case). *Let $\mathcal{H}$ be a finite hypothesis set. Then, for any $\delta > 0$, with probability at least $1 - \delta$, the following inequality holds:*

$$\forall h \in \mathcal{H}, \quad R(h) \leq \hat{R}_S(h) + \sqrt{\frac{\log |\mathcal{H}| + \log 2/\delta}{2m}} \tag{7}$$

Proof. This proof is based on parts of Mohri, Rostamizadeh, and Talwalkar 2012 and the section about PAC theorems. In the most general case, there may be no hypothesis in $\mathcal{H}$ consistent with the labelled training sample. This, in fact is the typical case in practice, where the learning problems may be somewhat difficult or the concept classes more complex than the hypothesis set used by the learning algorithm.

To derive the learning guarantees in the more general setting, we would use the following corollary, of which relates the generalization error and empirical error of a single hypothesis.

Corollary 3.3.1. *Fix $\epsilon > 0$. Then, for any hypothesis $h : X \to \{0,1\}$, the following inequalities hold:*

$$\underset{S\sim\mathcal{D}^m}{\mathbb{P}} \left[\hat{R}_S(h) - R(h) \geq \epsilon\right] \leq \exp\left(-2m\epsilon^2\right) \tag{8}$$

$$\underset{S\sim\mathcal{D}^m}{\mathbb{P}} \left[\hat{R}_S(h) - R(h) \leq -\epsilon\right] \leq -\exp\left(-2m\epsilon^2\right) \tag{9}$$

By the union bound, this implies the following two-sided inequality:

$$\underset{S\sim\mathcal{D}^m}{\mathbb{P}} \left[|\hat{R}_S(h) - R(h)| \leq \epsilon\right] \leq 2\exp\left(-2m\epsilon^2\right) \tag{10}$$

Proof. This can be proved using Hoeffding inequality. Recall that the inequality is stated for $Z_1, \dots, Z_n$ independent random variable, for range $z_i \in [a_i, b_i]$ with probability one, $S_n = \sum_{i=1}^{n} Z_i$ then for all $t > 0$:

$$\mathbb{P}(S_n - \mathbb{E}(S_n) \leq -t) \leq \exp\left(\frac{-2t^2}{\sum(b_i - a_i)^2}\right)$$

and

$$\mathbb{P}(S_n - \mathbb{E}(S_n) \geq t) \leq \exp\left(\frac{-2t^2}{\sum(b_i - a_i)^2}\right)$$

We can derive this for our case of the two empirical and generalization risk. Remember from 10.3 that we have $R(h) = \mathbb{E}[\hat{R}_S(h)]$, we can transform the phrase to be:

$$\underset{S\sim\mathcal{D}^m}{\mathbb{P}}\left[\hat{R}_S(h) - R(h) \geq \epsilon\right] = \underset{S\sim\mathcal{D}^m}{\mathbb{P}}\left[\hat{R}_S(h) - \mathbb{E}[\hat{R}_S(h)] \geq \epsilon\right]$$

with respect to $\mathcal{D}^m$. The form of $\hat{R}_S(h)$ has an additional $1/m$, hence, for simplification, we will take the form

$$\underset{S\sim\mathcal{D}^m}{\mathbb{P}}\left[\hat{R}_S(h)/m - \mathbb{E}[\hat{R}_S(h)]/m \geq \epsilon m\right]$$

By Hoeffding inequality, we have:

$$\underset{S\sim\mathcal{D}^m}{\mathbb{P}}\left[\hat{R}_S(h)/m - \mathbb{E}[\hat{R}_S(h)]/m \geq \epsilon m\right] \leq \exp\left(\frac{-2m^2\epsilon^2}{\sum(b_i - a_i)^2}\right) \tag{11}$$

which is

$$\underset{S\sim\mathcal{D}^m}{\mathbb{P}}\left[\hat{R}_S(h)/m - \mathbb{E}[\hat{R}_S(h)]/m \geq \epsilon m\right] \leq \exp\left(\frac{-2m\epsilon^2}{(b-a)^2}\right)$$

Here, our $\sum(b_i - a_i)^2$ was replaced by a more general bound which contains only the sufficient amount of m samples. This is because the sum follows from m samples, which are taken as random variable of choice, then, for $a \leq Z_i \leq b$

$$\frac{m^2}{\sum(b_i - a_i)^2} = \frac{m^2}{m(b-a)^2} = \frac{m}{(b-a)^2}$$

Since our hypothesis is mapping to $\{0, 1\}$, hence our result space is discretely bounded in the normal range, the inequality reduces to

$$\underset{S\sim\mathcal{D}^m}{\mathbb{P}}\left[\hat{R}_S(h)/m - \mathbb{E}[\hat{R}_S(h)]/m \geq \epsilon m\right] \leq \exp\left(-2m\epsilon^2\right) \tag{12}$$

Rearranging the left-hand-side to the original yields the inequality. Doing the same for the second case, and using the union bound, we get the two-sided inequality for $|\hat{R}_S(h) - R(h)|$. □

Such bound can be naturally extended toward any arbitrary range $[a, b]$, as for Hoeffding inequality being defined on the arbitrary closed interval instead. Then, we have the general bound

$$\underset{S\sim\mathcal{D}^m}{\mathbb{P}}\left[|\hat{R}_S(h) - R(h)| \leq \epsilon\right] \leq 2\exp -\frac{2m\epsilon^2}{(b-a)^2} \tag{13}$$

Setting the right-hand side of the final expression to be equal to δ and solving this for ϵ yields immediately the following bound for a single hypothesis.

Corollary 3.3.2 (Generalization bound - single hypothesis). *Fix a hypothesis $h : \mathcal{X} \to \{0, 1\}$ or more generally onto $h : \mathcal{X} \to [a, b] \subset \mathbb{F}$. Then, for any $\delta > 0$, the following inequality holds with probability at least $1 - \delta$:*

$$R(h) \leq \hat{R}_S(h) + \sqrt{\frac{\log 2/\delta}{2m}} \tag{14}$$

The desired bound can then be realized as extending the proved corollary bound, by noticing that,

$$\hat{R}_S(h) + \sqrt{\frac{\log 2/\delta}{2m}} < \hat{R}_S(h) + \sqrt{\frac{\log |\mathcal{H}| + \log 2/\delta}{2m}} \tag{15}$$

for all $|\mathcal{H}|$. □

For a looser bound, we can ultimately use an ϵ-consistent learner criteria in such case. PAC-learning setting raises some peculiar insight toward the problem itself. First, is the learning bound itself of which raises the importance of several factors — the dataset size, m, of which is used as the **total resources required** for approximating such function. Furthermore, the set complexity $\mathcal{H}$ is also used, albeit questionably inconsistent in the term $\log|\mathcal{H}|$, which implies at least consideration on the cardinality of the hypothesis class. In a finite domain, we notice that PAC-learning of consistent case is exactly the case of interpolation, assuming non-smooth and noisy data, and the inconsistent case describe the wider range of approximation-based case, but does not consider an approximation-aware bound. The design of $|\mathcal{H}|$ also does not allow for certain exact hypothesis class size measurement, except for combinatorial case where $\mathcal{H}$ is indeed finite. Nevertheless, the bound indeed indicates that for m increase, the hypothesis class often do much better in general. One particularly troublesome notion of which we will meet, in the inconsistent case, is the reversal of guarantee compared to bias-variance form. Specifically, the inconsistent, model-agnostic version of Theorem 3.4 is as followed:

Theorem 3.5 (Inconsistent case, model-agnostic, finite $\mathcal{H}$)**.** *For finite $\mathcal{H}$, then for any $\delta > 0$, up to ϵ-accuracy, with probability at least $1 - \delta$, the following inequality holds:*

$$R(h_{\mathcal{S}}) \leq \min_{h_{\mathcal{S}} \in \mathcal{H}} \hat{R}_{\mathcal{S}}(h_{\mathcal{S}}) + \sqrt{\frac{\log|\mathcal{H}| + \log 2/\delta}{2m}} \tag{16}$$

such that

$$\sqrt{\frac{\log|\mathcal{H}| + \log 2/\delta}{2m}} \leq \epsilon \tag{17}$$

for $m > 0$ samples. We also say that the error varies by $\mathcal{O}(\sqrt{1/m})$), such that

$$R(h_{\mathcal{S}}) \leq \min_{h_{\mathcal{S}} \in \mathcal{H}} \hat{R}_{\mathcal{S}}(h_{\mathcal{S}}) + \mathcal{O}\left(\sqrt{\frac{\log|\mathcal{H}|}{m}}\right) \tag{18}$$

We denote, again, $\hat{R}_{\mathcal{S}}$ the empirical risk given on set $\mathcal{S}$ of observations, and $h_{\mathcal{S}}$ the hypothesis obtained from dynamics regarding such observational set.

This theorem directly indicates that, for bigger $|\mathcal{H}|$, the rate of which m increases can also mitigate bias-variance, thus bigger dataset can mitigate such issues of overfitting. Indeed, we notice that $\mathcal{O}(\log|\mathcal{H}|) \leq \mathcal{O}(m)$ linear terms, thus if the algorithm $\mathcal{A}$ exhibits such dynamics in which dataset varies, bigger hypothesis space should not increase significantly of the error measure. However, this contradicts, in a sense, with the usual term of notion to bias-variance, of which suggest otherwise, hence overfitting. Or, rather insidiously speaking, the interpretation is wrong, such that bias-variance concern fixed m situation, while such bound indicates, or at least suggest, of variational m.

While this does not cover the case of which $|\mathcal{H}|$ is argued to be infinite, such is then also be bounded, as for any measure, temporary called $\mu : \mathcal{H} \to \mathcal{F}$ for field $\mathcal{F}$ of which can be evaluated to $\mathbb{R}$ or else, then such is bounded in the contribution terms, and thus of the framework in which we use to express the learning dynamics. It is also then to be concerned of different classes of $\{\mu\}$ itself, as a limiting factor. This, perhaps, will lead to a potential solution of double descent.

Indeed, a loose analogy, or rather speculation can follow. While criticisms of classical uniform-convergence PAC learning correctly observe that one-dimensional capacity sweep indeed, fail to predict or explain the modern double descent curve, such cannot be said in certain cases of the inconsistent setting. Suppose we introduce the parameter ρ of the growth regime on $\mathcal{M}(\mathcal{H}) : \mathcal{H} \to \mathbb{F}^k$ of the complexity measure or the size of the hypothesis space, in which usually $k = 1$ and the value is on the real field, for $\gamma = |\mathcal{S}|$ as the sample size, such that:

$$\tau = \mathcal{M}(\mathcal{H}), \quad \tau = \gamma \tag{19}$$

hence the escape trajectory is then controlled subsequently. The parameter τ then can be computed, accordingly for the optimization term as

$$\mathcal{O}\left(\sqrt{\frac{\log|\mathcal{H}|}{m}}\right) \equiv \sqrt{\frac{\log\tau + \log 2/\delta}{2\tau}} \tag{20}$$

This particular trajectory, can indeed, create certain cases where the right-half of the curve is similar to the trajectory of double descent. In general, investigation would see for

$$\mathcal{O}\left(\sqrt{\frac{\log|\mathcal{H}|}{m}}\right) \equiv \sqrt{\frac{\log q(\tau) + \log 2/\delta}{2q(\tau)}} \tag{21}$$

for arbitrary function $q(\tau), p(\tau)$ where it should exhibit double descent terminal terms. However, this task is really hard aside from simply guessing. A simple guess on structural reformulation would be that

$$\min_{h_{\mathcal{S}} \in \mathcal{H}} \hat{R}_{\mathcal{S}}(h_{\mathcal{S}}) \approx \frac{1}{m}\left(\log|\mathcal{H}| + \log\frac{1}{\delta}\right), \tag{22}$$

$$\mathcal{O} \approx I\sqrt{\frac{log(\tau - p) + \log(2/\delta)}{2(\tau - p)^{\epsilon} + 1}} \tag{23}$$

Thus the sum would be defined in a more orthodox way, as

$$\frac{1}{m}\left(\log|\mathcal{H}| + \log\frac{1}{\delta}\right) + I\sqrt{\frac{log(\tau - p) + \log(2/\delta)}{2(\tau - p)^{\epsilon} + 1}} \tag{24}$$

for arbitrary constant I, α, p, of which now the addition operator would be slightly adjust, such that the sum will contain only the LHS of the operator if the RHS is in its undefined domain, and otherwise they add on together.

3.4 Empirical and structural algorithms

Suppose that for a concept class $\mathcal{H}$, we can then partition it to $\mathcal{H} = \bigcup_{k\in\mathbb{N}} \mathcal{H}_k$ for finite k hypotheses. Assume that they satisfy the uniform convergence (Shalev-Shwartz et al. 2010), which means in the arbitrary path landscape, one can guarantee ℓ-reduction, of m samples on $\mathcal{S}$ to:

$$\lim_{m\to\infty} \sup_{\mathcal{D}} \mathop{\mathbb{E}}_{S\sim\mathcal{D}^m} \left[\sup_{h\in\mathcal{H}} [R(h) - R_S(h)] \right] = 0 \tag{25}$$

Then such is said to be possible as a ruling dynamic when for a learning problem, the empirical risks of hypotheses in the hypothesis class converges to their population risk uniformly, with a distribution-independent rate. The result of this is that a problem then can be considered learnable with the ERM rule, for any given characterization. We state the following definition.[b]

Definition 3.8. *A learning problem is learnable if there exist a learning rule $\mathcal{A}$ and a monotonically decreasing sequence $\epsilon_{cons}(m)$ such that $\epsilon_{cons}(m) \overset{m\to\infty}{\longrightarrow} 0$ and*

$$\forall\mathcal{D}, \quad \mathop{\mathbb{E}}_{S\sim\mathcal{D}^m} \left[F(\mathcal{A}(S)) - F^*\right] \leq \epsilon_{cons}(m), \tag{26}$$

with $F^* = \inf_{h\in\mathcal{H}} R(h)$. A learning rule $\mathcal{A}$ for which this hold is denoted as a universally consistent learning rule.

There are certainly a fundamental class of rule or algorithm $\mathcal{A}$ to resolve this particular uniform convergence of classes. One of the major assumption or observation from the point of the hypothesis class, is that the hypothesis cannot know the generalization error. Thereby, one strategy is the *empirical risk minimization* (ERM) method, minimizing only the empirical risk to the empirical best; then, certain measures to 'generalize' this heuristically is apprehended on the hypothesis.

Definition 3.9. *A rule $\mathcal{A}$ is an ERM (Empirical Risk Minimizer), denoted $\mathcal{A}_{ERM}$ if it minimizes the empirical risk $\hat{R}(h) = (1/m)\sum_{c\in\mathcal{C}} \ell(h, c)$, over the observational space S,*

$$\hat{R}_S(\mathcal{A}_{ERM}(S)) = \hat{R}_S(h_S) = \inf_{h\in\mathcal{H}} \hat{R}_S(h) \tag{27}$$

Hence, the ERM tries to obtain $\mathrm{ERM}(h') = \arg\min_{h\in\mathcal{H}} \hat{R}(h)$.

The problem of overfitting still somewhat resides in the setting of the method. Hence, usually, we have an implicit term in addition, called the *regularizer*, of which is as followed.

Definition 3.10. *A rule $\mathcal{A}$ is an SRM (Structural Risk Minimizer), denoted $\mathcal{A}_{ERM}$ if it minimizes the empirical risk $\hat{R}(h) = (1/m)\sum_{c\in\mathcal{C}} \ell(h, c)$, over the observational space S,*

$$\hat{R}_S(\mathcal{A}_{SRM}(S)) = \hat{R}_S(h_S) + \lambda r(h) = \inf_{h\in\mathcal{H}} \hat{R}_S(h) + \lambda r(h) \tag{28}$$

of the regularizing term $r(\cdot)$. Hence, the SRM tries to obtain $\mathrm{ERM}(h') = \arg\min_{h\in\mathcal{H}} \hat{R}(h)$*, while taking into consideration a forcing term r, usually defined on the model complexity.*

Hence, the SRM is inherently stronger than ERM. The regularizer $\lambda r(h)$, in specific case of convex loss landscape and reasonably well-behaved irregularities between domains, and of the structural regularization term $r_s(h) = \sum_{i,j} w_{ij}^2$ for weights of ij index, then we have the following theorem.

Conjecture 3.1 (ERM-SRM growth). *Consider the regulizer $\lambda r(h)$ for $h \in \mathcal{H}$. Assuming structural regularization $r_S(h) = \sum_{i,j} w_{ij}^2$ for linear-layering of ij index, for ERM and SRM optimizer classes,*

$$\begin{aligned} &\mathrm{ERM}\left(\inf_{h\in\mathcal{H}} \hat{R}_S(h), \ell\right) \\ &\qquad \approx o\left(\mathrm{SRM}\left(\inf_{h\in\mathcal{H}} \hat{R}_S(h) + \lambda r(h), [\ell, \lambda r(h)]\right)\right) \end{aligned} \tag{29}$$

Where the little-o notation is used for saying of the optimization strength per arbitrary time-similar notion of algorithmic evolution. The reason for this change is because of the restriction domain that SRM indicted on the model optimization landscape. Effectively, we introduce the structural bias $\lambda r(h)$ that forces the morph of the total loss landscape. In well-behaved case, this is usually effective to a point, however in realistic system and pathological cases, such can lead to the c' approximation in c-space, just as we introduced in the reformulation of learning system. In this paper, we would like to presume the ERM method in question as the famous *gradient descent*. We refer to Achlioptas, n.d.; Ruder 2017 for general view of the algorithm, and J. Zhang 2019 for a source on particularly gradient descent algorithm on deep learning models.

One particularly troublesome and should take serious consideration into interpreting it, is the fact of the insufficiency on ERM. The following proposition is rather questionable.

Proposition 3.6. *For any sample S, the following inequality holds for the hypothesis returned by ERM:*

$$\mathbb{P}\left[R(h'_S) - \inf_{h\in\mathcal{H}} R(h) > \epsilon\right] \leq \mathbb{P}\left[\sup_{h\in\mathcal{H}} |R(h) - \hat{R}_S(h)| > \frac{\epsilon}{2}\right] \tag{30}$$

We will prove this proposition in later section. For now, this proposition bounds the error of ERM, with respect to the probability of the generalization distance $|R(h) - \hat{R}_S(h)|$ between the two errors. As we might have suspected, the performance of ERM is typically very poor. This is because the algorithm disregards the complexity of the hypothesis set $\mathcal{H}$: in practice, either it is not complex enough, in which case the error can be large, or $\mathcal{H}$ is so rich, that the complexity makes the model shoot out of the complexity range. Additionally, in many cases, determining the ERM solution is computationally intractable.

b. The reason why $r(\cdot)$ is not characterized in this preliminary section, is because of their uncertainty of effect. Hypothetically, $r(\cdot)$ is an additional structure (bias) of which apply a new gradient field in conjunction to the loss gradient field itself. This addition is uncertain of its content and analysis, since it depends on various stochastic dynamics of the model. We reserve from using regularization for now because of such reason.

3.5 The PAC-ERM notion

Without restricting ourselves to specific ERM construction, we can still apply PAC on the set of ERM algorithms in its generality of itself. Within such understanding, the empirical risk minimization strategy is often motivated by the hope that *$R(h)$ and $\hat{R}(h)$ are not too far-distant apart*. Particularly, it also implicitly agree with statements conjectured with law of large number, for $n \to \infty$, $|R(h) - \hat{R}(h)| \approx \epsilon$ of irreducible noise. To what extent is this true, we have to take certain boundary on which might have been previously shown. We refer to authoritative introductory texts like Alquier 2024, for both PAC-ERM and later sections on PAC-Bayes variation.

Proposition 3.7 (Alquier 2024)**.** *For any $h \in \mathcal{H}$, for any $\delta \in (0,1)$,*

$$\mathbb{P}_S\left(R(h) > \hat{R}(h) + C\sqrt{\frac{\log(1/\delta)}{2n}}\right) \leq \delta \tag{31}$$

Proof. Applying Hoeffding's inequality onto $U = \mathbb{E}[\ell_i(h)] - \ell_i(h)$,

$$\mathbb{E}_S\left[\exp tn[R(h) - \hat{R}(h)]\right] \leq \exp\left[(nt^2C^2)/8\right] \tag{32}$$

For any $s > 0$, we have by Markov's inequality,

$$\begin{aligned}\mathbb{P}_S(R(h) - \hat{R}(h) > s) &= \mathbb{P}_S(\exp nt[R(h) - R] > \exp(nts)) \\ &\leq \frac{1}{\exp(nts)}\mathbb{E}_S\left[e^{nt[R(h)-\hat{R}(h)]}\right] \\ &\leq \exp\left(\frac{nt^2C^2}{8} - nts\right)\end{aligned} \tag{33}$$

We can make this bound as tight as possible, by optimizing the choice of t. For $t = 4s/C^2$, we have

$$\mathbb{P}(R(h) - \hat{R}(h) > s) \leq \exp\left(\frac{-2ns^2}{C^2}\right) \tag{34}$$

This means that, for a given h, the empirical risk $\hat{R}(h)$ cannot be much smaller than the risk $R(h)$. The order of this "much smaller" can be better understood by introducing $\delta = \exp(-2ns^2/C^2)$, of which substituting δ to s yields the required proposition. □

Proposition 3.7 bounds the asymptotic term such that $R(h)$ will exceed $\hat{R}(h)$ by certain margin, at worst $1/\sqrt{n}$. However, this is not enough to justify the use of ERM (which indeed use the near-space approximation from empirical to general risk), since the proof only applies for h that was fixed above in terms of complexity. For those that is implicit of the data itself, we cannot apply them[c]. The usual approach though, is to use the inequality

$$R(h_{ERM}) - \hat{R}(h_{ERM}) \leq \sup_{h\in\mathcal{H}}\left[R(h) - \hat{R}(h)\right] \tag{35}$$

c. This means models such ass *support vector machine* (SVM) or *Gaussian process regression* of which the empirical optimization function and the hypothesis itself, relies on the number of data points available.

and to prove a version of Proposition 3.7 that would hold uniformly on $\mathcal{H}$.

Theorem 3.8. *Assume that* $\mathrm{card}(\mathcal{H}) = M < +\infty$, *for any* $\delta \in (0,1)$,

$$\mathbb{P}_S\left(R(h_{ERM}) \leq \inf_{h\in\mathcal{H}}\hat{R}(h) + C\sqrt{\frac{\log(M/\delta)}{2n}}\right) \geq 1 - \delta \tag{36}$$

Proof. We upper bound the supremum in the inequality by

$$\begin{aligned}\mathbb{P}_S\left(\sup_{h\in\mathcal{H}}[R(h) - \hat{R}(h) > s]\right) &= \mathbb{P}_S\left(\bigcup_{h\in\mathcal{H}}\left\{[R(h) - \hat{R}(h)] > s\right\}\right) \\ &\leq \sum_{h\in\mathcal{H}}\mathbb{P}_S(R(h) - \hat{R}(h) > s) \\ &\leq M\exp\left(\frac{-2ns^2}{C^2}\right)\end{aligned} \tag{37}$$

This time, put $\delta = M\exp(-2ns^2/C^2)$ we gain

$$\mathbb{P}_S\left(\sup_{h\in\mathcal{H}}[R(h) - \hat{R}(h)] \leq C\sqrt{\frac{\log(M/\delta)}{2n}}\right) \leq \delta \tag{38}$$

Thus it satisfies

$$\mathbb{P}_S\left(\sup_{h\in\mathcal{H}}[R(h) - \hat{R}(h)] \leq C\sqrt{\frac{\log(M/\delta)}{2n}}\right) \geq 1 - \delta \tag{39}$$

Using the inequality

$$\mathbb{P}_S\left(R(h_{ERM}) \leq \hat{R}(h_{ERM})C\sqrt{\frac{\log(M/\delta)}{2n}}\right) \geq 1 - \delta \tag{40}$$

As $\mathcal{H}$ is finite, $\hat{R}(h_{ERM}) = \inf_{h\in\mathcal{H}}\hat{R}(h)$. □

This bound coincides for the *Probably Approximately Correct* PAC-bounds, specifically for algorithm of ERM principle, since it approximates $R(h_{ERM})$ within $C\sqrt{\log(M/\delta)/2n}$ with probability $1-\delta$, and once again, under very strong arguments.

3.6 Noise, Mohri, Rostamizadeh, and Talwalkar 2012

Using the notion of the Bayes classifier and Bayes error, one can define the *noise* of a given learning setting under PAC-learning consideration, as followed.

Definition 3.11 (Noise)**.** *Given a distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$, the noise at point $x \in \mathcal{X}$ is defined by*

$$noise(x) = \min_{x\in\mathcal{X}}\{\mathbb{P}[1 \mid x], \mathbb{P}[0 \mid x]\}$$

The average noise or the noise associated to $\mathcal{D}$ is then the minimum possible error aside from intrinsic model error, and is calculated by $\mathbb{E}[noise(x)]$

Thus, the average noise is precisely the Bayes error, that is, $noise = \mathbb{E}[noise(x)] = R^*$. The noise is a characteristic of the learning task indicative of its level of difficulty. A point $x \in \mathcal{X}$, for which noise is close to 1/2, is sometimes referred to as *noisy* and is of course a challenge for accurate prediction.

4. Problem statements

Before continuing further, we must review the problem statement that led to the problem of double descent. We will have a look at the theory, the original conflict, and the illustrative phenomenon in action. In general, the learning setting in consideration of statistical learning theory is often supervised learning, for guarantee of a plausible upper bound result.

4.1 Bias-variance tradeoff

In classical sources, James et al. 2009; I. Goodfellow et al. 2016; Hajek and Raginsky 2021; Mohri, Rostamizadeh, and Talwalkar 2012; Shalev-Shwartz and Ben-David 2014, the principle of bias-variance tradeoff is a classical principle used in the categorization of technique called *model selection*. Specifically, this refers to the progress of choosing the best predictor h^* that minimize the generalization error that is the main goal of a particular machine learning model, during the process of training and inference. Originally, the theory that bias and variance often come in to conjunction is **Estimation theory**, of which there exists an estimator $\hat{\theta}$ that use a set of observations, to estimate the concept, or the underlying mechanics of certain system that outputted the observation set, by the **probabilistic perspective** - that is, it is governed by appearance by an independently and identically distributed process of parameterized probability $\theta \in \Theta$. Here, parameterized means being expressed by a set, often finite, of parameters, by E. L. Lehmann 1998; Paninski 2005. It is only when Geman, Bienenstock, and Doursat 1992 introduced it to machine learning that we have an equivalent structure that is similar to bias-variance in machine learning.

4.1.1 Precursor (Geman, Bienenstock, and Doursat 1992)

The original definition of bias-variance tradeoff by Geman, Bienenstock, and Doursat 1992 is first constructed using the means-square error, which is regarded as a normal measure in the real encoding space. Their approach is to justify bias-variance via decomposition of the loss function ℓ, for such to find an alternative reasonable form of such loss landscape. Suppose of a regression problem to construct a hypothesis function $f(x)$ from $(x_1, y_1, \ldots, x_N, y_N)$ for the purpose of generalization - that is, predicting unseen variational values for different pair $(x_j, ?)$ such that $? = y_j + \epsilon$ for a conceivable implicit error. To be explicit about the relation of this problem, or f on the given data $\mathcal{D} = \{(x_i, y_i) \mid i \leq N\}$, denote $f(x;\mathcal{D})$ instead of f, the natural mean-square measure as a predictor is:

$$\mathcal{M}(f, y) = \mathbb{E}\left[((y - f(x;\mathcal{D})))^2 \mid x, \mathcal{D}\right] \tag{41}$$

for $\mathbb{E}[\cdot]$ the expectation wrt to a distribution P. Decomposing the right-hand side, we have:

$$\begin{aligned}\mathcal{M}(f, y) &= \mathbb{E}\left[((y - f(x;\mathcal{D})))^2 \mid x, \mathcal{D}\right] \\ &= \mathbb{E}\left[(y - \mathbb{E}[y \mid x])^2 \mid x, \mathcal{D}\right] + (f(x;\mathcal{D}) - \mathbb{E}[y \mid x])^2\end{aligned} \tag{42}$$

Here, $\mathbb{E}\left[(y - \mathbb{E}[y \mid x])^2 \mid x, \mathcal{D}\right]$ does not depend on $\mathcal{D}$, but simply the statistical variance of y given x. The term $(f(x;\mathcal{D}) - \mathbb{E}[y \mid x])^2$ is considered a natural measure of effectiveness on $\mathbb{R}^n$ as a singular predictor of y. Now, for $\mathbb{E}_{\mathcal{D}}\left[(f(x;\mathcal{D}) - \mathbb{E}[y \mid x])^2\right]$ which depends on the training set $\mathcal{D}$ in its computation, is decomposed into the form of *bias-variance decomposition* terms, by derivation:

$$\begin{aligned}&\mathbb{E}_{\mathcal{D}}\left[(f(x;\mathcal{D}) - \mathbb{E}[y \mid x])^2\right] \\ &\quad = \underbrace{\left\{\mathbb{E}_{\mathcal{D}}[f(x;\mathcal{D})] - \mathbb{E}[y \mid x]\right\}^2}_{\text{bias term}} \\ &\quad + \underbrace{\mathbb{E}_{\mathcal{D}}\left\{(f(x;\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(x;\mathcal{D})])^2\right\}}_{\text{variance term}}\end{aligned} \tag{43}$$

We summarize this in the following statement.

Theorem 4.1 (Bias-variance decomposition). *Suppose the model $f(x;\mathcal{D})$ for the data $\mathcal{D} = (x_i, y_i)$ and its parameter x is defined. For y_i of the target concept's responses y, and consider a regression problem with the loss measure $\mathcal{M}(f, y)$ of mean squared risk, the following statement is true:*

$$\begin{aligned}\mathbb{E}[\mathcal{M}(f, y)] &= \mathcal{B}(f, y) + \mathcal{V}(f, y) \\ &\quad + \mathbb{E}\left[\mathbb{E}\left[((y - f(x;\mathcal{D})))^2 \mid x, \mathcal{D}\right]\right]\end{aligned} \tag{44}$$

for $\mathbb{E}[\,\cdot \mid x, \mathcal{D}]$ any expression with dependencies on x and $\mathcal{D}$. The bias and variance term is subsequently expressed by

$$\mathcal{B}(f, y) = \underbrace{\left\{\mathbb{E}_{\mathcal{D}}[f(x;\mathcal{D})] - \mathbb{E}[y \mid x]\right\}^2}_{bias}, \tag{45}$$

$$\mathcal{V}(f, y) = \underbrace{\mathbb{E}_{\mathcal{D}}\left\{(f(x;\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(x;\mathcal{D})])^2\right\}}_{variance} \tag{46}$$

The above decomposition principle is often expressed into a form where there exists the intrinsic noise Brown and Ali 2024:

$$\begin{aligned}\mathbb{E}_D\left[\mathbb{E}_{xy}\left(y - \hat{f}(x)\right)^2\right] &= \mathbb{E}_x\left[\left(y^* - \mathbb{E}_D[\hat{f}(x)]\right)^2\right] \\ &\quad + \mathbb{E}_x\left[\mathbb{E}_D\left(\hat{f}(x) - \mathbb{E}_D[\hat{f}(x)]\right)^2\right] \\ &\quad + \mathbb{E}_{xy}\left[(y - y^*)^2\right]\end{aligned} \tag{47}$$

In all, almost all representation of bias-variance decomposition are the same. For simplicity of notation, we adopt the similar

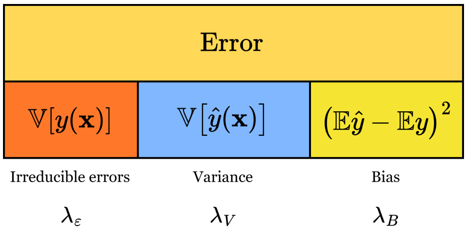


Figure 3. Decomposition of the error term into 3 parts. Respectively, the irreducible error $\mathbb{V}[y(\mathbf{x})]$, the variance (blue) and the bias (dark yellow). All of them are assumed to take up 100% of the error observed in such composition. The proportion is included using the three coefficient $\lambda_\epsilon, \lambda_V, \lambda_B$ where $\lambda_\epsilon + \lambda_V + \lambda_B = 1$.

form in the standard case of the paper Adlam and Pennington 2020:

$$\mathbb{E}\left[\hat{y}(\mathbf{x}) - y(\mathbf{x})\right]^2 = (\mathbb{E}\hat{y}(\mathbf{x}) - \mathbb{E}y(\mathbf{x}))^2 + \mathbb{V}\left[\hat{y}(\mathbf{x})\right] + \mathbb{V}[y(\mathbf{x})] \tag{48}$$

A main common theme of criticism toward bias-variance tradeoff is the fact that the decomposition is much more general, and intrinsic for the class of *mean squared loss*. However, when considering the naturalness of an error measure and then its direct applicant, mean square error on real space of sufficient support and measure comes of as a very natural choice of an error measure, at least in consideration of the regression setting; for that, we then can also define classification as another top-layer above a regression's similar real continuous space, i.e. a decision layer. Furthermore, it can also be shown Brown and Ali 2024; Pfau 2013 that it also holds for the class of Bregman divergence measure.

What does the decomposition say about model selection principle in general? In general, bias-variance is typically presented of its decomposition only, and the asymptotic prediction of its behaviours. In fact, one of the reason that it became the rule-of-thumb for ML practitioner, as well as generally statistical learning (Lafon and Thomas 2024 provides a quite rigorous treatment of bias-variance tradeoff in the section on statistical learning theory) solidify the trade-off as a particular model selection principle. Generally, this tradeoff can be summarized as followed:

Theorem 4.2 (Bias-variance tradeoff). *For the expected loss of any given hypothesis h, the bias $\mathcal{B}(f, y)$ and variance $\mathcal{V}(f, y)$ is inversely proportional, that is, $\mathcal{B}(f, y) \propto \lambda^{-1}\mathcal{V}(f, y)$ for some proportionality λ that may or may not be constant. In the most general case possible, $\lambda = -1$ on the entire error range.*

The tradeoff is then of inverse proportionality. Indeed, statistically, we have such tradeoff on a statistical framework in a more concrete sense. For the bias to increase, variance will increase, of which the criterion is inverse - we would like to have more bias but lower variance, and vice versa, according to such theory.

4.1.2 Expansion and variations

The bias-variance decomposition and tradeoff is not general. Indeed, works, by Geman, Bienenstock, and Doursat 1992; Sharma and Aiken 2014; P. M. Domingos 2000a; Adlam and Pennington 2020; Yang et al. 2020 and more all attempted to reconstruct and reformulate bias-variance, and it did leave out a picture of uncertainty regarding the concept. Specifically, there are certainly different way to do the bias-variance decomposition, as classified in Adlam and Pennington 2020 paper. Some of this are particularly specialized in *neural network* training, basing on the main structure of modern machine learning in neural network formalism.

1. **Semiclassical approach:** The semiclassical approach is fairly simple. We can rewrite to another variation of the classical approach to be

$$\begin{aligned}\mathop{\mathbb{E}}_{h\in\mathcal{H}}[\ell(h)] &= \mathbb{E}_x\mathbb{E}_\epsilon\left[\hat{y}(x) - y(x)\right]^2 \\ &= \mathbb{E}_x\left(\mathbb{E}_\epsilon[\hat{y}(x)] - y(x)\right)^2 + \mathbb{E}_x\mathbb{V}_\epsilon\left[\hat{y}(x)\right]\end{aligned} \tag{49}$$

and instead, consider the *structural conformation* of the decomposition, or rather, taking the expectation within $\theta \in \Theta$ of the structural description of $h \in \mathcal{H}$. Then, we shall have

$$\begin{aligned}\mathop{\mathbb{E}_{\mathrm{SC}}}_{h\in\mathcal{H}}[\ell(h)] &= \mathbb{E}_x\mathbb{E}_\theta\mathbb{E}_\epsilon\left[\hat{y}(x) - y(x)\right]^2 \\ &= \mathbb{E}_x\mathbb{E}_\theta\left(\mathbb{E}_\epsilon[\hat{y}(x)] - y(x)\right)^2 + \mathbb{E}_x\mathbb{E}_\theta\mathbb{V}_\epsilon\left[\hat{y}(x)\right]\end{aligned} \tag{50}$$

Do note that the order matters, since the expectation can be constant of a given ordering. And expected value over constant is just the constant.

2. **Symmetric decomposition**: This approach is fairly difficult, we refer to the paper itself of its own justification on the decomposition. The symmetric decomposition relies on the shape of the input, $\mathcal{X} = \{P, D, \epsilon\}$, of which you can remove the ϵ term and retain some decomposition on P and D per distribution.

4.1.3 Generalized decomposition

According to P. Domingos 2000b, 2000a, the bias-variance decomposition can usually be treated as being controlled by three factor error, with their respective coefficients $\lambda = (\lambda_1, \lambda_2, \lambda_3)$ for such:

$$\begin{aligned}&\mathbb{E}_\mathcal{D}\left[d((f, \mathbb{E}[y \mid x]))\right] \\ &\quad= \lambda_1\mathrm{Bias}(f, y) + \lambda_2\mathrm{Var}(f, y) + \lambda_3\epsilon(\mathcal{D}) \\ &\quad= \lambda_1 \underbrace{\{\mathbb{E}_\mathcal{D}[f(x;\mathcal{D})], \mathbb{E}[y \mid x]\}}_{\text{bias term}} \\ &\qquad+ \lambda_2 \underbrace{\mathbb{E}_\mathcal{D}\{(f(x;\mathcal{D}), \mathbb{E}_\mathcal{D}[f(x;\mathcal{D})])\}}_{\text{variance term}} \\ &\qquad+ \underbrace{\lambda_3\epsilon}_{\text{irreducible error}}\end{aligned}$$

where $\epsilon(\mathcal{D})$ is the irreducible error of the system, depends (theoretically) on the intrinsic imperfection of the dataset $\mathcal{D}$. This method of which the bias-variance-irreducible error are 'fit' into the total error by three coefficients $\lambda_1, \lambda_2, \lambda_3$ instead. This does not only scale up the B-V-IE triplet, but also leaves out the inexplainable gap between the scaling, and the actual normal measure.

4.1.4 Bias-variance algorithm

4.1.5 Functional approximation algorithm

Usually, the optimization problem is evaluated empirically of its bias-variance metric. That is, we test singular models successively, and then choose the generally best — usually within bias-variance tradeoff. This can be expressed as a function for theorem 4.2.

4.1.6 Approximation-estimation tradeoff

Another closely related notion to bias-variance is the concept of approximation-estimation tradeoff. We refer to Mohri, Rostamizadeh, and Talwalkar 2012; Lafon and Thomas 2024 for some analysis and mentions.

Let $\mathcal{H}$ be a family of functions mapping $\mathcal{X} \to \{1, -1\}$. This is the particular case of **binary classification**, in which can be straightforwardly extended to different tasks and loss functions. The *excess error* of a hypothesis $h \in \mathcal{H}$, is the difference between its error $R(h)$ and the Bayes error R^*. This can be decomposed to be the following:

$$R(h) - R^* = \left(R(h) - \inf_{h\in\mathcal{H}} R(h)\right) + \left(\inf_{h\in\mathcal{H}} R(h) - R^*\right) \quad (51)$$

The first bracket contains the **estimation error**, and the second bracket contains what is called the **approximation error**. The estimation error depends on the hypothesis h selected. It measures the error of h with respect to the infimum of the error achieved by hypotheses in $\mathcal{H}$, or that of the best-in-class hypothesis h^* when that infimum is reached. The approximation error measures how well the Bayes error can be approximated using $\mathcal{H}$. It is a property of the hypothesis set $\mathcal{H}$, a measure of its richness.

Model selection consists of choosing $\mathcal{H}$ with a favourable tradeoff between the approximation and the estimation error. However, in the most general case, this will be done, but not in practice, as it requires the underlying distribution $\mathcal{D}$ to be known to determine R^*, which is not possible. In contrast, the estimation error can be bounded, or can be analysed, using particular metric and analysis. It is however, worth to note that bias-variance and approximation-estimation have a very complicated relationship, for example, in Brown and Ali 2024 shown that they are indeed not the same, and is in fact two different decomposition, where one is the others' component. To see this, let us compare the two of them in details.

4.2 Double descent

Nevertheless, of bias-variance and the analogous statistical learning theory concept, the target is the same. It is the dilemma of which is presented in **Occam's razor**, for choosing the sufficient model of good complexity, or bias, for tradeoff of its generalization ability, or variance. Then there must exist a sweet spot between the axis of bias and variance, since they are as exhibited above inversely proportional to each other. However, double descent seemingly broke the status quo, and insists on an interesting phenomenon - under the same setting, if we 'crank' the complexity high enough, we will then reach a point then called the **interpolation threshold**, such that the trend reverse and the error rate, instead of being theorized to go up, goes down to a certain line of lower bound.

The first identification of the double descent phenomena dated back to the paper of Belkin - Belkin et al. 2019a, in which the title is literally "reconciling" modern machine learning practice and the bias-variance tradeoff. In modern machine learning practice, or state-of-the-art developments, models are now bigger than ever. If to notice, we will see that currently models are inherently large, for example, a normal large language model will have from 900 millions (900M) to a few billions, for example 10 billions (10B) parameters. That is not taking into account the overall dynamics and structure of the model, which dictates the operating range and efficiency of the model itself. These model, based on the neural network architecture are somewhat trained to exactly fit (or interpolate) the data, almost certainly so that it turn from a prediction setting to an estimation setting. By statistical learning theory, this would be considered overfitting, and yet, they often obtain very high accuracy on test data.

The main finding that Belkin found is a pattern for how the apparent performance on unseen data depends on model capacity and the mechanism underlying the emergence of double descent. When function class capacity is below the "interpolation threshold", learned predictors exhibit the classical U-shaped curve from Figure 4. The 'modern' interpolating regime marks the opposite trend to the right, where the risk starts to decrease up to a lower bound, which then can be called the *optimal descent bound*.

> The bottom of the U is achieved at the sweet spot which balances the fit to the training data and the susceptibility to over-fitting: to the left of the sweet spot, predictors are under-fit, and immediately to the right, predictors are over-fit. When we increase the function class capacity high enough (e.g., by increasing the number of features or the size of the neural network architecture), the learned predictors achieve (near) perfect fits to the training data—i.e., interpolation. Although the learned predictors obtained at the interpolation threshold typically have high risk, we show that increasing the function class capacity beyond this point leads to decreasing risk, typically going below the risk achieved at the sweet spot in the "classical" regime. (Belkin et al. 2019a)

At this point, we would like to also clarify the term interpolation threshold in this case. Usually, under the pretext of overfitting and underfitting, interpolation threshold means the following:

Definition 4.1 (Interpolation threshold). *Assuming a typical supervised learning setting, for a hypothesis h within class structure $\mathcal{H}$, and a supposed concept c of the class $\mathcal{C}$. Suppose* $[X, Y] \subset c \in \mathcal{C}$

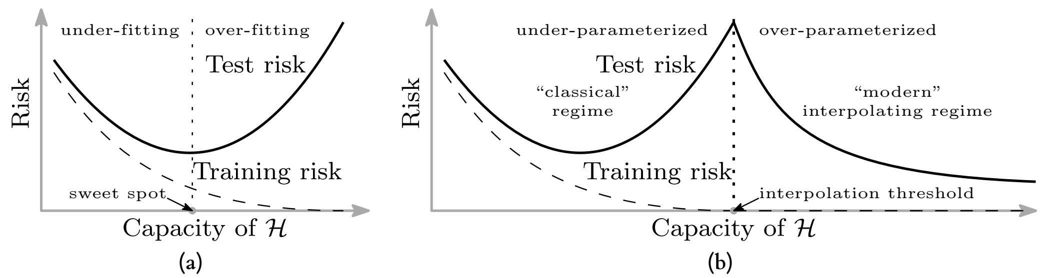


Figure 4. Curves for training risk (dashed line) and test risk (solid line). (a) The classical *U-shaped risk curve* arising from the bias-variance trade-off. (b) The *double descent risk curve*, which incorporates the U-shaped risk curve (i.e., the "classical" regime) together with the observed behaviour from using high capacity function classes (i.e., the "modern" interpolating regime), separated by the interpolation threshold. The predictors to the right of the interpolation threshold have zero training risk. Reproduced from Belkin et al. 2019a.

being the dataset, or observable space of c on which h is tested on. Under the ***empirical generalization learning*** *procedure, partitioning $[X, Y]$ into $[X, Y]_1$ and $[X, Y]_2$, for error evaluated on the first set, $\hat{R}_1(h)$ called training error, and the second set, $\hat{R}_2(h)$ called test error, with optimization applied on the first set, then evaluated statically on the second, the* ***interpolation threshold*** *with respect to a particular complexity measure of h, $\mathcal{M}(h)$, is the point $\mathcal{M}_0(h)$ where it is possible for $\hat{R}_2(h)/\hat{R}_1(h) \approx 1$, for $\hat{R}_1(h) \ll \epsilon$, $\hat{R}_2(h) \ll \epsilon$, with $\epsilon \geq 0$.*

This basically means that what is learnt, with respect to the model complexity that gauge *how much* and *what* the model can learn, perfectly fits the concept c itself, from the observation space on its own. We can say that, that specific sense, we gain the assumption that the observation space is *fully representative* of the concept class, and the concept class is then fully realized by the model. Such point where that is possible, or there then such behaviour is possible, is called the interpolation point. The definition relies on the statement 'where it is possible', hence it still, does not give us a particularly sound notion on to how and what constitute such possibility. The term empirical generalization learning comes from the fact that the procedure in practice of estimating and optimizing train-test-validation set in itself, is an imperfect empirical approximation of generalization testing. While the fact that we cannot gauge generalization error, in a purely blind black-box setting is settled (Mohri, Rostamizadeh, and Talwalkar 2012; Sugiyama 2015; Shalev-Shwartz and Ben-David 2014), the method is then to try to mimic such by using generalization-via-hidden observations. To facilitate this, deliberately, the empirical learning procedure is only dynamic during training session, and static throughout the whole testing and validation testing section, and further on.

By default, in general theory, such interpolating point is dangerous. Not only because it is inherently difficult to know exactly what the dataset constitute, whether it is actually representative of the concept that gives rise to it, or the structural information, noises, and interferences in between. It is hence impossible, to a certain point, to truly know the effectiveness of the model beside interpolating the current dataset which forms the observable details.

Another prominent discovery to look at is Nakkiran et al. 2019, on the double descent of deep learning models. The paper serves as the first observation upon which double descent is observed on modern, deep learning theoretic models, especially complex one like Transformer or ResNet networks.

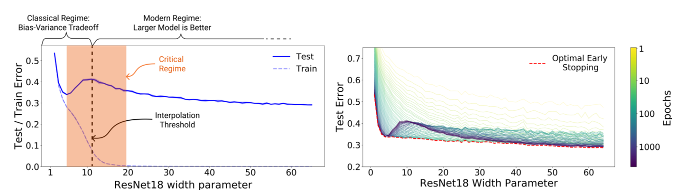


Figure 5. Left: Train and test error as a function of model size, for ResNet18s of varying width on CIFAR-10 with 15% label noise. **Right:** Test error, shown for varying train epochs. All models trained using Adam for 4K epochs. The largest model (width 64) corresponds to standard ResNet18. Resued from Nakkiran et al. 2019.

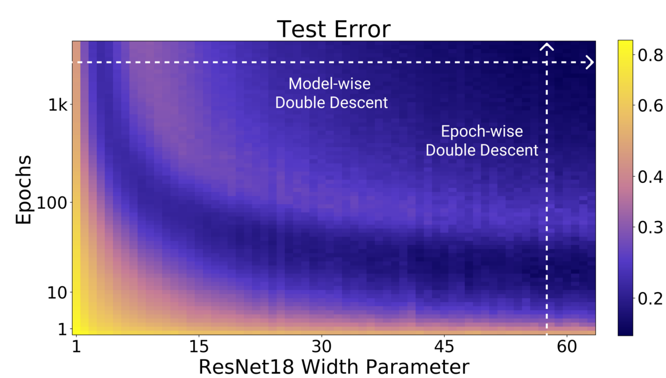


Figure 6. Test error as a function of model size and train epochs. The horizontal line corresponds to model-wise double descent–varying model size while training for as long as possible. The vertical line corresponds to epoch-wise double descent, with test error undergoing double-descent as train time increases. Reused from Nakkiran et al. 2019

They define *effective model complexity* of $\mathcal{T}$ (w.r.t. distribution $\mathcal{D}$) to be the maximum number of samples n on which $\mathcal{T}$ achieves on average ≈ 0 *training error*. This is an entirely empirical definition, similarly per definition as VC-dimension.

Definition 4.2 (Effective Model Complexity)**.** *The* Effective Model Complexity *(EMC) of a training procedure $\mathcal{T}$, with respect*

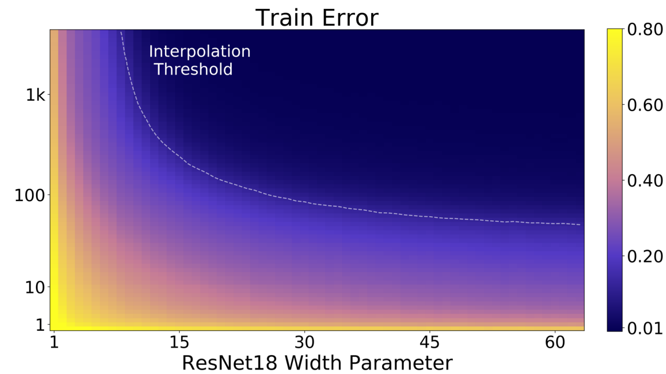


Figure 7. Train error of the corresponding models. All models are Resnet18s trained on CIFAR-10 with 15% label noise, data-augmentation, and Adam for up to 4K epochs. Reused from Nakkiran et al. 2019

to distribution $\mathcal{D}$ and parameter $\epsilon > 0$, is defined as:

$$\mathrm{EMC}_{\mathcal{D},\epsilon}(\mathcal{T}) := \max\left\{n \mid \mathbb{E}_{S\sim\mathcal{D}^n}[\mathrm{Error}_S(\mathcal{T}(S))] \leq \epsilon\right\}$$

where $\mathrm{Error}_S(M)$ *is the mean error of model M on train samples S.*

Using this definition, their main hypothesis can then be stated as the following three-fold regions:

Hypothesis 4.1 (Generalized Double Descent hypothesis, informal)**.** *For any natural data distribution $\mathcal{D}$, neural-network-based training procedure $\mathcal{T}$, and small $\epsilon > 0$, if we consider the task of predicting labels based on n samples from $\mathcal{D}$ then:*

Under-parameterized regime. *If $\mathrm{EMC}_{\mathcal{D},\epsilon}(\mathcal{T})$ is sufficiently smaller than n, any perturbation of $\mathcal{T}$ that increases its effective complexity will decrease the test error.*
Over-parameterized regime. *If $\mathrm{EMC}_{\mathcal{D},\epsilon}(\mathcal{T})$ is sufficiently larger than n, any perturbation of $\mathcal{T}$ that increases its effective complexity will decrease the test error.*
Critically parameterized regime. *If $\mathrm{EMC}_{\mathcal{D},\epsilon}(\mathcal{T}) \approx n$, then a perturbation of $\mathcal{T}$ that increases its effective complexity might decrease* **or increase** *the test error.*

It is to notice that this definition is also only observational. By that, it only outlines the specific region of interest where the paradigm shift is identified through experimental results. It has no effective predictive power, or rather descriptive power more than setting up hypothesis with respect to the effective model complexity, and some arbitrary perturbation. Nakkiran et al. 2019 also noticed this difficulty in providing a theoretical definition and theorem regarding such hypothesis, as said in their manuscript. The existence of the critical regime is also not defined.

The behaviour itself has been particularly investigated, in certain settings. For example, before Belkin et al. 2019a, Advani and Saxe 2017 investigated the generalization error in neural networks of high-dimensional measure. Of such, various behaviours that bear similarity to double descent can be observed. Belkin, Ma, and Mandal 2018 expanded on his previous research on the particular subset of kernel learning models, providing a theoretical analysis of specifically the Laplacian kernel on standard neural network. Mei and Montanari 2020 investigated random feature regression in such regard, and also found similar results.

The fact that double descent is not well-defined itself is a problem on its own. Up to the author's knowledge and of myself, there has been no effective definition or given description that outlines specifically how double descent can be structured. Instability in reproducing experiments and the effective range of the phenomena remains a question on its own.

5. Current researches

Ever since Belkin et al. 2019a, despite the general chaotic view of double descent that seems to hold up pretty well, many researches have been conducted throughout, trying to observe or explain it, as we briefly mentioned Nakkiran et al. 2019 as one of such main paper. Before attempting to do anything, and to further contribute with hypotheses forward, we will have to resolve to review previous coursework on such matter. Most of those works include some on theoretical analysis, some on dissection of optimization method then speculation of possible hindsight, some on experiments and observable effects, and so on.

As such, the first thing to do is to contrive the consensus on 'what' constitutes double descent, at least per observable results, by delivering a conjecture.

Conjecture 5.1. *On the landscape of double descent, model complexity versus generalization error, there exists a fundamental point $\mathcal{M}_0$ of model complexity axis, where $dR(h)/d\mathcal{M} < 0$ for $\mathcal{M} > \mathcal{M}_0$, which is the interpolation point.*

While this is not perfect, it provides us with a general idea on where to look for double descent. With this, we are ready to follow up the literatures in provision of said questions. Let us also recite what happens within the setting of many double descent results.

5.1 Discovery of double descent

While double descent is discovered by Belkin et al. 2019a and further on modern deep learning landscape of Nakkiran et al. 2019, it remains that the conclusion is not reached if there exists some fundamental models that would not have double descent, the consistency of double descent through different models entirely and so on. This is mainly because of a particular 'shock effect', since before double descent, bias-variance holds the candle as the main principle of choosing and optimizing models to its optimal saddle point. Suddenly by then we observe double descent, which introduces the interpolating threshold $\mathcal{M}_0$, that breaks this principle. There is then the open question on how far-reached is double descent in consideration of different architectures, and this is reflected in several papers above. As of now, the main structures that have been verified includes

- **Classical models** Yang and Suzuki 2024; Nakkiran 2019a;

Schaeffer et al. 2023; Lafon and Thomas 2024 (linear regression and classical polynomial), Li and Sonthalia 2024 (linear regression upon least square), Liu, Liao, and Suykens 2021, Y. S. Zhang 2024; Allerbo 2025 (kernel methods/regressions), Brellmann et al. 2024 (Classical MSBE-LSTD[d] optimization), Harzli et al. 2023 (Gaussian process). Some papers also have experiments on Random Fourier Feature Networks, and so on, but not substantial (including Belkin et al. 2019a paper).

- **Deep neural networks**: Luo and Dong 2023; Yang and Suzuki 2024; Pezeshki et al. 2021 (CNN, VGG, ResNet, DenseNet), Nakkiran et al. 2019 (Transformer), Shi et al. 2024 (GNN).

Most of the experiments are concerned of seeing and analysing double descent, basic analysis and experimental observations, test case scenarios for different models and setting and so on. All of the above models are then considered to be, concretely, reported of observing double descent, through various testing scenario on them. Review papers and literatures can be found as Schaeffer et al. 2023; Hu 2021, and Shi et al. 2024 on specifically, graph neural network.

5.2 Theoretical attacks

Because of its intrinsic nature, where double descent breaks the long-standing principle of learning selection, theoretical resolutions has been employed to tackle such problem. Indeed, for example, Cherkassky and Lee 2024 and Lee and Cherkassky 2022 aim toward such direction by understanding the behaviours of double descent using the classical VC-theory approach. However, such papers only introduce, or relate double descent to VC-theory via phenomenological approach - for example, reducing the VC-dimension while fixing or reducing training error, or by examining a potential proxy between the VC-bounds and potential factors encapsulated in the quantities involved. They however, do raise voices of certain inherent problems lies within the theory of learning and in general, learning models.

Beyond such, certain discovery and inquiries help in having a clearer picture on the behaviour of double descent through various structures. For example, Hastie et al. 2019, under pretext of studying high-dimensional ridgeless (ℓ_2 norm) interpolation, analyse the case of *isotropic features*, *correlated features*, misspecified model (feature observation inhibited), on ridgeless interpolation of linear model, and soon as nonlinear model and latent space model. While strong, their results are restricted to a specific class of assumptions and structure, and is doubtful to be extended further. Allerbo 2025 investigated and find a particular double descent trace when changing kernel during training, specified in the term of the minimum allowed bandwidth. The kernel bandwidth $\sigma \geq 0$ of a specific kernel filter determines the information smoothing, hence implies the complexity abides by the kernel structure, as well as determine the smoothness of information indicted by the term.

The sentiments that are shared in between researchers on the topic of double descent vary, but have some common roots, most of them coming from the difficulty to fully assess double descent and its factor components. While double descent concerns the evaluation of *generalization (test) error*, and the *model complexity* notions, the error in question is often implicitly defined or loosely considered in one way or another, and model complexity has no general consensus on its definition. In fact, this points out a very potential, if not already mismatch in the theoretical development. While optimizations and phenomenological aberration of models, as well as building and constructing models at will are well-developed, the definitions on which they are defined on is not fully understood, if not undefined. As of now, literatures on model complexities like al., Hu et al. 2021a; Janik and Witaszczyk 2021; Truong 2025; Luo, Wang, and Huang 2024 introduce their variations or a copula of different definitions altogether, where some are concerned with its effective complexity, some are concerned with its expressive or structural complexity with different ways to define the model underneath, from parameters to units, to random standard variables (Neal et al. 2018). There is also the rift in consideration between classical theory, of which mainly is for classical models, and modern deep neural network structures of which do not conform to such analysis.

5.3 High-dimensional ridgeless interpolation - Hastie et al. 2019

Aligning with our previous experiments, Hastie et al. 2019 conducted investigation into the class of ridgeless approximation criterion on linear-nonlinear classes disparity. The investigation itself is thorough in parameter-term. Specifically, they investigated the case of which we have **isotropic features**, where asymptotic risk depends on the weight w only through its norm $\|w\|_2$; **anisotropic features**, where the model risk depends on the geometry of the pair (Σ, w) for $\Sigma = \mathrm{Cov}(x_i)$ on the dataset $(y_i, x_i)^n_{i\leq n} \in \mathcal{S}$; and the case of nonlinear model, where the model itself is tested with added structure - x_i is obtained by passing $z_i \sim \mathcal{N}(0, I_d)$ through a one-layer neural network with random first layer weights.

5.4 Sample-wise double descent - Nakkiran 2019a

Regarding double descent, Nakkiran 2019a explored the problem, in which the increment of sample actually hurts linear regression process.

5.5 Multiscale feature learning dynamics - Pezeshki et al. 2021

5.6 Reinforcement learning and double descent - Brellmann et al. 2024

5.7 Out-of-Distribution Detection - Ammar et al. 2025

5.8 Gaussian processes and neural network - Harzli et al. 2023

5.9 Double descent on GNN - Shi et al. 2024

5.10 Weak feature double descent - Belkin, Hsu, and Xu 2020

5.11 Grokking, Emergence, and Double Descent - Huang et al. 2025

5.12 Variations on analysis

Many sub-variations and analysis alongside major substantiated discoveries listed above can help in analysis. Some of them

d. Mean-Squared Bellman Error and Least-Squared Temporal Difference are toward linking different existing dynamics to large neural network in non-linear fashion, or generalizing approximation of linear network toward nonlinear systems.

5.12.1 Dropout and double descent - Yang and Suzuki 2024

Dropout is a relatively well-established regularization technique for training deep neural networks, usually by exclusion of different neurons during processes, as seen i Srivastava et al. 2014a to control overfitting. Accordingly, Yang and Suzuki 2024 predicted or stated that regularization techniques can nullify double descent, and thus also dropout. The model considered is linear regression model, for p covariates $x \in \mathbb{R}^p$ and response or output $y \in \mathbb{R}$ are related by

$$y = x^\top \beta_0 + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2) \tag{52}$$

The situated dropout is as followed. Given training data $z^n = \{(x_i, y_i)\}_{i=1}^n$, for $X = [x_1, \ldots, x_n]^\top$ and $y = [y_1, \ldots, y_n]^\top$, the estimation of weight β by $\hat{\beta}(z^n)$ (model's weight), to minimizing the training error

$$\mathop{\mathbb{E}}_{r \sim \mathcal{B}er} \|y - (r \odot X)\beta\|_2^2 = \|y - \gamma X\beta\|_2^2 + (1-\gamma)\gamma\|\Gamma\beta\|_2^2 \tag{53}$$

for $\odot$ denoting the element-wise product on L^2 norm space, each element of $R \in \mathbb{R}^{n \times p}$ takes one and zero with probabilities γ and $1-\gamma$, for the distribution $\mathcal{B}er(\gamma)$ corresponding to such, and $\Gamma = \operatorname{diag}(X^\top X)^{1/2}$. The final description gives the form of the effective dropout objective. Originally from the paper, they consider it exactly of a Tikhonov regularization method. Set $\beta' = \gamma\beta$, then

$$\begin{aligned} &\|y - X\beta'\|_2^2 + \frac{1-\gamma}{\gamma}\|\Gamma\beta'\|^2, \\ &\hat{\beta}_{n\gamma} = \left(X^\top X + \frac{1-\gamma}{\gamma}\Gamma^\top\Gamma\right)^{-1} X^\top y \end{aligned} \tag{54}$$

minimization. While experiments regarding such regularization works in such domain and was illustrated as to be reasonably effective, fallacies in claims and insight remains, especially in the nullification of double descent. The problem here is that, any given amount of dropout is, accordingly to the author, similar to Tikhonov or ridge regularization on structural complexity. As explained in the c' optimization of c-space in preliminary sections, such induces a different, much smaller system in which the additional structural attractor points toward the region of low regularizing complexity throughout the domain. Such superposition between the $\{\lambda r(h)\}(\mathbb{R})$ landscape and the classical loss $\hat{R}$ only morph the dynamic setting to a different region, and thereby bypass double descent in such case for absolute non-overfitting convergence to $\min(\hat{R}_{\mathcal{S}}(h))$. This works well on well-behaved and loosely stable case, but in a lot of scenarios, such cannot be guaranteed, especially for pathological systems. Furthermore, Speaking in terms of the *inductive bias*, mismatch between model and effective regularization becomes a problem, as there exists no effective complexity measure for the regularization term $r(h)$. Some are ill-conditioned, for example, $r(h) = \sum_j w_{ij}^2$ for square terms ij-indexed weights only take into account the reasonable weight system, without the semantic itself, or meaningful weights. A forced dropout might potentially in certain less ideal cases, be more destructive, and introduces double descent back again, until the stochastic process shifts meaningful weight to a more resource constrained system. This means we have to induce, or at least formalize if stochastic process can recover meaningful semantic after dropout randomly. We can test this somewhat, by introducing a small toy test on the proposed dropout estimator:

Theorem 5.1 (Expected test error, Yang and Suzuki 2024). *Let $\alpha = C < \frac{1}{(1+\sqrt{p/n})^2}$, the expected test error is*

$$f(\alpha) = \left\{1 - a\alpha + 2\alpha^2\frac{p}{n}\right\}\|\beta_0\|^2 + \sigma^2\alpha^2(\alpha+1)\frac{p}{n} + O\left(\frac{1}{n^2}\right) \tag{55}$$

with $\alpha = \gamma/(1-\gamma)$.

While the theorem is proved, we then must know if such theorem correctly correlates to even small dynamic systems.

6. Problems

There are a lot of problems regarding how we treat the problem, how we see it, and how we actively analyze it. Most of them lies in the space of statistical learning theory, since bias-variance on itself has variations and justification that is seemingly very natural in statistical theory, like the other version name *approximation-estimation error tradeoff*. All of them contribute somewhat to the current outlook, and so far, it has been pretty bad, than what is being advocated of. Nevertheless, let us state them out, even if missing some particular details.

Problem 1. Authoritative statements

Despite its prevalence in particular research interest, double descent itself has no definite statement to describe its particular behaviour. This is also true for a list of other phenomena, such as *grokking* (Power et al. 2022), *emergence behaviours* (Wei et al. 2022, typically discussed in large language models, which is understandable of large-scale specific ruling system), *implicit bias* (Soudry et al. 2024), and *neural scaling law* (Kaplan et al. 2020; Wei et al. 2022). There is also *catastrophic forgetting* (Ven, Soures, and Kudithipudi 2024) which is still fairly native of large systems; nevertheless, those phenomena will be of target later on (or discussed in later chapters). The fact remains that we only *partially understand* these particular results with conjectures and theories to support the facilitation of new insights, new analytical lens, and nothing much. Without a clear definition of what is even double descent, sometimes, we cannot proceed, lest to say even solving it. This is particularly troublesome, since the current theory is so messy and not rigid that many even turns to exotic choices, like topological informatics learning theory, *Homotopy Theoretic and Categorical Models* (Manin and Marcolli 2024), and several more. So this is definitely one of the main problem to pushback on.

More catastrophic is that, the term double descent receives no formal definition aside from the principle empirical definition.

Problem 2. Model complexity

The notion of model complexity can be said to be often not so well-defined. There are attempts by Hu et al. 2021b; Luo, Wang, and Huang 2024; Barceló et al. 2020; Molnar, Casalicchio, and Bischl 2020; Janik and Witaszczyk 2021. The problem of model complexity is not so apparent until the inherent complex nature of models, typically in deep learning models (based on the MLP architecture) is observed to cannot be treated in similar way as classical models. In Hu et al. 2021b, most deep learning models are based on, and investigated of their complexity through measures like the expressive capacity, the effective capacity, and so on; aside from other teams and researchers investigated it by using computational complexity, or functional decomposition. One promising aspect of this is Molnar, Casalicchio, and Bischl 2020, where we are able to decompose the functional form a specific model into variations of its complexity.

Nevertheless, it is reasonable to say there are no conclusive measure or definition of a model's complexity. Considering that double descent also appears outside the range of deep learning model, and is somehow consistent in classical models (for example in models tested by Belkin et al. 2019a). This is particularly troublesome, as a theoretical analysis requires such definition, especially when the relation of interest is directly involved in such manner. This is a particular aspect needed to be resolved if there are to be progress made in this investigation. Additionally, the concept of overparameterization and underparameterization is also a problem, especially since its definition is not well-defined, hence become a problem in analysis by identifying the classification of the problem setting.

Problem 3. Model structure

In modern practice, the theoretical and formal treatment, rigorous consideration of different model structures and complexity is not realized. Partially, this is due to the bloom of machine learning and artificial intelligence from the early 2000s. Currently, there has been no conclusive theoretical definition or formulation about the structure of different models, typically can go under the name of mathematical modelling hypothesis, in a way that unify certain properties between architectures. Heuristics are also often employed in several newer models and widely-utilized structures typically seen in large-scale systems, leading to difficulty in formalization and factorization of affectants.

Problem 4. Theoretical insufficiency

The setting surrounding the learning theory, machine learning models is generally missing of its rigours and analytical properties. Most of the learning setting and model's training-testing setting are often narrowly stated in theoretical sense, and there are a lot of diffusing details that might or might not affect the model's perturbation without knowing. This in a certain way leads to the complexity in analysing different problem setting and experimental results, since the architecture used, the setting considered, model in questions, configuration (either custom or on system side) are not consistent overall.

Furthermore, a lot of terms, definitions are often hand-waived in papers or in discussions. This also led to the point that in Nakkiran et al. 2019, they have to somehow 'reinvent' another type of model complexity itself, albeit unsatisfying of a definition, is still a new kind of definition per insufficiently of such. while it is welcome to introduce new definitions, the fact is that it is subjectively inconclusive of terms, definitions and thereof leads to the perception and situation where multiple studies can be found, but all of them are defined on different ground.

Analysing statistical learning theory and overall landscape of statistical learning theory, and the learning theory as a whole, encountering new problems like double descent also revealed its weakness, and ultimately, perhaps one of the reason why it is ineffective against such problem. First, there are simply too many assumptions made, too many formulations made during said process. Furthermore, there are also unclear notions and concepts, of which make it even harder to analyse or fully formalize. Secondly, there are inherent conflicts and uncertainty within those theories, formulations and notions by itself. For example, in one sense, the No-free-lunch theorem is considered to be representative and true, whilst also simultaneously being considered the opposite of such. And amidst abnormality behaviours of the old bias-variance formulation, we also find distinctive weakness in our theory, for example, the concerning difficulty in defining the notion of *model complexity* in various contextual ways. To analyse double descent, perhaps we also need a new theory or formulation to support it. [e]

Another problem with classical analytical solution is that it is very *loose*, almost too loose to even considered of such. But sometimes it is too *rigid* in a sense such that it simply does not work at all. Indeed, basing our works and foundational assumptions on top of approximation theory is not quite good at all, because they are very limited. Overall, we can say it is not equipped for handling similar situation. For example, almost all bounds that is subjected of classical learning theory only

e. Furthermore, most of the general solution and bounds created by statistical learning theory is often in a very simplistic system. For example, if we are to utilize the Rademacher complexity measure (Mohri, Rostamizadeh, and Talwalkar 2012), most of the time we will have to compute it through the growth function $\Pi_{\mathcal{H}}(m)$ for m points, on the standard finite hypothesis class $\mathcal{H}$ such that

$$\Pi_{\mathcal{H}}(m) = \max_{x_1,\dots,x_m \in \mathcal{X}} \left|\left\{(h(x_1),\dots,h(x_m)) \mid h \in \mathcal{H}\right\}\right| \tag{56}$$

Most of our problems resolve to binary classification, or rather, the problem of pattern recognizing discrete, reduced classification form. It is not so sure for now if all problems can be reduced to such way, so we cannot draw conclusive analysis that is not diluted of mathematical formulation for complex systems. That is not to count the computationally intensive operation required to calculate the supposedly classical measure, while not entirely of itself holds any substantial reasonable information about the internal dynamics of the model itself.

guarantee particular existential theorem, under very simplified and specific situation without the virtue of extending it to a more general setting.

7. Hypothesis

In learning and researching of the progress and treatment of double descent, several hypotheses, most of the time observations and insights such, is formed. Hence, there are several hypotheses for what to look out for, as well as the formulation needed to interpret certain phenomena and observations. This ranges from topics of re-evaluating the statistical learning theory, estimation and refactor of assumptions (or *inductive bias* when designing models), to different interpretations and representations that can better explain, model, or gives certain insights.

7.0.1 Stability of bias-variance measure

Stability of bias-variance measure is questionable, as it is in literature (P. M. Domingos 2000a), loosely defined by defining a 3-dimensional vector $\lambda = (\lambda_1, \lambda_2, \lambda_3)$ for such:

$$\begin{aligned}
&\mathbb{E}_{\mathcal{D}}\left[d((f, \mathbb{E}[y \mid x])\right] \\
&= \lambda_1 \mathrm{Bias}(f, y) + \lambda_2 \mathrm{Var}(f, y) + \lambda_3 \epsilon(\mathcal{D}) \\
&= \lambda_1 \underbrace{\left\{\mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})], \mathbb{E}[y \mid x]\right\}}_{\text{bias term}} \\
&\quad + \lambda_2 \underbrace{\mathbb{E}_{\mathcal{D}}\left\{(f(x; \mathcal{D}), \mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})])\right\}}_{\text{variance term}} \\
&\quad + \underbrace{\lambda_3 \epsilon}_{\text{irreducible error}}
\end{aligned}$$

where $\epsilon(\mathcal{D})$ is the irreducible error of the system, depends (theoretically) on the intrinsic imperfection of the dataset $\mathcal{D}$. Assessment on stability of such decomposition is recommended.

7.0.2 Alternative measures

We believe that the bias-variance decomposition is a poor estimation and measure for controlling, validating and choosing model from, at least from the perspective of analysing missing details. Specifically, bias and variance term, as well as their supposed expression of decomposition in the standard loss measure of model effectiveness is not representative, nor expressive of the actual, underlying notion of *model complexity* $C(f)$ and its effectiveness. Furthermore, bias-variance decomposition is not straight-forward of others type of loss functions, except only for some classes of loss function (loss on Bregman divergences and exponential class). This sentiment is shared in Brown and Ali 2024.

7.0.3 Inductive bias

Inductive bias obviously plays a role, as mentioned by some literature [f], though we don't know exactly how the inductive bias can be formed to play any role in the central dynamic of the model operating process. Certain assumption, for example, the *convexity of the loss function* might actually affect the model as it is, and introduce unwanted patterns in the following process.

7.0.4 Randomness and probabilistic setting

Random initialization of parameters, as well as the initial conditions of the model initialization process might also be taken into account. Obviously, this is not a new insight, and is rather a very old one, however, this approach still has varied potential for interpreting double descent. Scaling up/down this effect might result in new insight, as well as inspecting the uncertainty and diffusion patterns in large-scale model deployment, in which from an end-to-end perspective requires several space transformation operations in successions.

7.0.5 Knowledge masking

The very ambiguous idea or treatment of statistical learning, as well as machine learning, is the idea of masked. Or, paraphrased, *hidden information* regarding the underlying process of estimation. Specifically, for example, we tend to assume things to not be able to grasp - either the probability $p_X(c)$ of the concept class, the distribution $\mathcal{D}$ specified to such input distribution, or the concept class $\mathcal{C}$, Markov properties, etc. We would like to formalize this notion as soon and as such in the most direct way, as it is one source of the many assumptions that we have in concession of theoretical results.

7.0.6 Intrinsic Data Factors

The effect that data has on the model should also be taken into account, as of fact. As far as we are concerned of, the shape, practical system that the dataset gives the setting, the factors running into the dataset, the choice of the dataset being sampled, selected in partitions, and other factors must also be analysed. Without such, we leave 'gaps' in which sensitive, potentially chaotic behaviours might occur frequently and would not be identified, or being classified into irreducible errors, static of the learning setting. As such, we might as well list a few potential points of interest.

1. **Data distribution**: the distribution of the dataset itself affects how the concept is represented in the space. If the representation space created by using the data is large, but sparse, then the condition would make for some behaviours to occur more easily. This is one degree of freedom of the system setting. If the dataset spread is low, yet the data is very volatile, then it will lead to inconclusive picture of the actual concept, hence introducing new behaviours on the now unknown region of the distribution. As such, the majority of those unknown spaces will treat every point there as a potential outlier instead.
2. **Data complexity**: The complexity of the data itself is of interest. In a controlled setting of response-inducing evaluation, identifying behaviours of the model with a glimpse on the data complexity - how the actual data works with the internal mechanics accompanied - is of specific interest, as it can affect some of the majorative effects and chaotic behaviours that occur in the learning process, especially stochastic process and

f. Most notably, Lafon and Thomas 2024, of which raised particular point about the need of learning with inductive bias as a structural presumption. Additionally, the paper also includes an analysis on the inductive bias of gradient descent, though solutions and particular application on double descent explanation is doubtful.

probabilistic optimization patterns.

3. **Data partitions**: The partitioning of data in the previously mentioned sample partitioning section might give rise to previously observed behaviours, as well as explaining particular volatility when changing the test partitioning set. The ordering of the dataset also helps in such regard.

It is also worth to note that since we emphasize the use of dataset partitioning to replicate the empirical-generalization guarantee, it is also then indicative that such partitioning would have its separate way of calculating test error on itself.

Intuitively, it can be illustrated as followed. Suppose of given two datasets $\mathcal{S}_1, \mathcal{S}_2$, we can trace a pattern, if, for assumption, the partition is $k = 10$. If the algorithm $\mathcal{A}$ attempt empirical learning optimization specifically, and best optimize on individual partition $\mathcal{S}_1^{(i)}$, then the path toward each and individual partition would potentially be very volatile, depends heavily on the partition randomization, and others factor leading to such. If the algorithm is justified in that sense, to 'decelerate' the optimization to a not so tight fit, the generalization would be much better. This is also the reasoning for the technique of **regularization**, and generalization task. However, again, such procedure cannot help but having volatile dataset itself as a factor of uncertainty. More specifically, the result might work for $\mathcal{S}_1$, but for another dataset $\mathcal{S}_2$, assuming no intersection, that is, $\mathcal{S}_1 \cap \mathcal{S}_2 = \varnothing$, then optimization based on one dataset would not be optimal in another, more so can be counterproductive. This is also used of the reasoning for **cross-validation** technique, of which aims to stabilize the partitioning. However, analytical analysis suggests a wider role of such, in the wider setting of learning theory, particularly in double descent later on.

8. Analysis on-claims

For understanding and analysing double descent and bias-variance trade-off, and furthermore in later section on identifying double descent, we would like to use test models specifically for exhibiting bias-variance, as well as testing hypothesis and forming theoretical conjectures. This is because of the troublesome nature of the phenomena itself, which proves difficult to analyse without sufficient experimental observations. Furthermore, specific analysis on these types of model might prove useful for generalization toward more difficult and complex structure.

Our major focus is on three types of model. First two are classical models and system, linear models on the real space $\mathbb{R}^n \to \mathbb{R}$. Second, is the type of **support vector machine** (SVM) models to highlight the analysis toward data-based fundamental analysis. The third one is the class and collection of different *neural network structures* (MLP, or variations). Our goal is then to expansively analyse, formalize, making conjectures and hypothesis and test them, to find, force double descent out, or identify it on the learning landscape. We note that structures of those models are not well-generalized, hence analysis from one might not apply or apply only in small conjunction. This approach is rather quite helpful in its own merit, as its choice and preference will gauge our parameters than we gauge the phenomena of the parameter it induces.

8.1 Setting: gradient-based optimization

In our usual experiment, we would most specifically, use a closely-regarded family of gradient-based algorithms, such as gradient descent (Cauchy 1847; Robbins and Monro 1951; Widrow and M. E. Hoff 1960; Hestenes and Stiefel 1952; Fletcher and Reeves 1964), mirror descent, natural descent, and so on (Nemirovski and Yudin 1983; Amari 1996, 1998; Yang and Amari 1998; Rattray, Saad, and Amari 1998; Gunasekar, Woodworth, and Srebro 2021; Luong et al. 2016; Xiao and Zhang 2014; Sohl-Dickstein 2012; Raskutti and Mukherjee 2013). While implementation and details of the specific algorithm can be different, they all shares some of the same concepts of which making them possible for optimization in the specified iterative domain. The most general formulation possible is the following algorithm on convexity:

Algorithm 1: Gradient descent on general convex function

Data: $(x_i, y_i) \in \mathcal{S}$, η, $\nabla(\cdot)$ operator.
Result: Expected optimization to the minimal error on $\ell(\cdot,\cdot)$.

```
begin
    x_1 ← randomly initialized points;
    for t ← 1 to T do
        x_{t+1} ← x_t − η∇f(x_t)
    end
end
return x̂ = 1/T ∑_{t=1}^T x_i
```

Nevertheless, one should be cautioned of the intrinsic assumption on which gradient descent of this type assumes. It assumes *convexity*, which can only be possible, if the landscape itself is supposedly, assumed to contain such global minima. Said global minima in a richer language can be described as the epistemic collapse of the learning structure, under the arbitrary optimization space. This is a *very serious assumption*, and should it fail, as seen in a lot of non-populated or general observation space, the entire procedure is invalid. It is also troublesome as to rely on the singular objective (numerically encoded) space in which it is trivial to realise that there are indeed, local minima. This scaling, with respect to both locality and global consideration, must be considered carefully, since *every single contribution* might distort the experiment in a very unexpected way.

8.2 Linear function class $h(u)$ - Lafon and Thomas 2024

Linear-linear scenario learning is one particularly well-studied case of representing double descent or not, usually through simplicity in analysis (of the similar ease of proof compared to binary classification). Models in this setting can be said to share the same *parameter space* with no additional structure or scaling mask (like in monomial basis). The model that is representative of this preliminary observation is from Lafon and Thomas 2024, specifically, linear regression (function approximation) with Gaussian noise, and of univariate case. Consider the family class $(\mathcal{H}_p)_{p\in[\![1,d]\!]}$ of linear functions $h : \mathbb{R}^d \mapsto \mathbb{R}$ where exactly p components are non-zero ($1 \leq p \leq d$), we have the following model.

Definition 8.1 (Lafon and Thomas 2024). *For $p \in [\![1, d]\!]$, $\mathcal{H}_p$ is the set of functions $h : \mathbb{R}^d \mapsto \mathbb{R}$ of the form:*

$$h(u) = u^T w, \quad \textit{for } u \in \mathbb{R}^d$$

With $w \in \mathbb{R}^d$ having exactly p non-zero elements. The complexity here itself can be interpreted as the effective random initialization, trimming down data of hidden details instead - this is indicated by the amount of information it observes given d dimension, and p-hypothesis dimension. [g]

Using such model, they consider the setting of minimizing the customized mean square loss,

$$\min_{w\in\mathbb{R}^d} \frac{1}{2}\|Xw - Y\|^2 \to \min_{w\in\mathbb{R}^p} \frac{1}{2}\|X_{\sim p}w - y\|^2 \tag{57}$$

of the sub-problem on p cardinality, where again p is the dimensionality or independent degree of freedom. This results in the following theorem that 'mimic' double descent on this linear regression model.

Theorem 8.1 (Double descent on linear regression - Lafon and Thomas 2024). *Let $(x, \epsilon) \in \mathbb{R}^d \times \mathbb{R}$ independent random variables with $x \sim \mathcal{N}(0, I)$ and $\epsilon \sim \mathcal{N}(0, \sigma^2)$, and $w \in \mathbb{R}^d$. we assume that the response variable y is defined as $y = x^T w + \sigma\epsilon$. Let $(p, q) \in [\![1, d]\!]^2$ such that $p + q = d$, $X_{\sim p}$ the randomly selected p columns sub-matrix of X. Defining $\hat{w} := \phi_p((\hat{w}), [\hat{w}])$ with $(\hat{w}) = X^+_{\sim p}y$ and $[\hat{w}] = 0$.*
The risk of the predictor associated to $\hat{w}$ is:

$$\mathbb{E}[(y - x^T\hat{w})^2] = \begin{cases} (\|w_{\sim q}\|^2 + \sigma^2)(1 + \frac{p}{n-p-1}) & p \leq n-2 \\ +\infty & n-1 \leq p \leq n+1 \\ \|w_{\sim p}\|^2(1 - \dfrac{n}{p}) + (\|w_{\sim q}\|^2 + \sigma^2)(1 + \dfrac{n}{p-n-1}) & p \geq n+2 \end{cases} \tag{58}$$

The point $p = n$ is representative of the interpolation threshold from underparameterization to overparameterization.

We present two parts - first are those fundamentals predicted by the theorem, and second are verifications of the theorem with our own version of testing. The first experimental setting is then conducted with fixed d, varying p for each varying n samples. Since this is a least square result, preliminary run is conducted using one-shot error-risk measure on least square closed-form approximation. Note that $p \leq d$, hence we are regarding the domain of the term underparameterized and equal parameterization setting. The result is presented in Figure 8

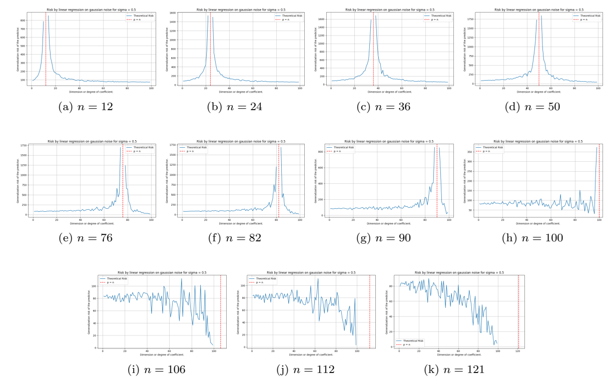


Figure 8. Theorem 8.1 behaviours on randomized setting. For this model, we have $d = 100$, $\sigma = 0.5$, and variational n. Full test cases are for $p = [1, 100]$, $n = \{12, 24, 36, 50, 76, 82, 90, 100, 106, 112, 121\}$ accordingly. The test function of the concept itself is the same linear model, but with different input only.

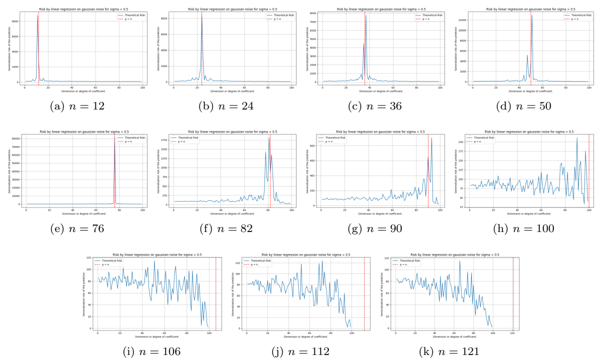
As we see from this particular experimental result, the theorem predicted a loose, left-peak absent, double descent curve estimation for a variety of test case with more and more samples in increment. There exist some fundamental conclusion that we have about the shift of the interpolation threshold upon to $p = n$, which is reflected in the piecewise prediction. However, this pattern breaks down for n higher than specific constant, for this case, starting from $n \approx 100$ to beyond. From here, at least from testing scenario, the test indicates that predictions suggest a breakdown of the double descent estimation. Beyond this, a simple semi-chaotic convergences is observed, from $n = 106, 112, 121$.

Interestingly, verifying this notion of easy enough to simplify it into what the statement holds under particular notion. Namely, as we find out, the above double-descent replicating formula can actually be realized by the parameter-wised error, comparing the parameter between the least-square solution and the actual concept. The formula for this is expressed by the exposed MSE on parameter function, for $\beta_{p,k}$ the coefficient value estimated for the k-th selected feature when the model is restricted for randomization of p features,

$$\text{MSE}_{\text{param}} = |\hat{w} - w|_2^2 = \sum_{j=1}^{d} \left(\hat{w}_j - w_j\right)^2 \tag{59}$$

where $\hat{w}_j$ is chosen such that it is equal 0 for $j \notin S$, otherwise $\beta_{p,k}$ for $S = \{i_1, \dots, i_p\}$ is the set of selected feature indices. Nevertheless, it is then found dubious why this is true, as for the original statement of Lafon and Thomas 2024 indicates correlation to the prediction error of the result, $x^\top\hat{w}$ instead. The result of evaluating $\text{MSE}_{\text{param}}$ is presented in Figure 9.

g. According to the *manifold hypothesis*, there are chances where the actual concept can be interpreted and mimicked by seeing that it lies on a lower-dimension manifold inside its actual expressive dimensional space.



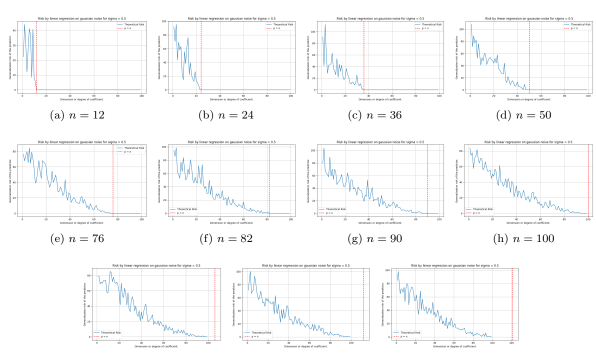
(a) $n = 12$ (b) $n = 24$ (c) $n = 36$ (d) $n = 50$
(e) $n = 76$ (f) $n = 82$ (g) $n = 90$ (h) $n = 100$
(i) $n = 106$ (j) $n = 112$ (k) $n = 121$

Figure 9. Contrary to Theorem 8.1, this is the behaviours on randomized setting for standard parameter-wised mean square error measure (MSE-on-parameter). For this model, we have $d = 100$, $\sigma = 0.5$, and variational n. Full test cases are for $p = [1, 100]$, $n = \{12, 24, 36, 50, 76, 82, 90, 100, 106, 112, 121\}$ accordingly. The test function of the concept itself is the same linear model, but with different input only.

If, however, we turn our attention to the statement itself, of which refers to the prediction error MSE_{pred}, defined by the equation

$$\text{MSE}_{\text{pred}} = \frac{1}{n}\left\|y - X_p\,\hat{\beta}\right\|_2^2 = \frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)} - x_p^{(i)\,T}\,\hat{\beta}\right)^2 \quad (60)$$

on h and c accordingly, a strong case of this happens to be not following the pattern indicted by the theorem. Rather than being the interpolation point of return, $p = n$ is now the convergence point of the prediction error instead. Test result is shown in Figure 10.



(a) $n = 12$ (b) $n = 24$ (c) $n = 36$ (d) $n = 50$
(e) $n = 76$ (f) $n = 82$ (g) $n = 90$ (h) $n = 100$
(i) $n = 106$ (j) $n = 112$ (k) $n = 121$

Figure 10. Contrary to Theorem 8.1 and both figure 9, this is the behaviours on randomized setting for standard parameter-wised mean square error measure (MSE-on-prediction). Especially, we do not see double descent pattern in this setting, as it is. For this model, we have $d = 100$, $\sigma = 0.5$, and variational n. Full test cases are for $p = [1, 100]$, $n = \{12, 24, 36, 50, 76, 82, 90, 100, 106, 112, 121\}$ accordingly. The test function of the concept itself is the same linear model, but with different input only.

Thence, the theorem gives us two things – the interpolation point, in this setting, refers to the parameter error for their double descent behaviour. However, on the prediction results' error measure, the interpolation point correlates to the point where the defined generalization risk in such case equals zero.

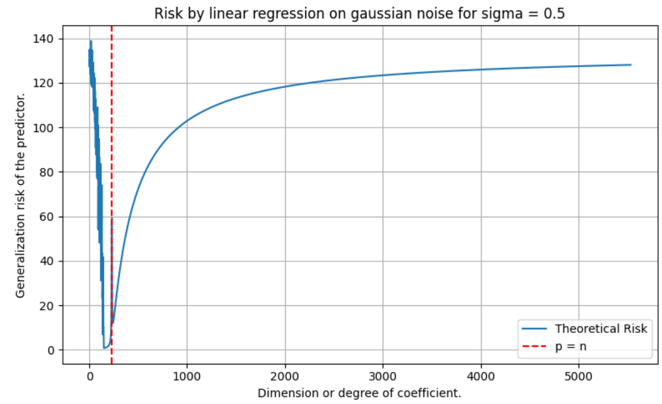


Figure 11. Similar experiment according to setting and result of Theorem 57, where p is in the range $[1, 5523]$. Here, we can see small double descent modelled exactly at the interpolation threshold, while the later region exhibit similar phenomenon as bias-variance tradeoff.

These patterns hold up to certain point. However, the test case is only up to $d = p$. An unexplored test case for $p > d$ indicate the surprising fact according still, to the theorem: the generalization risk takes double descent as a region with disturbance in the middle, but bias–variance appears afterward. This is illustrated in Figure 11, where the same theorem is applied but in the domain which is usually called as *overparameterized region*.

Such can then be seen as a prediction that double descent in certain case, can be extrapolated to be the transition between the two curvature of bias variance, or the abnormal point on the error curve measure. Such must then be verified, both in the experiment's correctness, and so on that lead to such result. Nevertheless, if true, it would mean that double descent indeed has another implication of itself.

8.2.1 Extension

The current theory prompted for a more general treatment of the linear function class, as in-line with exploring reasonable existing alternatives in interpretation (Nakkiran 2019b).

Let us states some hypothesis about the previous experimental cases. First, is the lack of the begin–tail of the prediction on double descent of linear random cutoff features model. This can be explained informally as to be resulted from the uniformity of the operating space. Specifically, all parameters of the $\mathbb{R}^n$ parameter space, with respect to the dual space on such $f : \mathbb{R}^n \to \mathbb{R}$ projection is defined, are of the same order of magnitude. That is, for $f_{i,k}(x), f_{j,k}(x)$ of arbitrary parameter indexing on the set k of shared parameter choice, and $i \neq j$ being the distinct choice, then we say $f_{i,k}(x) = \mathcal{O}(f_{j,k}(x))$ of the growth of both function on value ranges. Approximately speaking, the choice of parameter does not dramatically growth to a point of no return, like in later examples of polynomial approximation model. Such special model of multi-variable linear projection of scalar output also reinforces the numeric growth of the numerical encoding expression on the hypothesis model. The behaviour of the MSE-on-parameter, the internal risk, is also explainable using the same notion, such is that for a black-box case approximation, such linear model does not allow for average parameter deviation per approximation to be over-saturated and skewed (for example, the degree of polynomial directly counteract this, by separating expressive complexity by growth degree). How such hypothesis can be evaluated, we will have to make it for the general extension experiment, which we will have to conduct.

Let us define the more general class of linear model, denoted $\mathcal{H}_L$, in the following structure of the linear model. We also note that the previous linear class $\mathcal{H}_p$ only solves for the problem of underparameterized, hidden parameters p'. Let $k \in \mathbb{N}$ denote the number of input vectors, and let $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k$ represent the input vectors where each $\mathbf{u}_i \in \mathbb{R}^q$ for $i \in \{1, 2, \ldots, k\}$. Here, $q \in \mathbb{N}$ denotes the dimensionality of each input vector, with the convention that $q = 1$ corresponds to scalar inputs (i.e., $u_i \in \mathbb{R}$).

For a given dimension parameter $d \in \mathbb{N}$, we define $\mathcal{H}_L$ as the set of all functions $h : (\mathbb{R}^q)^k \to \mathbb{R}$ that can be expressed in the unified form:

$$h(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k) = \sum_{j=1}^{d} w_j \cdot \phi_j(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k) + b \tag{61}$$

where $\mathbf{w} = [w_1, w_2, \ldots, w_d]^\top \in \mathbb{R}^d$ is the weight vector, $b \in \mathbb{R}$ is the bias term, $\phi_j : (\mathbb{R}^q)^k \to \mathbb{R}$ is a feature map for $j \in \{1, 2, \ldots, d\}$ that transforms the input vectors into a scalar feature, we consider the following important special cases, distinguished by their choice of feature map ϕ_j. If we do not have such feature mapping of which preserves relative linearity, then we said it is the basic linear class dynamic.

Definition 8.2 (Linear class, $q = 1$)**.** *Consider class $h : (\mathbb{R}^{d+1})^q \to \mathbb{R} \in \mathcal{H}_{\mathsf{Lin}}$ of all functionals of the form as Equation 61. Then, when $q = 1$ of scalar observables, i.e., $u_i \in \mathbb{R}$ for $i \in \{1, \ldots, k\}$, then the function is simply as:*

$$h(u_1, u_2, \ldots, u_k) = \sum_{j=1}^{k} w_j \cdot u_j + b \tag{62}$$

for $\phi_j(u_1, \ldots, u_k) = u_j$ for $j \in \{1, 2, \ldots, k\}$, i.e. deactive feature map.

Such class can be extended rigorously, through a varied change in dynamics, specifically, considering the general feature representation on ϕ.

Definition 8.3 (General feature, kernel linear model)**.** *Consider class $\mathcal{H}_{\mathsf{Lin}}$. For maximum flexibility, allow arbitrary feature maps $\phi : (\mathbb{R}^{d+1})^q \times \mathbb{R}^k$ for $k = \{1, 2, \ldots\}$ as of hidden or feature mapping processor internally. The functional form is then expressed by*

$$h(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k) = \sum_{j=1}^{d} w_j \cdot \phi_j(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k) + b \tag{63}$$

for $j \leq d$ as the iterative form.

Common choices for such kernel mapping includes:

- *Polynomial features*: $\phi_j(\mathbf{u}) = u_{i_1} u_{i_2} \cdots u_{i_m}$ for various index combinations
- *Radial basis functions*: $\phi_j(\mathbf{u}_1, \mathbf{u}_2) = \exp(-\gamma|0\mathbf{u}_1 - \mathbf{u}_2|0^2)$
- *Fourier features*: $\phi_j(\mathbf{u}) = \sin(\boldsymbol{\omega}_j^\top \mathbf{u})$ or $\cos(\boldsymbol{\omega}_j^\top \mathbf{u})$
- *Neural network activations*: $\phi_j(\mathbf{u}) = \sigma(\mathbf{v}_j^\top \mathbf{u} + c_j)$ for activation σ,

thus basically hints at the generality of, at least being a *linear approximation* on various hypothesis spaces, while not actually being similar. Then, our setting of sparsity being synthetically induced introduces the basin of subclasses. For any $p \in \{1, 2, \ldots, d\}$, we can then simply define the sparse subclass $\mathcal{H}_{L,p} \subseteq \mathcal{H}_L$ as the set of all functions $h \in \mathcal{H}_L$ where the weight vector $\mathbf{w} \in \mathbb{R}^d$ has exactly p non-zero entries. Formally, per Lafon and Thomas 2024,

$$\mathcal{H}_{L,p} = \{h \in \mathcal{H}_L : |\{j : w_j \neq 0\}| = p\} \tag{64}$$

The subset $S \subseteq \{1, 2, \ldots, d\}$ of indices with $|S| = p$ corresponding to non-zero weights is assumed to be fixed throughout the learning process (i.e., the sparsity pattern is predetermined). This constraint yields:

$$h(\mathbf{u}_1, \ldots, \mathbf{u}_k) = \sum_{j \in S} w_j \cdot \phi_j(\mathbf{u}_1, \ldots, \mathbf{u}_k) + b \tag{65}$$

where only p features contribute to the final prediction. This effectively masks the data randomly, simulating the general observation-based information corruptions, without inducing specific bias like *dropout*, though potentially leads to strong perturbation in cases of meaningful information threshold for different function classes.

9. The polynomial transition problem

Polynomial space is one of those famous example that is often cited of the bias-variance tradeoff, for example, in I. Goodfellow et al. 2016; Sugiyama 2015. Indeed, it is considered, usually, a very effective and high-volatility construct class. Let us reiterate what we know about polynomial class.

There exists two type of elementary polynomial classification, based on their (i) variable count or (ii) its domain range. We first define the univariate polynomial function.

Definition 9.1 (Univariate polynomial)**.** *A function of a single variable t is a polynomial on its domain if we can put it in the form*

$$a_n t^n + a_{n-1} t^{n-1} + \cdots + a_1 t + a_0 t^0 \tag{66}$$

where $\{a_i\}$ are constants, and of a singular variable t.

From this, we can extend to polynomial of multiple variables. In general, a function $f(x, y)$ is a *multivariate polynomial* in x, y if and only if it can be represented as a finite sum of terms of the form $cx^k y^m$, where c is a coefficient and k an m are nonnegative integers. The number $k + m$ is called the degree of the term, and the degree of the polynomial $f(x, y)$ is equal to the highest degree of its own terms. We can then generalize multivariate polynomial and univariate polynomial in the same kind.

Definition 9.2 (Polynomial, multivariate generalization). *Let K be a field. We then denote $K[x_1, \ldots, x_n]$ the space of all polynomials in n variables, $x_1, \ldots, x_n$. A **multivariate monomial** is then an expression $x_1^{\alpha_1} \ldots x_n^{\alpha_n}$ where $\alpha_1, \ldots, \alpha_n$ are integers. The degree ò such multivariate polynomial is the largest degree of terms of any monomial term.*

Effectively, we can see that multivariate polynomial can take on many forms, and also unconventional forms. In order to capture any reasonable details, most of the time focus are on symmetric polynomials $f(x, y)$ in which $f(x, y) = f(y, x)$, or homogeneous polynomials in which all the terms are of the same degree.

Given such, operationally, polynomial is used for expressing other functions, due to their usefulness because of such simplicity and versatility. In fact, applications ranges, even to a polynomial basis on n-dimensional space, $(p_1, \ldots, p_n) \in K[x, \ldots, x_n]$; their growth rate being polynomial also added more to their strength of approximating fast increasing function, usually expressed and used in setting, for example, polynomial growth class $\mathcal{O}(p(x))$ in computer science. Two applications stand out for such properties: function approximation of arbitrary criteria ℓ, or polynomial interpolation on arbitrary points. Those two are interconnected, though their objective and object of concern is much different.

We can define the application of analytical polynomial on arbitrary function by using the criteria on approximation. Usually, this is expressed in the theory of approximation, and requires a guarantee that polynomial can indeed approximate to arbitrary accuracy, of arbitrary function on specific interval. This is the statement of Weierstrass theorem.

Theorem 9.1 (Weierstrass Approximation Theorem). *Let $f \in C[a, b]$ of all continuously differentiable functions on the closed interval $[a, b]$. Then, for every $\epsilon > 0$, there is a polynomial p such that $\|f - p\| < \epsilon$, for $\|f - p\| = \sum_{i=1}^{\infty} |(f(x_i) - p(x_i))|$* [h]

Proof. This is one of the many standard proofs for WAT (Schep 2007 to be specific). We begin by extending f to a bounded uniformly continuous function on $\mathbb{R}$, defining $f(x)$ as to be $f(a)(x-a+1)$ on the domain $[a-1, a)$, for $f(x) = -f(b)(x-b-1)$ on the domain $(b, b+1]$, and $f(x) = 0$ on $\mathbb{R}\setminus[a-1, b+1]$. There exists, in particular, $R > 0$ such that $f(x) = 0$ for $|x| > R$. Let $\epsilon > 0$ and M such that $|f(x)| \leq M$ for all $|x|$. let $\epsilon > 0$ and M such that $|f(x)| \leq M$ for all x. Then by the above theorem there exists $h_0 > 0$ such that for all $x \in \mathbb{R}$, we have

$$|S_{h_0} f(x) - f(x)| < \frac{\epsilon}{2} \tag{67}$$

Since we also have $f(u) = 0$ for $|u| > R$, we can write

$$S_{h_0} f(x) = \frac{1}{h_0\sqrt{\pi}} \int_{-R}^{R} f(u) \exp - \left(\frac{u-x}{h_0}\right)^2 du \tag{68}$$

On $\left[-\frac{2R}{h_0}, \frac{2R}{h_0}\right]$ the power series of e^{-v^2} converges uniformly, so there exists N such that

$$\left| \frac{1}{h_0\sqrt{\pi}} e^{-\left(\frac{u-x}{h_0}\right)^2} - \frac{1}{h_0\sqrt{\pi}} \sum_{k=0}^{N} \frac{(-1)^k}{k!} \left(\frac{u-x}{h_0}\right)^{2k} \right| < \frac{\varepsilon}{4RM}. \tag{69}$$

for all $|x| \leq R$ and all $|u| \leq R$, since in that case $|u - x| \leq 2R$. This implies that

$$\left| S_{h_0} f(x) - \frac{1}{h_0\sqrt{\pi}} \int_{-R}^{R} f(u) \sum_{k=0}^{N} \frac{(-1)^k}{k!} \left(\frac{u-x}{h_0}\right)^{2k} du \right| < \frac{\varepsilon}{2}. \tag{70}$$

for all $|x| \leq R$. If we put

$$P(x) = \frac{1}{h_0\sqrt{\pi}} \int_{-R}^{R} f(u) \sum_{k=0}^{N} \frac{(-1)^k}{k!} \left(\frac{u-x}{h_0}\right)^{2k} du, \tag{71}$$

then $P(x)$ is a polynomial in x of degree at most $2N$ such that $|S_{h_0} f(x) - P(x)| < \varepsilon/2$ for all $|x| \leq R$. This implies that $|f(x) - P(x)| < \varepsilon$ for all $x \in [a, b]$. □

Here, ϵ is defined to be the *global error* of the polynomial with respect to the arbitrary function, and with uniform approximation. Another variation works on the squared error, such that $\|f - p\| = \sum_{i=1}^{\infty} (f(x_i) - p(x_i))^2$, or the mean absolute

$$\|f - p\| = \lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} (f(x_i) - p(x_i))$$

Weierstrass Approximation Theorem provides tighter bound for the mean absolute error, but not the square error, or the mean square case. Nevertheless, they are natural measure on the uniform metric space $C(X)$ with measure $d(f, g) = \sup_{x\in X} |f(x) - g(x)|$. If said metric also use the L^p metric instead, which is defined by

$$d_p(f, g) = \left(\int_X |f(x) - f(g)|^p \, dx\right)^{1/p} \tag{72}$$

we can also see that it is weaker than the bound given by Weierstrass theorem. Generalization of Weierstrass theorem can be made by Stone–Weierstrass theorem instead, of which is state in real space as followed, though generally Weierstrass theorem is sufficed.

Theorem 9.2 (Stone–Weierstrass Theorem (real version)). *Let S be a locally compact Hausdorff space. Let A be the subset of bounded continuous real functions on X, or of subalgebras $\mathcal{C}(X, \mathbb{R})$ which contains a non-zero constant function, then A is uniformly dense in $\mathcal{C}(X, \mathbb{R})$ if and only if it separates points.*

h. We again note the disparity between $h(x) - c(x)$ or $c(x) - h(x)$ for clarity. In essence, $c(x) - h(c)$ is looking from the perspective of the concept itself. While $h(x) - c(x)$ is comparison on difference from the perspective of the hypothesis.

In the rigorous proof, or so is the proof for Theorem 9.1, the only objective is to establish the existence in which such polynomial might exist, but talk nothing effectively about how to obtain such polynomial, or the feasibility of such model. Indeed, in Bernstein's construtive proof (Sury 2019; Muresan 2023), the proof utilized the construction of the monomial-basis spanning sub-family of the Bernstein polynomial, and show that converges uniformly to f arbitrarily defined on the mapping space. Such is a sufficient proof, in which Bernstein's family is dense in $C[0, 1]$ and thus specify the polynomial structures itself; but overall, there exists no necessity proof on such matter. Such a necessity theorem does not exist, because WAT is usually expressed as *far too weak* to enforce such generality. A heavier version must be used, for example, Korovkin's on the sequence of positive linear operators, but otherwise this is inherently lacking. Expressively, this is the same for the dynamics and how different families of polynomial will emerge or moves under dynamical process and true approximation. However, the theorem is still useful in its sense of the universality, thus we will still use it, but **only** so.

The setting of polynomial interpolation is rather different from the setting of function approximation. The setting of interpolation gives an n samples set $\mathcal{S}$, of which in this case we restrict to $\mathbb{R} \to \mathbb{R}$. Then, we are required to fit exactly the entire dataset, such that the following setting is sufficed.

Definition 9.3 (Interpolation). *A function class $\mathcal{F}$ is said to **interpolate** a distinct dataset $\mathcal{S}_n = (x_n, y_n)$ of n samples, if there exists $f \in \mathcal{F}$ such that $\mathbb{E}\|S_i - |f(x_i)\| = 0$. Such f is then called the **interpolating function** of $\mathcal{S}_n$.*

Under such definition, the question is then if the polynomial class $\mathcal{P}_n$ of any given polynomial with the highest degree n is sufficed to approximate such dataset. The answer can be found using unisolvence theorem.

Theorem 9.3 (Unisolvence theorem). *Given $n + 1$ bivariate data points $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n) \in \mathcal{S}_n \subset \mathbb{R}^n$, with all x_i distinct, there exists a unique polynomial $p_n(x)$ of degree at most n such that $p(x_i) = y_i$ for $i = 0, 1, \ldots, n$.*

In determining such problem, we can also say that there exists a unique n-saddle point function represented here as polynomial, such that to fit n points perfectly by the definition of the saddle point. The process in obtaining such formula is difficult, however, and finding the perfect interpolation polynomial would be computationally significant, and erroneous. The following defintion specifies the concept of such saddle point.

Definition 9.4 (Saddle point,Nie, Yang, and Zhou 2021). *Let $X \subseteq \mathbb{R}^n$, $Y \subseteq \mathbb{R}^m$ be two sets (for dimension $n, m > 0$) and let $F(x, y)$ be a continuous function in $(x, y) \in X \times Y$. A pair $(x^*, y^*) \in X \times Y$ is said to be a saddle point of $F(x, y)$ over $X \times Y$ if*

$$F(x^*, y) \leq F(x^*, y^*) \leq F(x, y^*), \quad \forall x \in X, \forall y \in Y \tag{73}$$

Effectively speaking, each interpolation degree of the polynomial interpolation scheme admits a specific (x^*, y^*) of which that the convexity is absolved. Thus, the reason why you require n saddle points for n points, or d-degree polynomial for $d + 1$ points. The landscape of such interpolation is "wiggling", as indicted, since for any two interpolation point, the macroscopic details would be similar as Figure 12, of such extend for any local monotonicity on n saddle points, or basically a secant line.

x	a		x_i^*		x_{i+1}^*		b
$\mathcal{D}_1[p]$		$-$	0	$+$	0	$-$	
$p_1(x)$	$p(a)$	↘	y_i^*	↗	y_{i+1}^*	↘	$p(b)$
$\mathcal{D}_2[p]$		$+$	0	$-$	0	$+$	
$f_2(x)$	$p(a)$	↗	y_i^*	↘	y_{i+1}^*	↗	$p(b)$

Figure 12. Table of monotonicity for the saddle point landscape between $[a, b]$, for both cases of the landscape.

Polynomial regression is then one such analytical solution to function approximation task, using Weierstrass theorem as its basis. From such, we can also observe that polynomial regression, and hence polynomial interpolation, are correlated and can be reduced to each other. Polynomial regression can be reduced to interpolation via stating the criteria such that, denoting it as ℓ, then $\ell(f, p) = 0$, and interpolation can be reduced to regression via the condition that $p(x_i) - y_i \leq \epsilon$ for arbitrary $\epsilon > 0$. Hence, the regression formulae lie between the two extreme: the Weierstrass setting and the interpolation setting.

We then have to identify such extreme and what kind of extreme they represent. Fortunately, we know this as the problem of **transparency**, in the setting of approximation, which certainly correlates to the learning theory. According to such, the Stone-Weierstrass represents the total transparency case, where the function f is known to the approximator with polynomial class. Interpolation represents total black box, where the goal is only to interpolate the sample point by itself. In between such interpretation, there exists a fundamental assumption, hence the structural assumption.

Assumption 9.4 (Transparency). *For any bivariate pair (x, y), we assume each pair depends on and originate from a based concept $y = f(x)$.*

Two kinds of knowledge is then available, which correlates to the degree of freedom the observation space can have. First is the knowledge of f. This knowledge of f is encoded directly by the number of sample point and their distribution in certain case. Accordingly, the worst case is as followed.

Conjecture 9.1. *If $\forall x$ there exists y corresponding to such, and $(x, y) \in \mathcal{S}$, then we say we have total transparency to the concept f. If there exists only finite x that we can have y corresponding to, then we say we have limited transparency to the concept f. If we assume no knowledge of f, then we say the setting is called the **black-box setting**.*

Secondly is the action on the dataset itself. One of the main assumption that is usually observed is that

Assumption 9.5 (Fallacy). *We assume there exists every specific bivariate pair $(x_i, y_i) \in f[\mathcal{S}_n]$, such that $f(x_i) = y_i$.*

However, usually, this is not the case, as we have **noise** in addition from the dataset. The origin of the noise cannot be tracked. However, we can interpret what happens when such noise is introduced to the dataset. The noise transitions the transparency of the learned setting, on the field $\mathbb{R}$, by utilizing the dense property of the real space. Assuming we operate on $\mathbb{Z}$. Then, each discrete noise will introduce abnormality point such that no function on $\mathbb{Z}$ will be able to match, up to certain distance on $\mathbb{Z}$. This is because on such space, there exists no point x_1, x_2, for example, that lies between $d(x_1, x_2) = 1$. If the noise is randomized on ± 1, then there will exists no function that can approximate such disparity better than a random fluctuating function. However, on $\mathbb{R}$, the real number is inherently dense, such that there will always be certain point x_3 such that $x_3 \in [x_1, x_2]$, by the density of real number. By the distance metric on $\mathbb{R}$, we define $d(X, Y) = |X - Y|$ for distance between two data points. Noise intrinsically increases such distance, such that $d'(X, Y) > d(X, Y)$. Hence, the space in which density can appear increase, thus there then exists more fluctuation, and thus more uncertainty between such two points. This notion can be increased from local uncertainty to global uncertainty, hence the *noisy dataset*.

We then can measure or aggregate some properties of the dataset itself, including the **uncertainty** of the dataset. We define the simple uncertainty of the dataset as followed.

Definition 9.5 (Dataset uncertainty). *Given a closed interval $[a, b]$, for the sample set $\mathcal{S}_n = (X, Y) \in [a, b]$, the uncertainty $\mathcal{U}(\mathcal{S}_n)$ is defined as*

$$\mathcal{U}(\mathcal{S}_n) = \left[\mathop{\mathbb{E}}_{x_i \in X} \left(x_{i+1} - x_i \right)^2 + \mathop{\mathbb{E}}_{y_i \in Y} \left(y_{i+1} - y_i \right)^2 \right]^{1/2} \tag{74}$$

We assume X, Y are totally ordered sets.

Hence, the assumption of fallacy assumes $\mathcal{U}(\mathcal{S}_n) = 0$, though in reality it is often $\mathcal{U}(\mathcal{S}_n) > 0$. With this, we can quantify how the dataset is behaving on the scale of the bivariate form typically seen. Noise intrinsically increases this distance, usually by the y-axis. This dataset uncertainty increases with multivariate dataset, for example, $(\mathbf{x}, y)$ of $\mathbb{R}^n \to \mathbb{R}$. We call the new metric as the *empirical uncertainty metric*. From this, we can have the theorem:

Theorem 9.6 (Uncertainty tradeoff).

Thirdly, is the treatment on such information. Even though information is known, such as from the assumption of transparency and the fallacy conjecture, it relies on the method in which the model uses that make use of such information. Polynomial interpolation effectively removes information about transparency, and thus the interpolation setting is a kind of **black-box absolute setting**. This setting is *extremely brittle*, as to see that if y_i is slightly wrong due to measurement noise, the interpolant oscillates wildly, which is explained by Runge phenomenon (Runge 1901; Corless and Sevyeri 2018; Boyd 2009). Probability then arises from such setting naturally, though this particular degree of probabilistic setting must be quantified, which comes from the process that the model simulates upon. Regression is inherently not probabilistic, but treatment about the probabilistic assumption, for example, such that (X, Y) inherently as random variables. We can then separate this into three parts - the data-level probabilistic layer (data uncertainty), the model-level layer, and the algorithm-layer that determines the solution solver within such model. We can define that as followed.

Conjecture 9.2 (Probabilistic layer). *Given a learning setting with dataset $\mathcal{S}$ of bivariate pairs, $h \in \mathcal{H}$, $c \in \mathcal{C}$ as the hypothesis and concept, we have the following conceptual probabilistic setting:*[i]

1. *Data-level probability: Sampling assumptions (observation sample points) such that there exists the distribution $\mathcal{D} \sim (X, Y)$, intrinsic data uncertainty $\mathcal{U}(\mathcal{S}_n)$.*
2. *Model-level probability: Structure of the dataset is controlled by the concept structure $\{\theta\} \in c$, c is modelled as a target sampler $c \equiv \mathcal{D}$, knowledge about the structure of c, Bayesian prior-posterior assumptions, $\mathbb{P}(Y \mid X, \{\theta\})$. For Bayesian polynomial regression, the model-level probability is formulated as*

$$\mathbb{P}(Y \mid X) = \int \mathbb{P}(Y \mid X, \theta)\mathbb{P}(\theta)\, d\theta, \quad \mathbb{P}(\theta_j) = \mathcal{N}(0, \sigma^2) \tag{75}$$

   of which assumes aleatoric uncertainty and epistemic uncertainty from the weight.
3. *Algorithm-level probability: Stochastic method, decision conditional (dropout, MCMC), ensembles criteria, Monte-Carlo initialization, variational inference.*

Dynamically, in this paper, we would like to investigate this particular junction between uncertainties, to see what happens when the model transparency, the treatment of data, and so on, that exhibits the observations that we would receive from the model itself. The experimental details will infer such result.

Generally, we would be entitled to see particular trajectory specifically designed around the point of interpolating match, or $n = m$ of n degree of freedom, technically, and m data points. After this point, there exist two possibility of dynamic - either the model dilutes, in which case result in *double descent*, or the model fluctuate strongly afterward, leading to *bias-variance emergence* again later on. This conclusion however, would have to be analysed within the current framework of the possible interjection to its fluctuation strength. For example, in the monomial basis $\{x^i\}^n$, any given fluctuation contributes to the entire polynomial. In such case, fluctuation of the other $[0, n - k]$ of arbitrary k and large enough n would be unaccounted for by the scale. Conversely, restrict them to $[1, 0]$ range will reveal something about the system, however will destroy particular properties of the prediction, which is not

i. Currently, we have no way to **quantify** the 'probabilistic degree' of the system setting itself recommended. Nevertheless, we must conduct the test before giving in any reasonable conclusion.

Previously, we said that we did indeed focus on the interpolation problem. Now, assuming we have given the interpolating algorithm $\mathcal{A}$ to approximate the interpolation, what will this look like? Firstly, solving this is *very computationally expensive*, because it is a dense, $(n+1) \times (n+1)$ linear system, and even more so if we consider the overparameterized outcome. However, this can be reduced further down if we attempt to solve them using convergence bound, but will be still very expensive. That and incremental soft interpolation is somewhat out of the question since it will be kind of the same. Aside from the insight, we would still have to know what to do. Then there exists two ways - we have some version of this to optimize of simply evaluating by gradient descent optimization, or, we can use somewhat analytical fit for the case of soft interpolation modification.

A theory on such problem can be obtained of its mechanics, as will be a fairly intuitive way in consideration of the theory of interpolation, as well as the technique of interpolation by itself. For a fixed-range interpolation threshold problem, a hard-encoded interpolation solution requires n-degree polynomial, of which has $n+1$ degree of fitness, to fit $n+1$ distinct points. Suppose all points are distinct of the set S which is finite, then this is the case. If we switch to the problem of interpolation but induce a more forgiving bound, there exists a possibility that it induces a relative *density averaging path* for the points, especially if we are considering the case where $p > n$ for p saddle points or degree of fitness, and n the number of point to interpolate. In a more nuanced formulation, we have

$$\mathsf{Flatten}\left(\begin{bmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^p \\ 1 & x_1 & x_1^2 & \cdots & x_1^p \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_p & x_p^2 & \cdots & x_p^p \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^p \end{bmatrix}_{n\times p} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} - \begin{bmatrix} f(x_0) \\ f(x_1) \\ f(x_2) \\ f(x_3) \\ \vdots \\ f(x_n) \end{bmatrix}\right) \leq \epsilon \tag{76}$$

or equivalently, $\mathsf{Flatten}(\mathbf{X}_{n\times p}\mathbf{w}_n - \mathbf{Y}_n) \leq \epsilon$ where $\mathsf{Flatten}(\cdot)$ is the compression of vector to scalar, or so $\mathsf{Flatten} : \mathbb{R}^n \to \mathbb{R}$. This essentially is the problem of solving n points approximation, toward using only p saddle points available. If $\epsilon = 0$, it is well known that interpolation is impossible. However, if $\epsilon \neq 0$, there are possibilities, depends on the shape of the dataset, that there will exist a data-dependent property or more that indict a particular *density preference* toward those saddle points - or rather, the act of density averaging. This is reflected rather implicitly in the structure itself. In machine learning, the error measure is usually the mean average, such is to say that the criterion now would look more like $\mathbb{E}\Big[\mathsf{Flatten}(\mathbf{X}_{n\times p}\mathbf{w}_n - \mathbf{Y}_n)\Big] \leq \epsilon$. Formally, we have the following hypothesis

Hypothesis 9.1 (Polynomial double descent). *For a polynomial $\{a_i x^i\}_n$ on Vandermonde basis, approximating on the set $\{(x_k, y_k)\}_m$, then*

1. *For $n < m$, we expect the polynomial form factor to be inherently sparse, such that for an iterative approximation, for each p_i saddle point of the polynomial expansion form, there exists an ϵ-ball B_ϵ for $\epsilon > 0$ with at least one data point satisfying $(x_k, y_k) \in B_\epsilon$.*
2. *For $n = m$, we expect the polynomial to be able to fit all points, up to hard interpolation threshold such that $\hat{R}_S \approx 0$.*
3. *For $n > m$, we expect the polynomial form factor to be locally dense then turning to globally dense, over random ensembles. That is, for every point (x_k, y_k) there exists a β-ball B_β such that there is at least one saddle point of the polynomial belong to the ball.*

The shift that will be hypothesized to appear here is expected to be from sparse in the polynomial term sense, out to globally dense with enough model complexity, over an iterative process. One also main point of clarity is, at certain point right now, we haven't discovered the effect of error deviation, yet. For such, we theorize the following.

Conjecture 9.3 (Double descent). *Double descent happens, for a polynomial class model $h_n \in \mathcal{H}_p$, is upper bounded to certain error deviation $\epsilon > 0$ from the mean only of the generalization (test) dataset verification.*

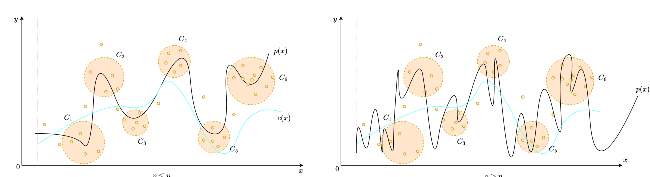


Figure 13. Intuition of Hypothesis 9.1. Effectively, in the overparameterized range ($p > n$), the problem turns into soft interpolation, where the polynomial is possible to be dense enough such that relative error remains internally low. This can also happen if the dataset itself is intrinsically large.

That is, the disparity between the test and training dataset helps in such regard. If the two dataset is partitioned in the sense that their deviation is practically the same (which can be done using synthetic generation for a control group), then this will amount to not much of it. After all, it will be the consequence of the algorithm and error landscape interpretation used.

For this, we propose another speculative observation beside such, as to consider adjourned transform of the Vandermonde basis to the Newtonian or similar basis, in which the effects and motion can be easily observed, as least easy. This is because aside from mapping down to $[-1, 1]$, normal polynomial on monomial basis gives suboptimal result because of intrinsic high-degree term domination. This will become more relevant later on in the analysis.

9.1 Structure

For mathematical analysis and experimental design, we would like to state the current main structure of a polynomial model.

Specifically, in such example, we are concerned of the shallow polynomial neural network, of which is stated as such in Arjevani et al. 2025. Notice that it is defined on the normal Vandermonde basis, where an expression of such polynomial model under Newtonian or Lagrangian basis is not so clear-cut of an analysis.

Definition 9.6 (Shallow polynomial neural network)**.** *A shallow polynomial is a neural network of one layer that are functions $f_{\mathcal{W}}$, defined such as: $\mathbb{R}^d \to \mathbb{R}$ of the form*

$$\begin{aligned} f_{\mathcal{W}}(x) &= \sum_{i=1}^{r} \alpha_i (w_i \cdot x)^d, \\ \mathcal{W} &= (\alpha, w_1, \ldots, w_k) \in \mathbb{R}^r \times (\mathbb{R}^n)^k, \end{aligned} \tag{77}$$

where $d \in \mathbb{N}$.

A modification is then available, in which we simply add $b(\cdot)$ for random initialization process. We note that this introduces certain flexibility to the model initial interpolation curvature.

Definition 9.7 (Polynomial neural network, general)**.** *The polynomial network based on Model 9.6 is modified as*

$$f_{\Theta}(x) = \sum_{i=1}^{d} \theta_i \left(\mathbf{w}^\top x\right)^i + b, \tag{78}$$

where $\Theta = (\{\theta_i\}, \{w_k\}, b)$.

From this, we can state the model within a testing scenario listed as such, to fully capture the structural foundation of the related testing model class. For the notion of complexity, naively, we employ two kinds of complexity measure. First is the usual complexity measure via parameter counts. Second, is the expressive complexity on the basis of the linear transformation space. That is, an expressive complexity that take into account the fact that $\mathcal{O}(wx^n)$ will always be stronger than $\mathcal{O}(wx^{n-1})$ for $x > 1$. All of which can be contained in a singular expression, as seen in definition 9.8. However, we would have to analyse what is happening inside the metric itself. The model itself is illustrated in Figure 14.

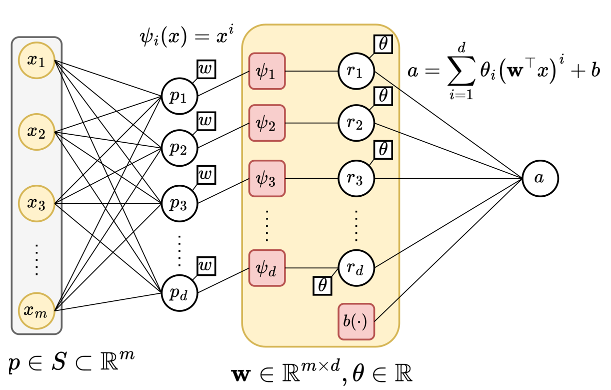


Figure 14. Illustration of polynomial model in definition 9.7.

In the equation for expressive complexity, the first term is the normalized linear parameters θ_i fixing and controlling the shift of the polynomial inning parameter, the second term is the sum of weight-aware complexity with degree factor, and the degree factor over operating range $\|x_{max} - x_{min}\|$ of the dataset, for the degree complexity. The final term is the last weight-aware complexity for the dominating degree n of the polynomial function. Let us begin with the gauge on coefficient norm, i.e. the term $\|\alpha_d\| + 1$. Here, we set 1 as baseline, and the arbitrary norm on α there coefficient effect. The terms coefficient in this expression matters because, for a function $f : [a, b] \to \mathbb{R}$, the Lipschitz constant L for polynomial f goes by

$$\begin{aligned} L_p &= \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|} && (79) \\ &= \sup_{x \neq y} \frac{|f_\Theta(x) - f_\Theta(y)|}{|x - y|} && (80) \\ &\leq \sup_{x \in [a,b]} \left\| f'_\Theta(x) \right\|_2 && (81) \\ &= \sup_{x \in [a,b]} \left\| \sum_{i=1}^{d} i\, \theta_i \left(\mathbf{w}^\top x\right)^{i-1} \mathbf{w} \right\|_2 && (82) \\ &\leq \|\mathbf{w}\|_2 \sup_{x \in [a,b]} \sum_{i=1}^{d} i\, |\theta_i|\, |\mathbf{w}^\top x|^{i-1} && (83) \\ &\leq \|\mathbf{w}\|_2 \sum_{i=1}^{d} i\, |\theta_i| \max\left\{|\mathbf{w}^\top a|, |\mathbf{w}^\top b|\right\}^{i-1}. && (84) \end{aligned}$$

Where we note that $f_\Theta(x) = \sum_{i=1}^{d} \theta_i \left(\mathbf{w}^\top x\right)^i + b$, and $\|\mathbf{w}\|_2$ is the Euclidean ℓ^2 norm. The magnitude of $f'_\Theta(x)$ depends directly on the coefficient size α_i, and also on how big is x on the domain. All of this mean for large expression and value range on f_Θ, the Lipschitz constant of that model is accordingly large. Defining α-norm that way is a cheap proxy for Lipschitz constant, since it gauges stability, and rate of change of the functional expression itself.

Definition 9.8 (Polynomial expressive complexity)**.** *For $x > 1$ or $\mathbf{x}_i > 1$ of $\mathbf{x} \in \mathbb{R}^n$ as input. Denote the polynomial class as $\mathcal{P}_n$ of their highest degree n, then we define the total expressive complexity of $p \in \mathcal{P}_n$ as followed.*

$$\begin{aligned} &\mathbb{C}_E(p) \\ &= \mathbb{E}\Bigg[\|\alpha\|_d + 1 \\ &+ \frac{\sum_{i \in S \setminus \{p_n\}} w_i\, i}{|0p|0 - p_n} \\ &+ \frac{1}{\|\overline{x} - \underline{x}\|} \left(\frac{\sum_{i \in S} i}{\|p\|} + w_n n \right) \Bigg], \end{aligned} \tag{85}$$

where $S = \{p_j\}$, $\|p\|$ denotes the amount of different degree separable terms and $\|x_{max} - x_{min}\|$ is the observable range of the dataset.

Table 1. Controllable parameters and hyperparameters for polynomial class analysis

Information	Notation	Details
Hypothesis model	$h[\mathcal{W}_h, \mathbf{H}_h] \in \mathcal{H}$	Hypothesis model. For simplicity, it contains $\mathcal{W}$ as the configuration parameter set, and $\mathbf{H}$ the fixed hyperparameters
Concept model	$c[\mathcal{W}_c, \mathbf{H}_c] \in \mathcal{C}$	Concept or target model. Similar configuration as the hypothesis model
Complexity measure(s)	$\{\mathbb{C}(\cdot)\}$	Set of complexity measure of choice for any arbitrary model taken in.
Sample count	m	The number of sample available for hypothesis model of c, for $m = \lvert\mathcal{S}\rvert$ dataset
Hypothesis dimension (hyperparameter)	$[d_h]$	An explicit hyperparameter contains the list of all in-testing polynomial degree for the hypothesis model.
Concept dimension (hyperparameter)	d_c	An explicit hyperparameter of the concept's true dimension. Theorized such that $d_c = d_{h,i}$ for arbitrary i will reduce to possible low-error problem.
Data noise	$\epsilon_{\mathcal{S}}$	Dataset-wide noise with relative strength to the value of the sample point. This is to reduce vanishing noise with higher-degree polynomial sample points.
Learning parameters	epoch, lr, bs	Two main parameters for gradient-descent, iterative-based algorithm - the epoch, the learning rate, and the batch size.

Notes: We output to $\{\mathbb{E}[\ell]\}$ as the expected value of the error measure on h, c. There are a few things possible, but under approximation setting, we use MSE-on-parameter and MSE-on-prediction for two different insights: internal representation and external representation.

9.2 *Scenario schema*

The scenario schema will be as followed. The following set of parameters are required for controlling the testing case, aside from different system measures. For parameter error, we reuse equation from Definition 9.8 but with degree aware terms. This is covered in Table 2.

We note a few problems with both interpolation interpretation and polynomial regression of higher-degree terms. Firstly, is the intrinsic interpolation error and numerical, software issue error. Higher degree polynomial can be represented as solving for a wide Vandermonde system of n degrees for m sample points. Using monomial basis for interpolation, either for soft interpolation also exhibits instability and numerical garbage within the operations, as indicated by Shen and Serkh 2025. They can also be a potential source of double descent, as the majority of models tested around this range are not coded effectively for handling large data operations as such, and artifacts from such choices might be taken as the indication of double descent in some particular cases.

Aside from it, and aside from the fully-connected case, we have the particular testing scenario that can be expected of the ease of implementation. First, is the mundane implementation that is of gradually increasing degree mismatch. That is, for polynomial of h_p and c_p, one requires only d_h and d_c to figure out the structure as there will exist exactly $d + 1$ with the constant term, terms of which of finite many degrees. The second case is where for p chosen randomly, such that aside from such p entry, then the other degree or nodes of input are simply scattered dependent nodes, linear algebraically. That is, for a uniformly constructed d-degree network, choose at random p particular degree to be fixed, while the other of $d - p$ are assigned randomly of belonging to some $p_i \in \{p\}$ nodes given the random choice on p above. Linearly algebraically, scattered and dependent nodes then mean for example, node 4 might use node 35's degree, and vice versa (if they were both non-fixed). The dependency structure is deliberately randomized rather than organized. For overparameterized case, we simply design the hypothesis by the following line of justification. For a mapping $\mathbb{R}^d \to \mathbb{R}$, the mapping $\mathbb{R}^{d+n} \to \mathbb{R}$ for $n = 1$ requires additional information, since the source provision does not contain information to handle additional n dimensional input. Hence, we design it the following. For the hypothesis h, such unit of which does not fit to the original input space, the *internal input space* that h 'hypothesize' contains, for the other n entries, $q \leq d$ per unit input. That is, each input of n overparameterized case receives at random, q inputs can be received, for example, $w_{n1}x_j + w_{n2}x_k$ for k, j specific input numbering. By the simplicity argument, then $q = 1$ for the simplest case. However, this creates a skewed polynomial network, of which informally, the provided structural bias or additional information, is the fact that there exists overparameterization. To prevent this, instead, the final case considers the most general structure — for each unit, each one can take q inputs controlled as hyperparameter of the network, then the set $\{p\}$ of chosen degree influence their degree choice, and is uniform if the sampling process it is not included.

10. Reformation of learning theory

Most of the paper will argue about different features of statistical learning theory (Sterkenburg 2024; Vapnik 1999; Hajek and Raginsky 2021). It is then imperative to discuss the background setting of such. On one hand, when talking of the details on statistical learning theory and the understanding of the encoded language to describe the learning action, typically we are given the observations, or dataset of the form $\mathcal{S} = (\mathcal{X}, \mathcal{Y}) \subset \mathbb{R}^n \times \mathbb{R}^m$, of all 2-tuple pairs, assumed to be sampled or observed and governed by a distribution $\mathcal{D}$. On the other hand, there is also the push toward a more general and sophisticated view, in which might as well reveal deeper or richer structures on the theoretical framework to realize. Note that this contains reformulation, or unification, reinterpretation or patchworks of existing structures too, and not simply claimed to be entirely new.

10.1 Domain and process

10.2 The learner's encoded setting

Previously, we say that we have the dataset, or rather called as **observations** of a given system c, $\mathcal{S}_m$ of size m, such that $(x, \cdot) \sim \mathcal{D}$ of a distribution, optionally of also y to be sampled from such distribution. The dataset itself contains pairs, as

$$\mathcal{S} = \{(x_1, y_1), (x_2, y_2), \dots (x_n, y_n)\} \subset \mathcal{X} \times \mathcal{Y}$$

where the dataset is assumed to be i.i.d. sampled according to $\mathcal{D}$, which is unknown by the priori. Hence, we have the first set of assumption.

Assumption 10.1. *The observational space is governed by a particular sampling distribution $\mathcal{D}$, for fixed y-response according to the concept $c \in \mathcal{C}$. The sampling process is further assumed to hold no further structure, hence is identically and independently distributed (i.i.d.).*

The learning problem is then formulated as followed. Given the machine learning model expressed a hypothesis h of the hypothesis class $\mathcal{H}$, the learning theory aims for creating a procedure to learn either elements of the concept class $\mathcal{C}$ of all concepts $c : \mathcal{X} \to \mathcal{X}$, or the function class $\mathcal{F}$ of all functions $f : \mathcal{X} \to \mathcal{Y}$, which is usually set to $\{0, 1\}$ or $[0, 1]$. This distinction is trivial, hence, if it is clear, we will talk about the concept class $\mathcal{C}$ only as representative form. The learner $\mathcal{L}(h)$ consider the set of possible hypothesis $\mathcal{H}$, in which might not coincide with $\mathcal{C}$. It receives a partial image of sample $S = (x_1, \dots, x_2, \dots, x_n)$ drawn i.i.d. according to $\mathcal{D}$ as well as the label $(c(x_1), \dots, c(x_n))$. This constitutes the dataset $\mathcal{S}$, which are based on specific concept $c \in \mathcal{C}$ for the hypothesis to learn. The task is then to use (or *extract*) meaningful information to select a hypothesis $h_S \in \mathcal{H}$ that accurately mimic c, with marginal error Θ. The notation h_S stands for all hypothesis that can be inferred from the range of the dataset (the first argument, $\mathcal{X}$).

The marginal error Θ is considered of two parameters, the empirical error $\hat{R}(h)$ and the generalization error $R(h)$. We give the following definition of them.

Definition 10.1 (Empirical risk). *Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and a sample $S = (x_1, \dots, x_m)$. For some particular $\epsilon > 0$, the **empirical error** or empirical risk of h is defined by*

$$\hat{R}_S(h) = \mathop{\mathbb{P}}_{x \in S \sim \mathcal{D}}[\ell\{h(x), c(x)\} \geq \epsilon] = \frac{1}{m} \sum_{i=1}^{m} \ell\{h(x_i), c(x_i)\} \tag{86}$$

The generalization risk is then defined as the extension of the empirical risk, of which instead of being evaluated on the set $\mathcal{S}$, it is evaluated on arbitrary infinite density sample space. The notation for the risk calculation can also be $R_{\mathcal{S}}$ instead, of which the need is to classify the set of which it is evaluated from.

Definition 10.2 (Generalization risk). *Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and an underlying distribution $\mathcal{D}$ on $\mathcal{X}$. For some particular $\epsilon > 0$, the generalization error or risk of h is defined by*

$$\begin{aligned} R(h) &= \mathop{\mathbb{P}}_{x \sim \mathcal{D}}[\ell\{h(x), c(x)\} \geq \epsilon] \\ &= \mathop{\mathbb{E}}_{x \sim \mathcal{D}}[\ell\{h(x), c(x)\}] \\ &= \int_{x \in \mathcal{D}} \ell\{h(x), c(x)\}\, dP(x) \end{aligned} \tag{87}$$

In essence, the empirical risk measure the hypothesis to observation error, and the generalization risk captures the difference of the hypothesis to the actual concept. Notice that by doing this, we have implicitly assumed that the observations and the concept is not the same thing, under the majority of scenarios. For the observations to be *approximately equal* or sufficiently captures the concept, there then would have to be various criteria to fulfil. This can be illustrated as Illustration 15. For fixed $\mathcal{H}$, for fixed and sufficiently large $\mathcal{S}$, and no observation (data) errors, the empirical risk is the generalization risk. These two measures between h and c constitute the learning problem, which can also be separated into both cases - either empirical learning or generalization learning, one to minimize $\hat{R}(h)$, and one to minimize $R(h)$. Here, we emphasize such hierarchy by defining the generalization goal as the **nested criteria** for empirical risks minimization.

Definition 10.3 (Empirical learning problem). *We present the formal form of the empirical learning. Suppose we have a **target**, $c \in \mathcal{C}$, where $\mathcal{C}$ is an arbitrary concept class that captures targets of the same type. Suppose we are provided a set of observations $\mathcal{S}$. The problem is to use certain algorithm $\mathcal{A}$ using $\mathcal{S}$, to obtain a hypothesis h^* for a fixed $\mathcal{H}$ such that:*

$$R(h^*) = \min_{h \in \mathcal{H}} \hat{R}(h) = \min_{h \in \mathcal{H}} \mathop{\mathbb{E}}_{x \sim \mathcal{D}, x \in \mathcal{S}} \ell\{h(x), c(x)\} \tag{88}$$

The generalization best $\boldsymbol{h}$ is also called in literature as the *Bayes model*, such that it reduces the Bayes risk that is the infimum of the generalization risk, $R^* = \inf_{h \in \mathcal{H}} R(h)$. The hypothesis h^* is often called the *empirical best*, for it being the minimal,

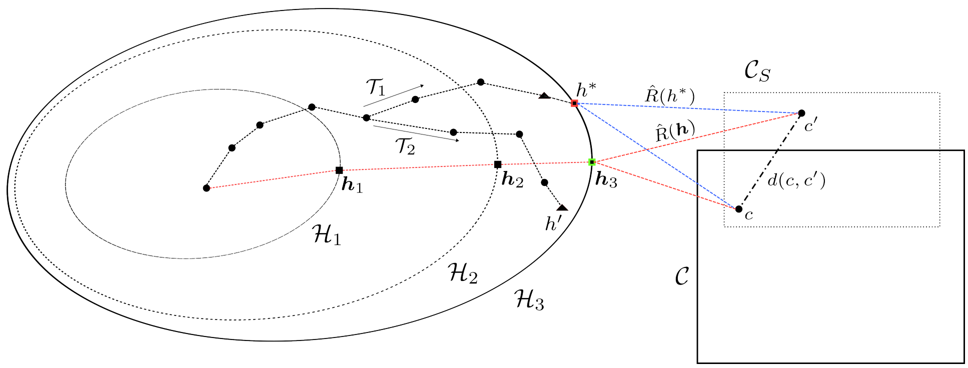


Figure 15. **Illustrative dynamic of the learning problem.** For an incremental hypothesis space sequence $\mathcal{H}_1, \mathcal{H}_2, \mathcal{H}_3$, we aim to obtain the procedure $\mathcal{T}$ that would either reach the *generalization solution* $\boldsymbol{h}$, or the *empirical solution* h^* for a particular hypothesis class, with respect to the concept c and its observed concept c'. Model selection hence dictates how an algorithm or procedure than choose the best possible hypothesis to approximate the generalization solution from the empirical solution set. Do notice that here we explicitly state that the data would create a **proxy concept** that overlaps with the concept class and of some arbitrary 'distance' from the true concept by $d(c, c')$. In case the two classes overlap, there still exists the arbitrary distance. Also, we can also observe the intuitive notion of increasing hypothesis class - while it indeed can help in getting closer to the concept class, the somewhat intrinsic property of current learning theory being the relative random procedure class make it probabilistically unstable.

finite hypothesis of the lowest loss evaluation on the entire observation space $\mathcal{S}$. There exists no certified assumption regarding whether h^* aligns with the minimal generalization error.

Definition 10.4 (Generalization learning problem). *We present the formal form of the generalization learning problem. Suppose we have a **target**, $c \in \mathcal{C}$, where $\mathcal{C}$ is an arbitrary concept class that captures targets of the same type. Suppose we are provided a set of observations $\mathcal{S}$. Supposed we have an algorithm $\mathcal{A}$ that for fixed hypothesis space $\mathcal{H}$, equation 88 holds true. The problem is to use certain algorithm $\mathcal{A}'$ such that, under limited availability, to obtain $\boldsymbol{h}$, satisfies:*

$$R(\boldsymbol{h}) = \min_{h\in\mathcal{H}} R(\boldsymbol{h}) \leq \{\epsilon\}, \quad \epsilon > 0 \tag{89}$$

For a set of risk bounds ϵ. If the setting is deterministic, then there exists $\epsilon = 0$ accordingly.

Then, we can partially say that generalization problem is an advancement from empirical solution. As such, empirical solution imposes the observed concept, and not the concept itself. This is reflected in modern machine learning landscape, by modelling the concept as distributions, and then using notions such as KL-divergence to measure such relative disparity.

The process of generalization goal can be, described in certain form as the global optimization objective in which empiricism can obtain from reasonable set of assumptions and conditions. The following algorithmic framing indeed aims to focus and extradite such sense of operational goal.

These two stages of learning procedure comes of as the intrinsic design of machine learning setting. As opposed to interpolation, where empirical learning problem would have to be put into first priority to achieve the best possible point-through model, machine learning leans on approximation more and a separated criterion that is only visible in the learning setting - the fact that we are using c' to extract c. This disparity is exactly the **generalization problem**, of which is then also the goal of

Optimization problem 2: Empirical-generalization learning dynamics (non-proxy)

Data: $(x_i, y_i) \in \mathcal{S}, \Gamma \supset \mathcal{S}$, objective $\ell(y, f(x))$, $\nabla(\cdot)$ differentiable gradient field on ℓ.
Result: Expected optimized model to the minimal error on the objective $\ell(\cdot,\cdot)$, dependent on $\mathcal{S}$.
begin
  $\Gamma \supset \mathcal{S}$, the general set;
  $R(\cdot), \hat{R}(\cdot) \leftarrow \ell$ as objective functional;
  for *Fixed $\mathcal{H}$, with $h \in \mathcal{H}$, for $\epsilon > 0$* **do**
    Find $h \rightarrow \boldsymbol{h}'$, using algorithm $\mathcal{A}$ for minimization objective, such that

$$\mathcal{A}\left(\min_{h\in\mathcal{H}} \hat{R}_{\mathcal{S}}(h) + R(h)\right) \leq \epsilon \tag{90}$$

    for procedure $\mathcal{A}(\cdot,\cdot)$ of structural constraints.;
  end
end
return *Optimized model h on arbitrary objective space* (ℓ, ∇).

machine learning, to approximate the general solution using the observed solution. One then can ask what is the difference between the two notions, and when we can guarantee that both will 'converge' to the same point. The process of **model selection** is then the procedure that will optimize $h \in \mathcal{H}$ to a given target fixture, such as the empirical best.

One of the major assumption or observation from the point of the hypothesis class, is that the hypothesis cannot know the generalization error. Thereby, the more efficient and often used procedure/algorithm in training, is then the *empirical risk minimization* (ERM) method, minimizing only the empirical risk to the empirical best; then, certain measures to 'generalize' this heuristically is apprehended on the hypothesis. The process of model selection is also the place that we get the concept of **bias-variance tradeoff**, and its longstanding form to be a foundational metric on model selection.

10.3 Overfitting and underfitting

Following the above framework of statistical probabilistic learning theory, one understanding about the concept of underfitting and overfitting (similar to James et al. 2009) can be reached, using the notion of *partitioning* of observation space. Such work is typically done because of **practical condition**, in which the following constraints is applied.

Assumption 10.2. *No knowledge of $c \in \mathcal{C}$ is provided.* [j]

Without such knowledge, $R(h)$ cannot be computed, accordingly. We specify such approach as *heuristic*, as it is only an approximation, and for better definition, we define it as followed.

Definition 10.5 (Heuristic empirical risk)**.** *Given a dataset $\mathcal{S}$ of size m, supposedly of the concept $c \in \mathcal{C}$, and the hypothesis $h \in \mathcal{H}$, then the **heuristic empirical risk** take a partitioning $k < m$ with $k/m \geq 1/2$, hence defining the heuristic empirical loss on k, denoted $\hat{R}_k(h)$.*

Definition 10.6 (Heuristical generalization risk)**.** *Given a dataset $\mathcal{S}$ of size m, supposedly of the concept $c \in \mathcal{C}$, and the hypothesis $h \in \mathcal{H}$. Fixed $\hat{R}_k(h)$, for the final optimization of algorithm $\mathcal{A}$, then the **heuristic generalization risk** take the remaining 2-partition of portion $p = k - m$, hence defining the heuristic generalization loss on p, denoted $\hat{R}_p(h_{\mathcal{A}})$.*

The two heuristic definition however, can be noted to not be equal to the true generalization risk - empirical risk pair in reality. Nevertheless, one can then define overfitting and underfitting in such manner.

Definition 10.7 (Underfitting)**.** *A model h is called **underfitting** if for a constant $\epsilon > 0$ chosen for the setting, then $\hat{R}_k(h_{\mathcal{A}}) > \epsilon$, $\hat{R}_p(h_\alpha) > \epsilon$ over the entire observation space.*

Usually, underfitting is defined to be such that test error is smaller than training error partition, however, we do not think that is a good definition. Especially since, at least in this setting, the empirical error under algorithm $\mathcal{A}$ is reduced to the representative smallest error that $\mathcal{A}$ can achieve on that particular model, within particular setting of h, then come the heuristic generalization error. Considering also that $k/m \geq 1/2$, that is, at least the training partition is half of the observation set, then only under very specific case will $\hat{R}_p(h_\alpha)$ be smaller than $\hat{R}_k(h)$, statistically speaking. Thence this line of thought also reinforces the idea that both of them will be larger than the threshold ϵ. Usually, heuristically, this threshold of error is 0.3, normalized. Next, we define the notion of overfitting.

Definition 10.8 (Overfitting)**.** *A model h is called **overfitting** if there exists a specific constant $\epsilon > 0$ chosen for the setting, such that $\hat{R}_k(h) < \epsilon$, $\hat{R}_p(h_\alpha) > \epsilon$ over the entire observation space.*

Namely, this mean that the static learning makes the generalization error worse. This draws the analogy of the student - studying explicitly for the past paper tests, yet unaware of the larger margin of questions in the large set of the test paper pool, leading to subpar performance. Similarly, overfitting is theorized to be fitting too fit to the reduced observation space - hence when it comes to the generalization observation space, it fails within such heuristic. The notion works on the mask of the risks, $\hat{R}_k$ and $\hat{R}_p(h_\alpha)$.

In practice, Assumption 10.2 means we cannot have $c \in \mathcal{C}$, or its probabilistic interpretation $\mathcal{D}$ such that $x \sim \mathcal{D}$, or in the extreme non-structured case, $(x, y) \sim \mathcal{D}$. In such case, the typical practical fix relies on the partitioning of the available partitioning space, that is, the empirical-generalization proxy. For large enough singleton count of $|\{(X_k, Y_k)\}|$, we assume that $\hat{R}_{\mathsf{Gen}}(h) \approx R(h)$. Thus, the algorithm can be stated in a form as Algorithm 3.

Optimization problem 3: Empirical-generalization learning dynamics (proxy)

Data: $(x_i, y_i) \in \mathcal{S}, y_i = \{\pm 1\}$, hinge objective $\ell(y, f(x))$, $\nabla(\cdot)$ differentiable gradient field on ℓ.
Result: Expected optimized model to the minimal error on the objective $\ell(\cdot,\cdot)$, dependent on $\mathcal{S}$.
begin
  $((X_1, Y_1), \dots)_n \leftarrow \mathsf{Partition}_n(\mathcal{S})$;
  $\{(X_j, Y_j)\} \leftarrow \mathsf{Train}$;
  $\{(X_k, Y_k)\} \leftarrow \mathsf{Gen}$ for $k \neq j$;
  for *Fixed $\mathcal{H}$, with $h \in \mathcal{H}$, for $\epsilon > 0$* **do**
    (Optimistically) achieve $h \to h'$, using algorithm $\mathcal{A}$ for minimization objective, such that

$$\mathcal{A}\left(\min_{h\in\mathcal{H}} \hat{R}_{\mathsf{Train}}(h) + \hat{R}_{\mathsf{Gen}}(h)\right) \leq \epsilon \qquad (91)$$

    for procedure $\mathcal{A}(\cdot,\cdot)$ of structural constraints.;
  end
end
return *Optimized model $h_{\mathcal{A}}$ on arbitrary objective space* (ℓ, ∇).

The assumption above of the generalization set to the generalization truth error is apparent in the usual system theorem, Theorem 10.3.

Theorem 10.3 (Mohri, Rostamizadeh, and Talwalkar 2012)**.** *For fixed $\mathcal{H}$, for fixed and sufficiently large $\mathcal{S}$, and no observation*

j. Few points need to be said about this assumption. This assumption represents the *worst-case possible* of a system dynamic. From the perspective of a designer, inherent leakage of information can be provided in the design of a model. In such, case we are operating in the mode of a *resource-constrained, closed* game, in which the model with further assumptions of fixed class, operates on. The problem space itself is inherently large and penalize any additional structural changes. If one is to occur, we usually mention it as *implicit bias*, but another name that would be more fitting would be *structural attractor*, after the notion of gradient attractor in chaotic systems.

errors, the empirical risk is the generalization risk:

$$\mathop{\mathbb{E}}_{\mathcal{S}\sim\mathcal{D}^m}[\hat{R}_{\mathcal{S}}(h)] = R(h) \tag{92}$$

Proof. By linearity of the expectation, and the fact that we assume the sample is given i.i.d., we can write:

$$\begin{aligned}\mathop{\mathbb{E}}_{\mathcal{S}\sim\mathcal{D}^m}[\hat{R}_{\mathcal{S}}(h)] &= \frac{1}{m}\sum_{i=1}^{m}\mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}\left[\mathbb{1}_{h(x_i)\neq c(x_i)}\right] \\ &= \frac{1}{m}\sum_{i=1}^{m}\mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}\left[\mathbb{1}_{h(x)\neq c(x)}\right],\end{aligned} \tag{93}$$

for any x in sample S. Thus, we result in

$$\begin{aligned}\mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}[\hat{R}_S(h)] &= \mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}\left[\mathbb{1}_{h(x)\neq c(x)}\right] \\ &= \mathop{\mathbb{E}}_{x\sim\mathcal{D}}\left[\mathbb{1}_{h(x)\neq c(x)}\right] \\ &= R(h).\end{aligned} \tag{94}$$

which yields the desired result. □

This theorem can be transferred to a more generalized conjecture, of which fit the generalized notion of the empirical proxy and so on.

Conjecture 10.1. *Supposed $\mathcal{H}, \mathcal{C}$ are fixed, for $\mathcal{S}$ is the empirical c^* for $c \in \mathcal{C}$ the true concept. Then, for algorithm $\mathcal{A}(\cdot,\cdot)$ enforcing minimization apparatus, and the specific objective on the second argument, we have:*

$$\begin{aligned}&\mathop{\mathcal{A}}_{j\to\infty,\mathcal{S}_i\neq\mathcal{S}_k}\left(\min_{h\in\mathcal{H}}\hat{R}_{\mathsf{Train}}(h) + \hat{R}_{\mathsf{Gen}}(h), [\ell]\right) \\ &\approx \mathcal{A}\left(\min_{h\in\mathcal{H}}\hat{R}_{\mathcal{S}}(h) + R(h), [\ell]\right) + \epsilon_{\mathcal{S}}\end{aligned} \tag{95}$$

For $\mathcal{S}_i \neq \mathcal{S}_j$ such that there are only distinct samples, j the observation set size, $[\ell]$ the set of structural or empirical objectives enforced on the problem landscape, and $\epsilon_{\mathcal{S}}$ is the empirical minimum that the algorithm result can achieve assuming sufficient 'perfect interpolation'.

By the end of this section, we would also want to note that we deliberately choose an expression basis of $\mathcal{A}$ via the couple tuplet $(\min, [\ell])$ as a choice for *algorithm-agnostic class* of algorithm satisfies the dynamic only, not just implementations. The algorithm, in certain sense, depends much more on structural assumptions, of which is not accessible via observations but via design, and intrinsic information specified in the observation class itself. This is a requirement for fine-tuning analysis in the long-run, as there exists a myriad of different approaches. Indeed, the classical theory of learning simply does not have much to offer in terms of a unified perspective, in which important factors are analysed of its realization. Specifically speaking of such, classical theory does do worse in many parts, for example, the aspect of transparency given the model, supervisor and the concept, of which their roles are not fully understood or formulated correctly, both in controlled setting and extension to replication of real life examples. Thereby, the problem requires a newer theory of learning to come forward. A new learning theory must extend beyond what constitute the basic learning model, such as the idealized PAC-learning, and statistical learning setting in general. More specifically, we want to design a learning theory that is more akin to **modelling theory** of its internalization, but also attain the learning of machine learning theory.

Within the constraints of such theoretical framing, we can then construct a different notion in which the generalization and the empirical learning problem can possibly relate to each other, at least from the statistical framework. Let us consider the problem of mathematical modelling in such sense. Beside the model class construction, which will be discussed in further sections, the entire idea of this structuralization is to refine the framework of which learning and so on can be defined itself.

11. System-wide structuralization

From which we can draw out of the above discussions, concurrent theories are inherently unstable and with many variations, making them ill-suited for instant application on analysis of different models without losing major information about the structural details and expressions. Hence, we propose a novel analysis on the basis of the axiomatization of the learning setting, under various structural assumptions.

11.1 The idea of neural network formalism

Previously, we have been discovering, examining past results and testing on models of different types, generally classical models. While decisive, each modelling setting requires different analysis and is often not so streamline, typically fragmented as per different functional schemes, probabilistic schemes, optimization and algorithmic-based choice of designs, and so on. They have barely any unified view or general patterns for extension, thus making analysis inherently difficult. Within such, our interest, and discovery, posit that the formalism or the idea of neural network (accordance to designs and patterns of McCulloch and Pitts 1943; Hebb 1949; Rosenblatt 1958a; Widrow and T. Hoff 1960; Minsky and Papert 1969; Fukushima 1980; Kohonen 1982; Hopfield 1982; Werbos 1974; Rumelhart, Hinton, and Williams 1986; Cybenko 1989; Hornik 1991; Hochreiter and Schmidhuber 1997; LeCun et al. 1998; Bishop 1995; Haykin 1994) and some of its modern understanding (as of I. Goodfellow et al. 2016; T. Zhang 2023; Hinton, Osindero, and Teh 2006; LeCun, Bengio, and Hinton 2015; I. J. Goodfellow et al. 2014; Srivastava et al. 2014b; Szegedy et al. 2014; Simonyan and Zisserman 2014; Vaswani et al. 2017; He et al. 2016; Devlin et al. 2018; Brown et al. 2020) practiced and exhibited, to be, in its general form, the correct but ill-defined and poorly developed framework. Thus is our goal in this section to layout such task, and the development that will be described in this section.

11.1.1 Basic understanding

In modern form, a neural network (I. Goodfellow et al. 2016; T. Zhang 2023) N is typically specified by the dual $(L, [S])$ where L is the number of processing layer of a neuron, and $[S]$ is the list of neurons configuration per layer - for example, the number of neurons per layer, and so on. For such network, the definition can be simple as followed

Definition 11.1 (Standard multilayer network, **zhang2023divedeeplearning**)**.** *We define a K-layer fully-connected deep neural network with real-valued output. Let $m^{(0)} = d$ and $m^{(K)} = 1$ for m the width of the network, and d the shape of the input vector. We then recursively define:*

$$x_j^{(0)} = x_j \quad (j = 1, \dots, m^{(0)}), \tag{96}$$

$$x_j^{(k)} = h\left(\sum_{j'=1}^{m^{(k-1)}} \theta_{j,j'}^{(k)} x_{j'}^{(k-1)} + b_j^{(k)}\right) \quad (j = 1, \dots, m^{(k)}), \tag{97}$$

$$f(x) = x_1^{(K)} = \sum_{j=1}^{m^{(K-1)}} u_j x_j^{(K-1)} \tag{98}$$

where $k = 1, 2, \dots, K-1$, for the model parameters can be represented by $w = \{[u_j, \theta_{j,j'}^{(k)}, b_j^{(k)}] : j, j', k\}$ with $m^{(k)}$ being the number of hidden units at layer k; $\theta \in \mathbb{R}^m$ the weight of the neuron.

Here, each neuron is denoted by their expression as $f(\mathbf{W}^\top \mathbf{x} + b)$ for the activation function f. We can simplify a lot of those structures above into the neural network schemes. For example, the linear function class case can be simplified to a single $(2, [n, 1])$ neural network, such as:

$$x_j^{(0)} = x_j \quad (j = 1, \dots, n) \tag{99}$$

$$x^{(1)} = \sum_{j'=1}^{n} w_{j'} x_{j'}, \tag{100}$$

without the activation function h. The polynomial neural network can be represented by simply a shallow, $[3, p]$ neural network. More concretely, if we take the *input channel* as a (pre-)layer, then

$$x_j^{(0)} = x_j \quad (j = 1, \dots, m^{(0)}), \tag{101}$$

$$x_j^{(1)} = h\left(\sum_{j'=1}^{m^{(0)}} \theta_{j,j'}^{(1)} x_{j'}^{(0)}\right) + b_j^{(k)} \quad (j = 1, \dots, m^{(1)}), \quad h(x_j) = x^j \tag{102}$$

$$f(x) = x_1^{(2)} = \sum_{j=1}^{m^{(1)}} u_j x_j^{(1)} \tag{103}$$

for a neural network of shape $(3, [n, n, 1])$, and of polynomial custom scaling function. For support vector machine, this is even harder to do, since its definition varies between different notion of the model itself. However, it is interesting because we can treat it as an arbitrary generation of a neural network structure. There are papers about formalizing SVM in the sense of neural network, including making it to be an infinitely-wide shallow neural network, for example, Chen et al. 2021; Wang, Feng, and Wu 2019; Zhang and Wang 2014, however, the structure that is formed from the SVM learner can be expressed as followed:

Setting 11.1 (Support vector machine)**.** *For any given optimization problem on input space $x_i \in \mathbb{R}^d$, and label $y_i = \{\pm 1\}$, the shallow neural network resultant from the training procedure is always of the form:*

$$x_j^{(0)} = x_j \quad (j = 1, \dots, m^{(0)}), \tag{104}$$

$$x_j^{(1)} = \sum_{j' \in m^{(1)}} \theta^{(1)} x_{j'}^{(0)} K(x_{j'}^{(0)}, x) \tag{105}$$

$$f(x) = \operatorname{sign}(x_1^{(2)}) \tag{106}$$

for a specific kernel $K(x, x')$.

Those three fundamental conversions to neural network raise interesting insight into the neural network formalism. For linear and polynomial shallow neural network, we design the network and then provide it with a training scenario, evaluated of the proxy generalization. SVM however can be considered an adaptive data construction, where the neural network is constructed from the dataset or observation set itself, using minimization on RKHS and with constraints. This prompts exactly as such, on one side the pre-optimization construction, and one is the post-construction optimization of a neural network architecture. This interpretation is surprisingly useful when talking about neural network's unit-like structure. Hence, from such evidence prompts us to analyse a neural network, not in only the sense of managing its layers, but also by its unit components together.

This way of analysing unit-wise gives us much more interpretability, and also potential for a newer kind of network specification. As far as we are concerned of, a neural network can represent an active *mathematical modelling system*. This then indicates, for example in adversarial learning case between the target neural network tailored to specific concept, and the learning hypothesis network, then the process of learning itself can be loosely interpreted as a concept of mathematical modelling representation learning, even in the case of black-box model, and not just simply phenomenological modelling of statistical data. To do this, we will examine the structure from the perspective of a **generator**.

11.1.2 The SVM generator

Support vector machine, first formally originated from Vapnik 1999, is a model inherently, specifically defined for pattern recognition task, and binary classification via an *optimal separating hyperplane*. There are two variants for SVM, namely, for linear and nonlinear hyperplane, but the main general aspect of SVM is its reversal of the direction on model building -

instead, it aims to minimize the structure of a model, using the data-dependent structure, or *support vectors*.

The SV machine implements the following two precursor ideas: It maps the input vectors x, supposed of the setting, into a high-dimensional feature space Z through some nonlinear mapping, chosen a priori. In this space, an optimal separating hyperplane is constructed. By statistical learning theory, Vapnik restricted the function class (as for infinite hypothesis it is impossible to learn) to the class of hyperplanes by

$$\langle \mathbf{w} \cdot \mathbf{x} \rangle + b = 0; \quad \mathbf{w} \in \mathbb{R}^n, b \in \mathbb{R} \tag{107}$$

which $\mathbf{w}$ is the controlling weight, $\mathbf{x}$ is the input space to the space of all binary $\{-1, +1\}$ category. Hence, this basically divide the input space into two: one part containing vectors of the class -1 and the others being $+1$. If there exists such a thing, then it is said to be *linearly separable*. To find the class of a particular vector $\mathbf{x}$, we use the following decision function

$$f(\mathbf{x}) = \operatorname{sgn}[\langle \mathbf{w} \cdot \mathbf{x} \rangle + b] \tag{108}$$

This is called the **hyperplane classifier** class[k]. The above form of finding final classification can then be presented in dual form, which then depends only on dot products between vectors, as

$$f(\mathbf{x}) = \operatorname{sgn}\left(\sum_{i=1}^{\ell} y_i \alpha_i \langle \mathbf{x} \cdot \mathbf{x}_i \rangle + b\right) \tag{109}$$

where $\alpha_i \in \mathbb{R}$ is a real-valued variable that can be viewed as a *measure* of how much informational value $\mathbf{x}_i$ has (for x_i the support vectors we get from training)[l] The second idea is of the *kernel method*. This particular method, useful and well-known of the time SVM was created, gain a more prominent position among other techniques, including neural network. Specifically, this method is characterized by the mapping of the input vectors into a richer (usually high-dimensional) feature space where they satisfy *linear separable* criteria. This prompted the possibility to solve nonlinear problem through such mapping, and indeed yields a nonlinear decision surface in the input space, which is linear in the feature space. A **kernel** (function) is then a function $k(\mathbf{x}, \mathbf{y})$ that given two vectors in input space, return the dot product of their images in feature space such that $k(\mathbf{x}, \mathbf{y}) = \langle \phi(x), \phi(y) \rangle$ for the nonlinear mapping ϕ. The general form of SVM is then usually cited as

$$f(\mathbf{x}) = \operatorname{sgn}\left(\sum_{i=1}^{\ell} y_i \alpha_i k\langle \mathbf{x} \cdot \mathbf{x}_i \rangle\right) \quad \ell \leq n \tag{110}$$

k. As can be understood simply from such, there exists many hyperplanes that can correctly classifies the classes. It has been then shown that the hyperplane that guarantees the best generalization problem performances is the one with the maximal margin of separation between two classes (**Cristianini2000AnIT**).

l. The *support vector* of a particular hyperplane $\mathbf{y}$ is the collections of all points that lies closest to the hyperplane, which is specified as condition for maximization on both side of maximal margin vector machine.

for any particular final categorization. One of the major problem for analysing SVM, is in the properties of it potentially having infinitely many parameters, depends on the size of the dataset. This makes the curve of both bias-variance and double descent not applicable in such regard - infinitely many parameters' complexity, yet still finite complexity class[m].

With respect to the observable space, SVM and neural network, we can call the construction scheme of SVM a type of *backward construction* from data $\mathcal{S}$. Or rather, the following algorithm represents SVM construction problem:

Factory 4: Support Vector Machine - backward construction.

Data: $(x_i, y_i) \in \mathcal{S}, y_i = \{\pm 1\}$, hinge objective $\ell(y, f(x))$, $\nabla(\cdot)$ differentiable gradient field on ℓ.
Result: Expected optimized model to the minimal error on the objective $\ell(\cdot, \cdot)$, dependent on $\mathcal{S}$.
begin
$d \leftarrow \dim |\mathcal{S}|$;
$n \leftarrow |\mathcal{S}|$;
$\ell(y, f(x)) = \max(0, 1 - yf(x))$;
Kernalization $\phi : \mathbf{x} \to \mathbf{x}'$;
$f(\mathbf{x}) \leftarrow \mathbf{w}^\top \phi(\mathbf{x}) + b$;
Objective
Attain the minimization

$$\min_{w,b \in f} \frac{1}{2} \|w\|^2 + C \sum_{1 \leq i \leq n} \ell(y_i, f(x_i)) \tag{111}$$

Ideally for $y_i(\mathbf{w}^\top \phi(\mathbf{x}_i) + b) \geq 1, \ \forall i, y_i \in \mathcal{S}$, penalized by ℓ.;
end
return *Optimized model* $\mathsf{SVM}(f)$ *on objective* ℓ *of* ∇ *field.*

Such a factory is reversed inherently, and with its formulation constrained in the form $\mathbf{w}^\top \phi(\mathbf{x}) + b$, of which is equivalent to a non-linear shallow neural network with infinite width (**wang2017research**; Chen et al. 2021), but is not a full-fledged neural network in technicality, hence it is only a functional equivalence. That is, such result only indicates that for $N \to \infty$ of the number of neurons of a single-hidden layer neural network, $\sum_{j=1}^{N} a_j (w_j^\top x + b_j)$, then

$$\mathsf{NN}(x) \approx \mathsf{SVM}(x) = \sum_{i}^{\ell} \alpha_i k\langle x, x_i \rangle, \ell \leq n \tag{112}$$

The support vector, ℓ list, is inherently implicit of the factory itself. On the other hand, typical deep learning or neural network constructions are rather *forward-constructed*, that is,

m. From such observation, it would seem imperative that the notion of complexity based solely on the parameter counts of certain model construction is ill-equipped for certain flavours of machine learning model, in this case support vector machine (in the same type is the Gaussian Mixture Model). No double descent has been observed by empirical reports record per the author's knowledge.

the model is in a sense, prefabricated then tested of its efficacy, and thus the effective performance of the model is limited by iterations of its restricted frame. We can easily visualize the factory of which neural network are constructed, different of the SVM factory case:

12. Complexity notion

Assume the working space is the real field $\mathbb{R}^n$. Let us recall a model $h \in \mathcal{H}$ of certain hypothesis class $\mathcal{H}$. Then, the **complexity** refers to the complexity of the hypothesis h itself; there is also the notion of class maximal complexity for $\mathcal{H}$. Complexity in this case refers to the mass of the model, or rather, to answer the following questions:

1. What is the effective expression range of a hypothesis?
2. What is the hypothesis structured in?
3. If the model is constructed by parameters, how many parameters are there?
4. How many components are there in specific model h, and how would another model $h' \in \mathcal{H}$ differs from h?

Those questions are not so easily answered. Specifically, analysis often puts the hypothesis class $\mathcal{H}$ as a class of functions. So, there exists the class $\mathcal{H}_n$ of n-th degree polynomials, or else. Effective complexity in such case means the range of expression a function can give – for example, for n-th polynomial class. For the class $\mathcal{H}_n$ of polynomials of degree n on a closed interval $[a, b] \subset \mathbb{R}$, the Weierstrass approximation theorem guarantees that, as $n \to \infty$, any continuous function $f : [a, b] \to \mathbb{R}$ can be approximated arbitrarily closely in the uniform norm. Hence, the expressive complexity of $\mathcal{H}_n$ grows with n, but for fixed n only a finite-dimensional subset of $C([a, b])$ is exactly representable. This distinction highlights the difference between *approximation capacity* of a hypothesis class and the *exact representational capacity* captured by its parameterization. These expressions depend on the field or space the hypothesis class lives in, for example, if the hypothesis is configured to work in the set of all functions in $\mathbb{F} = \mathbb{R}$ of algebraic field, then we call it the algebraic family. There are various hypothesis classes in such case, and usually, not a lot of them can be analysed and represented in the same way, and hence also the analysis of the 'size' of the hypothesis, also between different hypothesis classes, prompted the usage of a more general notion, called **representation complexity**[n].

The above complexity (and so is in Table 2), aside from sample space-related measure, can also be applied onto the concept class's members itself. We also take the class complexity in such case, as the argument that maximize those measure: that is, the largest value of the measure, taken on the class as the class's complexity.[o][p][q][r][s]

The next complexity measures that can be taken of, comes from the learning process structure itself. Usually, in statistical learning theory (Hajek and Raginsky 2021; Mohri, Rostamizadeh, and Talwalkar 2012; Shalev-Shwartz and Ben-David 2014), we often use the setting that is related to **supervised process**. This usually means the process can be configured in a feedback loop, or induced response toward the model - often come from the observational space, where the concept that is set to be the target of the process - then called as learning process - is conjured. For now, we reserve this to the classical learning theory and process, and will not inquire further of such direction.

The learning process assumes an operational space containing the structure, for example, $\mathbb{R}$-encoding space of real values, and a set of **samples** living in such space, denoted $\mathcal{S}$. Those samples are often referred to as the **observation set** of a particular objective of learning. As mentioned above, $\mathcal{S}$ belongs to the concept c, assumed to be of certain concept class $\mathcal{C}$. $\mathcal{S}$ is obtained by using an observing process to create or generate the result. This process can be determined or considered in multiple way, however, generally can be considered by having a deterministic interpretation (using an explicit function $c(x)$) or by probabilistic approach (modelling a distribution $\mathcal{D}$). Hence, we gain another type of complexity, with regard to the process called **sample complexity** – usually defined of cardinality – the amount of sample needed for particular purpose, for particular objective. For the complexity category up to the structure of the learning setting by itself, we call it the **sample space complexity**.

One final complexity that is worth mentioning is then the **optimization complexity**, measure on the action of the learning algorithm. By such, it investigates the construction of an optimization algorithm solely dependent on its size of the algorithm, and other non-process related measures. Optimization complexity can also be related or expressed, in situation of time complexity, which is typical in computing system.

One might ask why we need to determine and categorize complexity measures and the need to generalize and differentiate of each one. Indeed, while in usual use cases, such complexity notion can be considered sufficient in controlling and gauging the model itself. However, just as James et al. 2009 indicated in the section on deep learning double descent, such complexity

n. Representation complexity refers to the number of components needed to express h and generally, any hypothesis in $\mathcal{H}$. This is often the parameters count, since the parameters themselves are used in constructing the function in algebraic system. Thus, again, an n-th degree polynomial might have $2(n + 1)$ parameters, for $n+1$ variational input (or variable), and $n+1$ coefficients. Notice that we specifically at the beginning, use the notion of algebraic structure $\mathbb{R}$ because of it. If we restrict ourselves to the expression, for example, for $\mathbb{B} = \{0, 1\}$, then it can be expressed in various ways in logic theory, such as conjunctive normal form (CNF), disjunctive normal form (DNF), and else. This forbid the analysis based on the effective process itself, but rather the number of components required, hence taking them as complexity of construction.

o. Expressive complexity measures the variety of input-output behaviours the model can implement.

p. Structural complexity is decomposable into component-wise measures (e.g., per-layer, per-module) and useful for architectural comparisons.

q. Sample complexity depends on hypothesis class, loss, and desired error/confidence. Further clarification must be made of a grounded definition for utilization.

r. Sample space complexity infers the statistical measure over the sample space; affects learnability and effective sample size.

s. We also note that the notion of optimization is usually hard to pin down analytically, often characterized empirically or by worst-case bounds.

Table 2. Descriptions of complexity measures applied to models and learning processes, from Hajek and Raginsky 2021; Sugiyama 2015; Mohri, Rostamizadeh, and Talwalkar 2012; Kearns and Vazirani 1994 and Vapnik 1999 with some custom notions.

Type	Notation	Purposes	Notes
Representation	$\mathcal{M}_r$	Complexity of encoding a model (size of representation, alphabet-dependent).	Model-level measure.
Expressive	$\mathcal{M}_e$	Complexity of the model's observable behaviour (operational expressivity of the mapping it implements).	Model-level measure.
Structural	$\mathcal{M}_s$	Complexity of the model's internal architecture (how components compose and interact).	Model-level measure.
Structural mass	$\mathcal{M}_\Sigma$	Aggregate representation size derived from representation complexity across the model.	Sum of $\mathcal{M}_r, \mathcal{M}_e, \mathcal{M}_s$
Sample	$\mathcal{M}_{p\in\mathcal{S}}, \mathcal{M}_p$	Number of samples required to reach a prescribed performance level (statistical sample-count).	Process-level measure.
Sample space	$\mathcal{M}_\mathcal{S}$	Intrinsic complexity of the data distribution / sample domain (richness, diversity, structured-ness).	Global process measure.
Optimization	$\mathcal{M}_\epsilon$	Complexity of the learning algorithm (convergence rate, computational/iteration cost, sensitivity) or worst-medium-best learning cost.	Process-level measure.

Notes: The notion of structural mass is convoluted. In certain setting, it is simply the parameter space $\theta \in \Theta$, but in some setting, there exists fundamental difference than identically weighted parameter space complexity, as seen in multi-basis polynomial models, not taking into account homogeneous deep learning networks. Optimization complexity coincides with PAC-learning bounds for polynomial runtime, and *sample complexity* sometimes can only be the dimension of the input/observation space.

might eventually, of some regions, inadequate to express and evaluate the complexity of the model in its operations any more. Such transition is unanswered, and while the founding of complexity notion is still fragmented, it is indeed difficult to not paying attention specifically to those notions themselves. Considering such of the learning framework and the more sophisticated goal that we push forth of reforming the learning theory, it is then natural to define such.

References

[PDF] A Unifeid Bias-Variance Decomposition and its Applications | Semantic Scholar. Accessed July 16, 2024. https://www.semanticscholar.org/paper/A-Unifeid-Bias-Variance-Decomposition-and-its-Domingos/e1ed9d24db5e8f7ab326aeb797e965a94f5ad6d3.

[PDF] A Unifeid Bias-Variance Decomposition and its Applications | Semantic Scholar. Accessed July 16, 2024. https://www.semanticscholar.org/paper/A-Unifeid-Bias-Variance-Decomposition-and-its-Domingos/e1ed9d24db5e8f7ab326aeb797e965a94f5ad6d3.

Achlioptas, Panos. n.d. Stochastic Gradient Descent in Theory and Practice [in en]. *Lecture note, Stanford's AI.*

Adlam, Ben, and Jeffrey Pennington. 2020. *Understanding double descent requires a fine-grained bias-variance decomposition.* arXiv: 2011.03321 `[stat.ML]`. https://arxiv.org/abs/2011.03321.

Advani, Madhu S., and Andrew M. Saxe. 2017. *High-dimensional dynamics of generalization error in neural networks.* arXiv: 1710.03667 `[stat.ML]`. https://arxiv.org/abs/1710.03667.

al., Molavi et. *Model Complexity, Expectations, and Asset Prices | The Review of Economic Studies | Oxford Academic.* Accessed July 16, 2024. https://academic.oup.com/restud/article-abstract/91/4/2462/7222145?redirectedFrom=fulltext&login=false.

Allerbo, Oskar. 2025. *Changing the kernel during training leads to double descent in kernel regression.* arXiv: 2311.01762 `[stat.ML]`. https://arxiv.org/abs/2311.01762.

Alquier, Pierre. 2024. User-friendly introduction to pac-bayes bounds. *Foundations and Trends in Machine Learning* 17 (2): 174–303. ISSN: 1935-8245. https://doi.org/10.1561/2200000100. http://dx.doi.org/10.1561/2200000100.

Amari, Shun-ichi. 1996. Neural learning in structured parameter spaces - natural riemannian gradient. In *Advances in neural information processing systems 9 (nips'96),* 287–293. Also cited as "Natural Riemannian Gradient".

———. 1998. Natural gradient works efficiently in learning. *Neural Computation* 10 (2): 251–276. https://doi.org/10.1162/089976698300017746.

Ammar, Mouïn Ben, David Brellmann, Arturo Mendoza, Antoine Manzanera, and Gianni Franchi. 2025. *Double descent meets out-of-distribution detection: theoretical insights and empirical analysis on the role of model complexity.* arXiv: 2411.02184 `[stat.ML]`. https://arxiv.org/abs/2411.02184.

Arjevani, Yossi, Joan Bruna, Joe Kileel, Elzbieta Polak, and Matthew Trager. 2025. *Geometry and optimization of shallow polynomial networks.* arXiv: 2501.06074 `[cs.LG]`. https://arxiv.org/abs/2501.06074.

Ascoli, Stéphane d', Levent Sagun, and Giulio Biroli. 2020. Triple descent and the two kinds of overfitting: where & why do they appear? In *Advances in Neural Information Processing Systems,* 33:3058–3069. Curran Associates, Inc. Accessed August 26, 2024. https://proceedings.neurips.cc/paper/2020/hash/1fd09c5f59a8ff35d499c0ee25a1d47e-Abstract.html.

Barceló, Pablo, Mikaël Monet, Jorge Pérez, and Bernardo Subercaseaux. 2020. *Model interpretability through the lens of computational complexity.* arXiv: 2010.12265 `[cs.AI]`. https://arxiv.org/abs/2010.12265.

Belkin, Mikhail, Daniel Hsu, Siyuan Ma, and Soumik Mandal. 2019a. Reconciling modern machine learning practice and the bias-variance trade-off. ArXiv:1812.11118 [cs, stat], *Proc. Natl. Acad. Sci. U.S.A.* 116, no. 32 (August): 15849–15854. ISSN: 0027-8424, 1091-6490, accessed July 16, 2024. https://doi.org/10.1073/pnas.1903070116. http://arxiv.org/abs/1812.11118.

———. 2019b. Reconciling modern machine learning practice and the bias-variance trade-off [in en]. ArXiv:1812.11118 [cs, stat], *Proc. Natl. Acad. Sci. U.S.A.* 116, no. 32 (August): 15849–15854. ISSN: 0027-8424, 1091-6490, accessed July 16, 2024. https://doi.org/10.1073/pnas.1903070116. http://arxiv.org/abs/1812.11118.

Belkin, Mikhail, Daniel Hsu, and Ji Xu. 2020. Two models of double descent for weak features. *SIAM Journal on Mathematics of Data Science* 2, no. 4 (January): 1167–1180. ISSN: 2577-0187. https://doi.org/10.1137/20m1336072. http://dx.doi.org/10.1137/20M1336072.

Belkin, Mikhail, Siyuan Ma, and Soumik Mandal. 2018. *To understand deep learning we need to understand kernel learning.* arXiv: 1802.01396 `[stat.ML]`. https://arxiv.org/abs/1802.01396.

Bias-variance tradeoffs in program analysis | Proceedings of the 41st ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages. Accessed July 16, 2024. https://dl.acm.org/doi/10.1145/2535838.2535853.

Bishop, Christopher M. 1995. *Neural networks for pattern recognition.* Oxford University Press.

Bousquet, Olivier, Steve Hanneke, Shay Moran, Ramon van Handel, and Amir Yehudayoff. 2020. *A theory of universal learning.* arXiv: 2011.04483 `[cs.LG]`. https://arxiv.org/abs/2011.04483.

Boyd, JP. 2009. Divergence (runge phenomenon) for least-squares polynomial approximation on an equispaced grid and mock-chebyshev subset interpolation. Study of divergence behavior and mitigation by Mock-Chebyshev grid, *Applied Mathematics and Computation* 213 (1): 292–303. https://doi.org/10.1016/j.amc.2008.12.087.

Brellmann, David, Eloise Berthier, David Filliat, and Goran Frehse. 2024. On double descent in reinforcement learning with LSTD and random features. In *The twelfth international conference on learning representations.* https://openreview.net/forum?id=9RIbNmx984.

Brown, Gavin, and Riccardo Ali. 2024. Bias/variance is not the same as approximation/estimation. *Transactions on Machine Learning Research,* ISSN: 2835-8856. https://openreview.net/forum?id=4TnFbv16hK.

Brown, Tom B., Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, et al. 2020. Language models are few-shot learners. *arXiv preprint,* https://arxiv.org/abs/2005.14165.

Buschjäger, Sebastian, Lukas Pfahler, and Katharina Morik. 2020a. *Generalized Negative Correlation Learning for Deep Ensembling.* ArXiv:2011.02952 [cs, stat], December. Accessed July 16, 2024. https://doi.org/10.48550/arXiv.2011.02952. http://arxiv.org/abs/2011.02952.

———. 2020b. *Generalized Negative Correlation Learning for Deep Ensembling* [in en]. ArXiv:2011.02952 [cs, stat], December. Accessed July 16, 2024. http://arxiv.org/abs/2011.02952.

Cauchy, Augustin-Louis. 1847. General method for the resolution of simultaneous systems of equations. *Comp. Rend. Sci. Paris* 25:536–538.

Chen, Yilan, Wei Huang, Lam M. Nguyen, and Tsui-Wei Weng. 2021. On the equivalence between neural network and support vector machine. *arXiv preprint arXiv:2111.06063,* https://arxiv.org/abs/2111.06063.

Cherkassky, Vladimir, and Eng Hock Lee. 2024. To understand double descent, we need to understand vc theory. *Neural Networks* 169:242–256. https://doi.org/10.1016/j.neunet.2023.10.014. https://doi.org/10.1016/j.neunet.2023.10.014.

Corless, Robert M., and Leili Rafiee Sevyeri. 2018. The runge example for interpolation and wilkinson's examples for rootfinding. Modern analysis of Runge's phenomenon and conditioning, *arXiv preprint* 0 (arXiv:1804.08561). https://arxiv.org/abs/1804.08561.

Curth, Alicia, Alan Jeffares, and Mihaela van der Schaar. 2023. *A u-turn on double descent: rethinking parameter counting in statistical learning.* arXiv: 2310.18988 `[stat.ML]`. https://arxiv.org/abs/2310.18988.

Cybenko, G. 1989. Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals and Systems* 2:303–314. https://doi.org/10.1007/BF02551274.

Davies, Xander, Lauro Langosco, and David Krueger. 2023. *Unifying Grokking and Double Descent* [in en]. ArXiv:2303.06173 [cs], March. Accessed August 26, 2024. http://arxiv.org/abs/2303.06173.

Devlin, Jacob, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: pre-training of deep bidirectional transformers for language understanding. *arXiv preprint,* https://arxiv.org/abs/1810.04805.

Domingos, Pedro. 2000a. A unified bias-variance decomposition and its applications. In *Proceedings of the 17th international conference on machine learning (icml-2000),* 231–238. Morgan Kaufmann. https://homes.cs.washington.edu/~pedrod/papers/mlc00a.pdf.

———. 2000b. A unified bias-variance decomposition for zero-one and squared loss. In *Proceedings of the 17th national conference on artificial intelligence (aaai-2000),* 564–569. AAAI Press. https://homes.cs.washington.edu/~pedrod/papers/aaai00.pdf.

Domingos, Pedro M. 2000a. A Unifeid Bias-Variance Decomposition and its Applications. In *Semantic scholar.* June. Accessed July 16, 2024. https://www.semanticscholar.org/paper/A-Unifeid-Bias-Variance-Decomposition-and-its-Domingos/e1ed9d24db5e8f7ab326aeb797e965a94f5ad6d3.

———. 2000b. A unified bias-variance decomposition for zero-one and squared loss. In *Aaai/iaai.* https://api.semanticscholar.org/CorpusID:2063488.

E. L. Lehmann, George Casella. 1998. *Theory of Point Estimation* [in en]. Springer Texts in Statistics. New York: Springer-Verlag. ISBN: 978-0-387-98502-2, accessed May 25, 2025. https://doi.org/10.1007/b98854. http://link.springer.com/10.1007/b98854.

Fletcher, R., and C. M. Reeves. 1964. Function minimization by conjugate gradients. *The Computer Journal* 7 (2): 149–154. https://doi.org/10.1093/comjnl/7.2.149.

Fukushima, Kunihiko. 1980. Neocognitron: a self-organizing neural network model for a mechanism of pattern recognition unaffected by shift in position. *Biological Cybernetics* 36:193–202. https://doi.org/10.1007/BF00344251.

Geman, Stuart, Elie Bienenstock, and Rene Doursat. 1992. Neural networks and the bias/variance dilemma. *Neural Computation* 4 (1): 1–58. https://doi.org/10.1162/neco.1992.4.1.1.

Goodfellow, Ian, Yoshua Bengio, Aaron Courville, and Yoshua Bengio. 2016. *Deep learning.* Vol. 1. MIT Press.

Goodfellow, Ian J., Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. 2014. Generative adversarial nets. In *Advances in neural information processing systems (neurips).* https://arxiv.org/abs/1406.2661.

Gunasekar, Suriya, Blake Woodworth, and Nathan Srebro. 2021. Mirrorless mirror descent: a natural derivation of mirror descent. In *Proceedings of the 24th international conference on artificial intelligence and statistics (aistats 2021),* 130:2305–2313. PMLR.

Hajek, Bruce, and Maxim Raginsky. 2021. *Statistical learning theory.* Vol. 1. https://maxim.ece.illinois.edu/teaching/SLT/.

Harzli, Ouns El, Bernardo Cuenca Grau, Guillermo Valle-Perez, and Ard A. Louis. 2023. *Double-descent curves in neural networks: a new perspective using gaussian processes.* arXiv: 2102.07238 [stat.ML]. https://arxiv.org/abs/2102.07238.

Hastie, Trevor, Andrea Montanari, Saharon Rosset, and Ryan J Tibshirani. 2019. Surprises in high-dimensional ridgeless least squares interpolation. *arXiv preprint arXiv:1903.08560.*

Haykin, Simon S. 1994. *Neural networks: a comprehensive foundation.* Macmillan.

He, Kaiming, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the ieee conference on computer vision and pattern recognition (cvpr),* 770–778. https://arxiv.org/abs/1512.03385.

Hebb, D. O. 1949. *The organization of behavior: a neuropsychological theory.* New York: John Wiley & Sons.

Heckel, Rebekka, and Murat Yilmaz. 2020. *Early stopping in deep networks: double descent and how to eliminate it.* arXiv preprint. Discusses epoch-wise double descent and mitigation strategies. https://arxiv.org/abs/2007.10099.

Hellström, Thomas, Virginia Dignum, and Suna Bensch. 2020. *Bias in Machine Learning – What is it Good for?* ArXiv:2004.00686 [cs], September. Accessed July 16, 2024. https://doi.org/10.48550/arXiv.2004.00686. http://arxiv.org/abs/2004.00686.

Hestenes, Magnus R., and Eduard Stiefel. 1952. Methods of conjugate gradients for solving linear systems. *Journal of Research of the National Bureau of Standards* 49 (6): 409–436. https://doi.org/10.6028/jres.049.044.

Hinton, Geoffrey E., Simon Osindero, and Yee-Whye Teh. 2006. A fast learning algorithm for deep belief nets. *Neural Computation* 18:1527–1554. https://www.cs.toronto.edu/~fritz/absps/ncfast.pdf.

Hochreiter, Sepp, and Jürgen Schmidhuber. 1997. Long short-term memory. *Neural Computation* 9 (8): 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735.

Hopfield, J. J. 1982. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences* 79 (8): 2554–2558. https://doi.org/10.1073/pnas.79.8.2554.

Hornik, Kurt. 1991. Approximation capabilities of multilayer feedforward networks. *Neural Networks* 4 (2): 251–257.

Hu, Erdong. 2021. An overview of double descent and overparameterization. *independent,* no. 0.

Hu, Xia, Lingyang Chu, Jian Pei, Weiqing Liu, and Jiang Bian. 2021a. *Model Complexity of Deep Learning: A Survey.* ArXiv:2103.05127 [cs], August. Accessed July 18, 2024. https://doi.org/10.48550/arXiv.2103.05127. http://arxiv.org/abs/2103.05127.

———. 2021b. *Model complexity of deep learning: a survey.* arXiv: 2103.05127 [cs.LG]. https://arxiv.org/abs/2103.05127.

Huang, Baihe, Shanda Li, Tianhao Wu, Yiming Yang, Ameet Talwalkar, Kannan Ramchandran, Michael I. Jordan, and Jiantao Jiao. 2025. *Sample complexity and representation ability of test-time scaling paradigms.* arXiv: 2506.05295 [cs.LG]. https://arxiv.org/abs/2506.05295.

Jacot, Arthur, Francois Gabriel, and Clement Hongler. 2018. Neural tangent kernel: convergence and generalization in neural networks. In *Advances in neural information processing systems (neurips).* Kernel view of wide-network behavior.

James, Gareth, Trevor Hastie, Robert Tibshirani, and Jerome H. Friedman. 2009. *The elements of statistical learning: data mining, inference, and prediction.* 2nd. Springer Series in Statistics. Springer.

Janik, Romuald A., and Przemek Witaszczyk. 2021. *Complexity for deep neural networks and other characteristics of deep feature representations.* arXiv: 2006.04791 [cs.LG]. https://arxiv.org/abs/2006.04791.

Kaplan, Jared, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. 2020. *Scaling laws for neural language models.* arXiv: 2001.08361 [cs.LG]. https://arxiv.org/abs/2001.08361.

Kearns, Michael J., and Umesh V. Vazirani. 1994. *An introduction to computational learning theory.* Cambridge, MA, USA: MIT Press. ISBN: 0262111934.

Khan, Mohammad Emtiyaz, and Håvard Rue. 2024. *The Bayesian Learning Rule.* ArXiv:2107.04562 [cs, stat] version: 4, June. Accessed July 16, 2024. https://doi.org/10.48550/arXiv.2107.04562. http://arxiv.org/abs/2107.04562.

Kohonen, Teuvo. 1982. Self-organized formation of topologically correct feature maps. *Biological Cybernetics* 43:59–69. https://doi.org/10.1007/BF00337288.

Krizhevsky, Alex, Ilya Sutskever, and Geoffrey E. Hinton. 2012. Imagenet classification with deep convolutional neural networks. In *Advances in neural information processing systems (neurips).* https://papers.nips.cc/paper/4824-imagenet-classification-with-deep-convolutional-neural-networks.pdf.

Lafon, Marc, and Alexandre Thomas. 2024. *Understanding the Double Descent Phenomenon in Deep Learning* [in en]. ArXiv:2403.10459 [cs, stat], March. Accessed August 26, 2024. http://arxiv.org/abs/2403.10459.

LeCun, Yann, Yoshua Bengio, and Geoffrey Hinton. 2015. Deep learning. *Nature* 521:436–444. https://www.cs.toronto.edu/~hinton/absps/NatureDeepReview.pdf.

LeCun, Yann, Leon Bottou, Yoshua Bengio, and Patrick Haffner. 1998. Gradient-based learning applied to document recognition. *Proceedings of the IEEE* 86 (11): 2278–2324. http://yann.lecun.com/exdb/publis/pdf/lecun-01a.pdf.

Lee, Eng Hock, and Vladimir Cherkassky. 2022. *Vc theoretical explanation of double descent.* arXiv: 2205.15549 [stat.ML]. https://arxiv.org/abs/2205.15549.

Li, Xinyue, and Rishi Sonthalia. 2024. Least squares regression can exhibit under-parameterized double descent. In *Advances in neural information processing systems,* edited by A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, 37:25510–25560. Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2024/file/2d43f7a61b57f83619f82c971e4bddc0-Paper-Conference.pdf.

Liu, Chris Yuhao, and Jeffrey Flanigan. 2023. *Understanding the role of optimization in double descent.* arXiv: 2312.03951 [cs.LG]. https://arxiv.org/abs/2312.03951.

Liu, Fanghui, Zhenyu Liao, and Johan A. K. Suykens. 2021. *Kernel regression in high dimensions: refined analysis beyond double descent.* arXiv: 2010.02681 [stat.ML]. https://arxiv.org/abs/2010.02681.

Luo, Jing, Huiyuan Wang, and Weiran Huang. 2024. *Investigating the impact of model complexity in large language models.* arXiv: 2410.00699 [cs.LG]. https://arxiv.org/abs/2410.00699.

Luo, Yifan, and Bin Dong. 2023. *Double descent of discrepancy: a task-, data-, and model-agnostic phenomenon.* arXiv: 2305.15907 [cs.LG]. https://arxiv.org/abs/2305.15907.

Luong, Duy V. N., Panos Parpas, Daniel Rueckert, and Berc Rustem. 2016. A weighted mirror descent algorithm for nonsmooth convex optimization problem. *Journal of Optimization Theory and Applications* 170 (3): 900–915. https://doi.org/10.1007/s10957-016-0963-5.

Manin, Yuri, and Matilde Marcolli. 2024. Homotopy theoretic and categorical models of neural information networks. *Compositionality* Volume 6 (2024) (September). ISSN: 2631-4444. https://doi.org/10.46298/compositionality-6-4. http://dx.doi.org/10.46298/compositionality-6-4.

McCulloch, Warren S., and Walter Pitts. 1943. A logical calculus of the ideas immanent in nervous activity. *Bulletin of Mathematical Biophysics* 5:115–133.

Mehrabi, Ninareh, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. 2022. *A Survey on Bias and Fairness in Machine Learning.* ArXiv:1908.09635 [cs], January. Accessed July 16, 2024. https://doi.org/10.48550/arXiv.1908.09635. http://arxiv.org/abs/1908.09635.

Mei, Song, and Andrea Montanari. 2019a. The generalization error of random features regression: precise asymptotics and double descent curve. *arXiv preprint,* https://arxiv.org/abs/1908.05355.

———. 2019b. The generalization error of random features regression: precise asymptotics and double descent curve. *arXiv preprint arXiv:1908.05355.*

———. 2020. *The generalization error of random features regression: precise asymptotics and double descent curve.* arXiv: 1908.05355 [math.ST]. https://arxiv.org/abs/1908.05355.

Minsky, Marvin, and Seymour Papert. 1969. *Perceptrons: an introduction to computational geometry.* MIT Press.

Mohri, Mehryar, Afshin Rostamizadeh, and Ameet Talwalkar. 2012. *Foundations of machine learning.* The MIT Press. ISBN: 026201825X.

Molavi, Pooya, Alireza Tahbaz-Salehi, and Andrea Vedolin. 2024. Model Complexity, Expectations, and Asset Prices. *The Review of Economic Studies* 91, no. 4 (July): 2462–2507. ISSN: 0034-6527, accessed July 16, 2024. https://doi.org/10.1093/restud/rdad073. https://doi.org/10.1093/restud/rdad073.

Molnar, Christoph, Giuseppe Casalicchio, and Bernd Bischl. 2020. Quantifying model complexity via functional decomposition for better post-hoc interpretability. In *Machine learning and knowledge discovery in databases,* 193–204. Springer International Publishing. ISBN: 9783030438234. https://doi.org/10.1007/978-3-030-43823-4_17. http://dx.doi.org/10.1007/978-3-030-43823-4_17.

Muresan, Mircea. 2023. Bernstein approximation and beyond: proofs by means of elementary probability theory. *arXiv preprint arXiv:2307.11533.*

Nakkiran, Preetum. 2019a. *More data can hurt for linear regression: sample-wise double descent.* arXiv:1912.07242. ArXiv preprint. https://arxiv.org/abs/1912.07242.

———. 2019b. *More data can hurt for linear regression: sample-wise double descent.* arXiv: 1912.07242 [stat.ML]. https://arxiv.org/abs/1912.07242.

Nakkiran, Preetum, Gal Kaplun, Yamini Bansal, Tristan Yang, Boaz Barak, and Ilya Sutskever. 2019. *Deep Double Descent: Where Bigger Models and More Data Hurt.* ArXiv:1912.02292 [cs, stat], December. Accessed July 16, 2024. https://doi.org/10.48550/arXiv.1912.02292. http://arxiv.org/abs/1912.02292.

Nakkiran, Preetum, Prayaag Venkat, Sham Kakade, and Tengyu Ma. 2020. *Optimal regularization can mitigate double descent (openreview entry).* OpenReview forum submission. https://openreview.net/forum?id=7R7fAoUygoa.

———. 2021. *Optimal regularization can mitigate double descent.* arXiv: 2003.01897 [cs.LG]. https://arxiv.org/abs/2003.01897.

Neal, Brady. 2019. *On the bias-variance tradeoff: textbooks need an update.* arXiv: 1912.08286 [cs.LG]. https://arxiv.org/abs/1912.08286.

Neal, Brady, Sarthak Mittal, Aristide Baratin, Vinayak Tantia, Matthew Scicluna, Simon Lacoste-Julien, and Ioannis Mitliagkas. 2018. A modern take on the bias-variance tradeoff in neural networks. *arXiv preprint arXiv:1810.08591.*

Nemirovski, Arkadii S., and David B. Yudin. 1983. *Problem complexity and method efficiency in optimization.* Wiley-Interscience Series in Discrete Mathematics. Chichester, UK: John Wiley / Sons. ISBN: 9780471103455.

Nie, Jiawang, Zhi Yang, and Guoyin Zhou. 2021. The saddle point problem of polynomials. *Foundations of Computational Mathematics* 22 (5): 1133–1169. https://doi.org/10.1007/s10208-021-09526-8. https://link.springer.com/article/10.1007/s10208-021-09526-8.

Olmin, Amanda, and Fredrik Lindsten. 2024. *Towards understanding epoch-wise double descent in two-layer linear neural networks.* arXiv: 2407.09845 [stat.ML]. https://arxiv.org/abs/2407.09845.

Oono, Kenta, and Taiji Suzuki. 2020. Graph neural networks exponentially lose expressive power for node classification. In *International conference on learning representations.* https://openreview.net/forum?id=S1ldO2EFPr.

Paninski, Liam. 2005. *Statistics 4107: Intro to Math Stat (fall 2005).* Accessed May 25, 2025. https://sites.stat.columbia.edu/liam/teaching/4107-fall05/.

Pezeshki, Mohammad, Amartya Mitra, Yoshua Bengio, and Guillaume Lajoie. 2021. *Multi-scale feature learning dynamics: insights for double descent.* arXiv: 2112.03215 [cs.LG]. https://arxiv.org/abs/2112.03215.

Pfau, David. 2013. *A generalized bias-variance decomposition for bregman divergences.* Technical report.

Power, Alethea, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. 2022. *Grokking: generalization beyond overfitting on small algorithmic datasets.* arXiv: 2201.02177 [cs.LG]. https://arxiv.org/abs/2201.02177.

Quétu, Victor, and Enzo Tartaglione. 2023a. Can we avoid Double Descent in Deep Neural Networks? [In en]. In *2023 IEEE International Conference on Image Processing (ICIP),* 1625–1629. ArXiv:2302.13259 [cs]. October. Accessed August 26, 2024. https://doi.org/10.1109/ICIP49359.2023.10222624. http://arxiv.org/abs/2302.13259.

———. 2023b. Can we avoid Double Descent in Deep Neural Networks? [In en]. In *2023 IEEE International Conference on Image Processing (ICIP),* 1625–1629. ArXiv:2302.13259 [cs]. October. Accessed August 26, 2024. https://doi.org/10.1109/ICIP49359.2023.10222624. http://arxiv.org/abs/2302.13259.

Raskutti, Garvesh, and Sayan Mukherjee. 2013. The information geometry of mirror descent. ArXiv:1310.7780, *arXiv Preprint.*

Rattray, M., D. Saad, and Shun-ichi Amari. 1998. Natural gradient descent for on-line learning. *Physical Review Letters* 81:5461–5464. https://doi.org/10.1103/PhysRevLett.81.5461.

Robbins, Herbert, and Sutton Monro. 1951. A stochastic approximation method. *Annals of Mathematical Statistics* 22 (3): 400–407. https://doi.org/10.1214/aoms/1177729586.

Rosenblatt, Frank. 1958a. The perceptron: a probabilistic model for information storage and organization in the brain. *Psychological Review* 65 (6): 386–408. https://doi.org/10.1037/h0042519.

———. 1958b. The perceptron: a probabilistic model for information storage and organization in the brain. *Psychological review* 65 6:386–408. https://api.semanticscholar.org/CorpusID:12781225.

Ruder, Sebastian. 2017. *An overview of gradient descent optimization algorithms.* ArXiv:1609.04747 [cs], June. Accessed June 28, 2025. https://doi.org/10.48550/arXiv.1609.04747. http://arxiv.org/abs/1609.04747.

Rumelhart, David E., Geoffrey E. Hinton, and Ronald J. Williams. 1986. Learning representations by back-propagating errors. *Nature* 323:533–536. https://doi.org/10.1038/323533a0.

Runge, Carl. 1901. Über empirische funktionen und die interpolation zwischen äquidistanten ordinaten. The original Runge example showing divergence with equispaced interpolation, *Zeitschrift für Mathematik und Physik* 46:224–243.

Schaeffer, Rylan, Mikail Khona, Zachary Robertson, Akhilan Boopathy, Kateryna Pistunova, Jason W. Rocks, Ila Rani Fiete, and Oluwasanmi Koyejo. 2023. *Double Descent Demystified: Identifying, Interpreting & Ablating the Sources of a Deep Learning Puzzle.* ArXiv:2303.14151 [cs, stat], March. Accessed July 16, 2024. https://doi.org/10.48550/arXiv.2303.14151. http://arxiv.org/abs/2303.14151.

Schep, Anton R. 2007. *Weierstrass' proof of the weierstrass approximation theorem.* Technical Report / Preprint. PDF, May 1. University of South Carolina. https://people.math.sc.edu/schep/weierstrass.pdf.

Shalev-Shwartz, Shai, and Shai Ben-David. 2014. *Understanding machine learning: from theory to algorithms.* USA: Cambridge University Press. ISBN: 1107057132.

Shalev-Shwartz, Shai, Ohad Shamir, Nathan Srebro, and Karthik Sridharan. 2010. Learnability, stability and uniform convergence. *Journal of Machine Learning Research* 11 (90): 2635–2670. http://jmlr.org/papers/v11/shalev-shwartz10a.html.

Sharma, Rahul, and Alex Aiken. 2014. Bias-variance tradeoffs in program analysis. In *Proceedings of the 41st ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages,* 127–137. POPL '14. New York, NY, USA: Association for Computing Machinery. ISBN: 978-1-4503-2544-8, accessed July 15, 2024. https://doi.org/10.1145/2535838.2535853. https://doi.org/10.1145/2535838.2535853.

Shen, Zewen, and Kirill Serkh. 2025. On polynomial interpolation in the monomial basis. *SIAM Journal on Numerical Analysis* 63, no. 2 (March): 469–494. ISSN: 1095-7170. https://doi.org/10.1137/23m1623215. http://dx.doi.org/10.1137/23M1623215.

Shi, Cheng, Liming Pan, Hong Hu, and Ivan Dokmanić. 2024. *Homophily modulates double descent generalization in graph convolution networks.* arXiv: 2212.13069 [cs.LG]. https://arxiv.org/abs/2212.13069.

Simonyan, Karen, and Andrew Zisserman. 2014. Very deep convolutional networks for large-scale image recognition. *arXiv preprint,* https://arxiv.org/abs/1409.1556.

Sohl-Dickstein, Jascha. 2012. The natural gradient by analogy to signal whitening, and recipes and tricks for its use. ArXiv:1205.1828, *arXiv Preprint.*

Soudry, Daniel, Elad Hoffer, Mor Shpigel Nacson, Suriya Gunasekar, and Nathan Srebro. 2024. *The implicit bias of gradient descent on separable data.* arXiv: 1710.10345 [stat.ML]. https://arxiv.org/abs/1710.10345.

Spiess, Jann, Guido Imbens, and Amar Venugopal. 2023. *Double and single descent in causal inference with an application to high-dimensional synthetic control.* arXiv: 2305.00700 [econ.EM]. https://arxiv.org/abs/2305.00700.

Srivastava, Nitish, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. 2014a. Dropout: a simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research* 15 (56): 1929–1958. https://jmlr.org/papers/volume15/srivastava14a/srivastava14a.pdf.

———. 2014b. Dropout: a simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research* 15:1929–1958. https://www.jmlr.org/papers/v15/srivastava14a.html.

Sterkenburg, Tom F. 2024. Statistical learning theory and occam's razor: the core argument. *Minds and Machines* 35, no. 1 (November). ISSN: 1572-8641. https://doi.org/10.1007/s11023-024-09703-y. http://dx.doi.org/10.1007/s11023-024-09703-y.

Sugiyama, Masashi. 2015. *Introduction to statistical machine learning.* San Francisco, CA, USA: Morgan Kaufmann Publishers Inc. ISBN: 9780128023501.

Sury, B. 2019. Weierstrass' theorem — leaving no stone unturned. Expository article comparing different proofs including Bernstein, *Lecture Notes / Expository Review.*

Szegedy, Christian, Wei Liu, Yangqing Jia, Pierre Sermanet, Scott Reed, Dragomir Anguelov, Dumitru Erhan, Vincent Vanhoucke, and Andrew Rabinovich. 2014. Going deeper with convolutions. *arXiv preprint,* https://arxiv.org/abs/1409.4842.

Transtrum, Mark K., Gus L. W. Hart, Tyler J. Jarvis, and Jared P. Whitehead. 2025. *Egad! double descent is explained by generalized aliasing decomposition.* arXiv: 2408.08294 [math.ST]. https://arxiv.org/abs/2408.08294.

Truong, Lan V. 2025. *On rademacher complexity-based generalization bounds for deep learning.* arXiv: 2208.04284 [stat.ML]. https://arxiv.org/abs/2208.04284.

Valiant, L. G. 1984. A theory of the learnable. *Commun. ACM* (New York, NY, USA) 27, no. 11 (November): 1134–1142. ISSN: 0001-0782. https://doi.org/10.1145/1968.1972. https://doi.org/10.1145/1968.1972.

Vapnik, Vladimir. 1999. *The nature of statistical learning theory.* Springer: New York.

Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Advances in neural information processing systems (neurips).* https://arxiv.org/abs/1706.03762.

Ven, Gido M. van de, Nicholas Soures, and Dhireesha Kudithipudi. 2024. *Continual learning and catastrophic forgetting.* arXiv: 2403.05175 [cs.LG]. https://arxiv.org/abs/2403.05175.

Wang, Jingyuan, Kai Feng, and Junjie Wu. 2019. Svm-based deep stacking networks. In *Proceedings of the twenty-eighth international joint conference on artificial intelligence (ijcai-19),* 5262–5268. https://doi.org/10.24963/ijcai.2019/731. https://www.ijcai.org/proceedings/2019/731.

Wei, Jason, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, et al. 2022. *Emergent abilities of large language models.* arXiv: 2206.07682 [cs.CL]. https://arxiv.org/abs/2206.07682.

Werbos, Paul J. 1974. Beyond regression: new tools for prediction and analysis in the behavioral sciences. PhD diss., Harvard University.

Widrow, Bernard, and Marcian E. Hoff. 1960. Adaptive switching circuits. In *1960 ire wescon convention record, part 4,* 96–104. New York: Institute of Radio Engineers.

Widrow, Bernard, and Ted Hoff. 1960. *Adaline: an adaptive 'adaline' neuron and adaptive switching circuits.* Technical report. Stanford University.

Xiao, Lin, and Tong Zhang. 2014. A proximal stochastic gradient method with progressive variance reduction. *SIAM Journal on Optimization* 24 (4): 2057–2075. https://doi.org/10.1137/140961791.

Yang, H. H., and Shun-ichi Amari. 1998. Complexity issues in natural gradient descent method for training multilayer perceptrons. *Neural Computation* 10 (8): 2137–2157. https://doi.org/10.1162/089976698300017007.

Yang, Tian-Le, and Joe Suzuki. 2024. Dropout drops double descent. *Japanese Journal of Statistics and Data Science* 7, no. 2 (March): 615–632. ISSN: 2520-8764. https://doi.org/10.1007/s42081-024-00242-5. http://dx.doi.org/10.1007/s42081-024-00242-5.

Yang, Zitong, Yaodong Yu, Chong You, Jacob Steinhardt, and Yi Ma. 2020. *Rethinking Bias-Variance Trade-off for Generalization of Neural Networks* [in en]. ArXiv:2002.11328 [cs, stat], December. Accessed July 16, 2024. http://arxiv.org/abs/2002.11328.

Zhang, Jiawei. 2019. *Gradient Descent based Optimization Algorithms for Deep Learning Models Training.* ArXiv:1903.03614 [cs], March. Accessed June 28, 2025. https://doi.org/10.48550/arXiv.1903.03614. http://arxiv.org/abs/1903.03614.

Zhang, Rui, and Hong Wang. 2014. The equivalence relationship between kernel functions based on svm and four-layer functional networks. In *Intelligent computing methodologies,* edited by De-Shuang Huang, Vito Bevilacqua, Prashan Premaratne, and Praveen Gupta, 8589:81–89. Lecture Notes in Computer Science. Springer. https://doi.org/10.1007/978-3-319-09339-0_10. https://doi.org/10.1007/978-3-319-09339-0_10.

Zhang, Tong. 2023. *Mathematical analysis of machine learning algorithms.* Cambridge University Press. https://doi.org/10.1017/9781009093057. https://www.cambridge.org/core/books/mathematical-analysis-of-machine-learning-algorithms/EB9BABB05A5C312F19C38E5A01A5ECFC.

Zhang, Ya Shi. 2024. *Manipulating sparse double descent.* arXiv: 2401.10686 [cs.LG]. https://arxiv.org/abs/2401.10686.

Appendix 1. Reevaluation of theorem 57

To verify the theorem itself, we test it with a standard setting without the theorem results. This is reflected in Figure 8. The result is somewhat consistent with the theorem and its latter parameter error and prediction error results, however, this time, it correlates exactly as the prediction error instead. Additionally, peculiarity occurs in the non-double descent region, with much less volatility than in the prediction of the theorem.

Algorithm 5: Linear regression with Gaussian white noise, full control.

Data: n, p, d, σ, r, fixed range $[a, b]$, $u \in [0, 1]$
Result: Threefold error measure - MSE on parameters p, d, MSE on prediction Y', Y, and theoretical risks measure by prescription.

begin
- Randomizer$(\cdot) \leftarrow$ PCG64$(n \leq 100)$; `Randomizing pseudo-random engine`
- $\sigma \leftarrow 0.5$, $d \leftarrow 150$; `Common presetting for parameters`
- $nl \leftarrow \{12, \ldots, 280\}$; `Common full-list` n `count of datapoints`
- $p \leftarrow 150$; `Underparameterization to equal`
- $\mathbf{p} \leftarrow$ range$(1, p)$;
- $[a, b] \leftarrow [-100, 100]$;
- $w_c \sim$ PCG64.Uniform$([1, d], 150)$;
- $b_c \sim$ PCG64.Uniform$([1, 10])$;
- $w_p \sim$ PCG64.Uniform$([1, p], p)$;
- $b_p \sim$ PCG64.Uniform$([1, 10])$;
- $X \sim$ PCG64.Uniform(range $\leftarrow [a, b]$, size $\leftarrow [n, d]$); n `samples of` d `dimensions.`
- concept$(\cdot, \cdot, \cdot) \leftarrow (d, w_c, b_c)$;
- $Y \leftarrow$ concept(X, w_c, b_c);
- $\sigma_{\text{scaled}} \leftarrow \sigma \cdot \mathbb{E}[|Y|]$;
- $Y \leftarrow Y$ + PCG64....Normal $(\mu \leftarrow 0,\ \sigma \leftarrow \sigma_{\text{scaled}},\ \text{size} \leftarrow |Y|)$; `Noisy featuring`
- **for** $n \in nl$ **do**
  - $w_p^* \leftarrow$ PCG64.Permutation(d, p);
  - **for** $i \leftarrow 1$ **to** $|w_p|$ **do**
    - **if** $i \neq w_p^*$ **then**
      - $w_p[i] \leftarrow 0$, fixed zero point.
    - **end**
  - **end**
  - model$(\cdot, \cdot, \cdot) \leftarrow (d, w_p, b_p)$;
  - $Y \leftarrow$ model(X, w_p, b_p);
  - approx $\leftarrow$ LeastSquare(model, Y', X, Y), fixed w_p.
- **end**
- MSE$_{\text{param}}(\cdot, \cdot, \cdot) \leftarrow$ MSE$(Y, \{\text{approx}\}, \text{model})$;
- MSE$_{\text{pred}}(\cdot, \cdot)$ = MSE(Y, Y') $\hat{R}_{\text{Thm}}$ = TR(w_c, σ, p, d, n);

end
return MSE$_{\text{param}}$, MSE$_{\text{pred}}$,$\hat{R}_{\text{Thm}}$

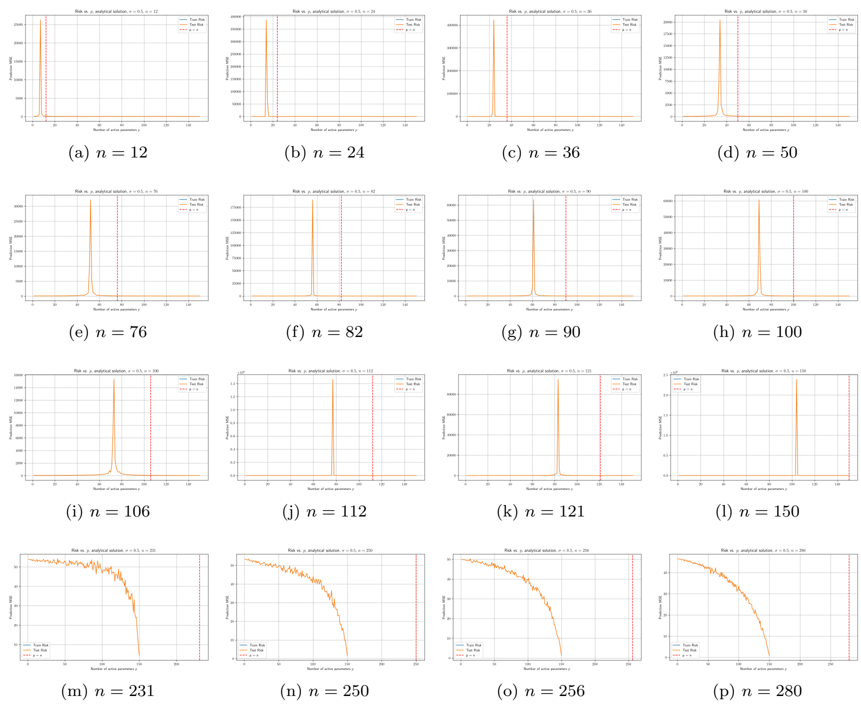
We use the Algorithm 5 for verification, including three modes of error measure, which will then be calculated accordingly. For gradient descent, we refer to another experiment. For such algorithm, we notice that the operational space of the model is $\mathbb{R}^n \to \mathbb{R}$. The masking of parameters then is not well-received, since the output space is far more skewed - single dimension resultant with no scaling difference because the model in question is supposed to be linear. Regarding such, it makes sense to realize that the model itself can still attain relatively stable error, because its feature compounds on the scaling parameters

To test the stability of the model, we run for a stable section of the permutations done on p. The result is outlined in 8. Interestingly, while the interpolation threshold theorized being $p = n$ does not hold, the pattern of double descent without head descent still fundamentally exists, albeit in a much more constrained region, as can see from $n = 12$ to $n = 150$. The pattern from then onward follows a reduced trend, without the chaotic behaviours seen. The result is shown in Figure 16.

Appendix 1.1 Remark

At least from this experiment, overall, results gathered using Theorem 8.1 and variations of it suggests inconsistent results. The supposed interpolation threshold has different interpretation and indication in different scenario, including both parameter-wise error and prediction-only error measure. Furthermore, verification also shows inconsistency with the interpolation threshold, albeit given the fact that the margin of error from the highest peak to the interpolation point is consistent up to certain n. Breakdown of the result from theorem, as well as double descent behaviour starts with high n counts, as well as interesting outlook of bias-variance reappearance for $p \sim 5000$, far more than what is tested in normal testing cases. Such result suggests further experiments to validate the observations, as well as statistical analysis of the entire system.

Specifically, the interpretation of the result is not perfect. Under such setting, then for the concept of the same class, so $d = 150$, any concept of $p > d$ would result in the metric taking zero value of the 'truncated region' where d is literally flat on the $d + 1$ onward space. Hence, the error being added on top of such error measure is the component distance of p to the flat sub $p - d$ space itself - which explains the curve going upward, as more and more risks are configured. Nevertheless, the amount of data provided, at least in terms of this model, does indeed have non-trivial effect on the model itself, and the least square solution of best fit which gives the potential minimal, hence the empirical best. Considering the results, we can see regions where the theorem 'breaks' of its bias-variance pattern. Further experiment to verify such claim and reconfiguration is needed. Worryingly, in this particular case, there exists no saddle point that was theorized by bias-variance trade-off, which must then be explained accordingly. Furthermore, we also notice the troublesome fact that this particular pattern works on *same operational space* models, that is, linear models approximating linear models. This setting is fairly limited, and would not generalize well. However, we also do not know what kind of measure to take in such case. Conflicting notions and reports aid into the confusion of such topic more than can be normally observed, and the anomalous behaviours in



(a) $n = 12$ (b) $n = 24$ (c) $n = 36$ (d) $n = 50$

(e) $n = 76$ (f) $n = 82$ (g) $n = 90$ (h) $n = 100$

(i) $n = 106$ (j) $n = 112$ (k) $n = 121$ (l) $n = 150$

(m) $n = 231$ (n) $n = 250$ (o) $n = 256$ (p) $n = 280$

Figure 16. Risk curves for varying sample size n (dimension $d = 100$, noise $\sigma = 0.5$). Double descent is determined in similar sequence, even though the interpolation theoretical $p = n$ is actually shifted over to the right from the actual interpolation observables, in parameterization. Afterward, correlation fails, and the theoretical p is ineffective, similar to previous theorem-based test run. Sample set size range is $n \in \{12, 24, 36, 50, 76, 82, 90, 100, 106, 112, 121, 150, 231, 250, 256, 280\}$. One additional remark is that the error landscape is relatively thin in either side, making it abnormal in comparison to actual bias-variance curvature.

some researches (for example, in Shi et al. 2024, we received that in GNN, there exists no trace of double descent) makes it even harder. As such, both the status of bias-variance being false, albeit in a given range only, and double descent in masking certain behaviours of the modelling process, have taken a mysterious stance in the modern machine learning community.

While discovering the double descent phenomena, it also exposes theoretical concepts that are not organized or formalized, portions of the theory and empirical observations that are within conflict of each others, issues with definitions, theorems, and interpretation of them. Those problems would then ultimately hinder further research in such direction, as well as broader theoretical requirement of the theory of machine learning, and specifically to a larger and more complex system of deep learning.

Appendix 2. Observing gradient descent decomposition

While the interpretation of what the bias-variance tradeoff, the usual training-versus-test curve, and statistical learning consideration, another interesting aspect to consider (as well for double descent), is how bias-variance decomposition acts on gradient descent and its components. This claim is perhaps reinforced by the work of Adlam and Pennington 2020 which indicates that different bias-variance decompositions effects differently on the total system mechanism.

The main term of a gradient descent algorithm is the gradient of the loss function $\mathcal{L}(h(x), y)$. For the hypothesis h, the number of instances supplied denoted by k has three main cases: $k = 1$ for stochastic descent, $k \in (1, m)$ for batch gradient descent, and $k = m$ for standard gradient descent, for a given space of m samples. The gradient in such case is calculated using the empirical risk on m size, that is:

$$\mathcal{L}_k(h(x), y) = \hat{R}_{S[k]}(h) = \frac{1}{k}\sum_{i=1}^{k} \ell(h(x_i), y_i) \tag{113}$$

We use the classical bias-variance tradeoff at first. Since the gradient is a linear operator, we have:

$$\begin{aligned}\nabla_k(\mathcal{L}) &= \nabla_k\left((\mathbb{E}\hat{y}(\mathbf{x}) - \mathbb{E}y(\mathbf{x}))^2 + \mathbb{V}\left[\hat{y}(\mathbf{x})\right] + \mathbb{V}[y(\mathbf{x})]\right)\\ &= \nabla_k\left(\mathbb{E}\left[\hat{y}(\mathbf{x})\right] - \mathbb{E}\left[y(\mathbf{x})\right]\right)^2 + \nabla_k\mathbb{V}\left[\hat{y}(\mathbf{x})\right] + \nabla_k\mathbb{V}\left[y(\mathbf{x})\right]\\ &= \nabla_k\left(\mathbb{E}\left[\hat{y}(\mathbf{x})\right] - \mathbb{E}\left[y(\mathbf{x})\right]\right)^2 + \nabla_k\mathbb{V}\left[y(\mathbf{x})\right]\end{aligned} \tag{114}$$

This effectively decompose the gradient into three parts that influence the overall gradient calculation. Because $\nabla_k\mathbb{V}\left[y(\mathbf{x})\right]$ calculates the gradient of irreducible error, which is supposed to be constant of the sample space, it is then equal 0. Suppose a sample $\mathbf{x}_k$ of size km, of partition k. Hence, we have gradient descent based on kth run over the entire dataset, permuted.

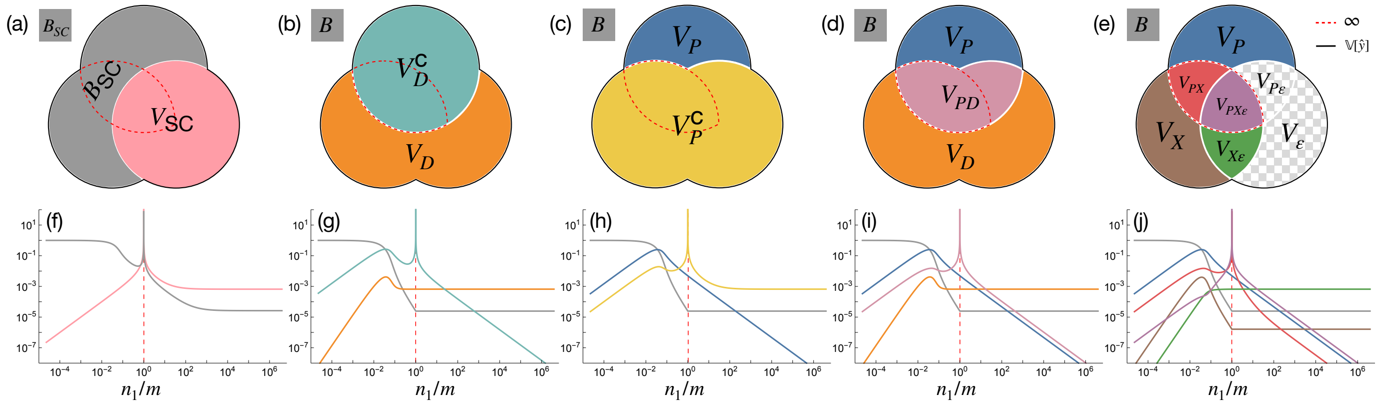


Figure 17. **(a-e) The different bias-variance decompositions.** (f-j) Corresponding theoretical predictions for $\gamma = 0$, $\phi = 1/16$ and $\sigma = \tanh$ with SNR $= 100$ as the model capacity varies across the interpolation threshold (dashed red). (a,f) The semi-classical decomposition of Hastie et al. 2019; Mei and Montanari 2019b has a nonmonotonic and divergent bias term, conflicting with standard definitions of the bias. (b,g) The decomposition of Neal et al. 2018 utilizing the law of total variance interprets the diverging term V_D^c as "variance due to optimization". (c,h) An alternative application of the law of total variance suggests the opposite, *i.e.* the diverging term V_P^c comes from "variance due to sampling". (d,i) A bivariate symmetric decomposition of the variance resolves this ambiguity and shows that the diverging term is actually V_{PD}, *i.e.* "the variance explained by the parameters and data together beyond what they explain individually." (e,j) A trivariate symmetric decomposition reveals that the divergence comes from two terms, V_{PX} and $V_{PX\varepsilon}$ (outlined in dashed red), and shows that label noise exacerbates but does not cause double descent. Since $V_\varepsilon = V_{P\varepsilon} = 0$, they are not shown in (j). Taken from Adlam and Pennington 2020

Then, the loss function calculates in the form

$$\begin{aligned}\nabla_{k,n}(\mathcal{L}) &= \nabla_{k,n}\left[\frac{1}{n}\sum_{i\le n}\hat{y}(x_{kn+i}) - \frac{1}{n}\sum_{i\le n} y(x_{kn+i})\right]^2 \\ &+ \nabla_{k,n}\left[\frac{1}{n}\sum_{i=1}^{n} y(x_i)^2 - \left(\frac{1}{n}\sum_{i=1}^{n} y(x_i)\right)^2\right]\end{aligned} \tag{115}$$

The indices is now changed to (k, n) for $m/k = n$ as the ith partitioned, ordered run, thereby $kn + i$ runs for the index. If we assume a simplified structure θ_t of parameters $y(x_i)$ depends on, we can reduce it further.

$$\begin{aligned}&\nabla_{k,n}(\mathcal{L}) \\ &= 2\left(\frac{1}{n}\sum_{i=1}^{n}\hat{y}(x_{kn+i}) - \frac{1}{n}\sum_{i=1}^{n} y(x_{kn+i})\right) \\ &\times \left(\frac{1}{n}\sum_{i=1}^{n}\nabla_k\hat{y}(x_{kn+i}) - \frac{1}{n}\sum_{i=1}^{n}\nabla_k y(x_{kn+i})\right) \\ &+ \frac{2}{n}\sum_{i=1}^{n} y(x_i)\,\nabla_k y(x_i) - \frac{2}{n^2}\left(\sum_{i=1}^{n} y(x_i)\right)\left(\sum_{i=1}^{n}\nabla_k y(x_i)\right) \\ &= 2\Big[\left(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y]\right)\left(\mathbb{E}_n[\hat{G}_k] - \mathbb{E}_n[G_k]\right) \\ &\qquad + \mathbb{E}_n[YG_k] - \mathbb{E}_n[Y]\,\mathbb{E}_n[G_k]\Big] \\ &= 2\Big[\left(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y]\right)\left(\mathbb{E}_n[\nabla_k\hat{Y}] - \mathbb{E}_n[\nabla_k Y]\right) \\ &\qquad + \mathrm{Cov}_n(Y, \nabla_k Y)\Big] \\ &= 2\left(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y]\right)\left(\mathbb{E}_n[\nabla_k\hat{Y}] - \mathbb{E}_n[\nabla_k Y]\right) \\ &\qquad + 2\,\mathrm{Cov}_n(Y, \nabla_k Y)\end{aligned}$$

for the expectation over Z_i, and covariance,

$$\mathbb{E}_n[Z] = \frac{1}{n}\sum_{i=1}^{n} Z_i, \quad \mathrm{Cov}_n(U, V) = \mathbb{E}_n[UV] - \mathbb{E}_n[U]\,\mathbb{E}_n[V]. \tag{116}$$

Here, we can see two terms multiplication, for the first one being the correlation between the expected error $\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y]$ and the gradient $\mathbb{E}_n[\nabla_k\hat{Y}] - \mathbb{E}_n[\nabla_k Y]$. This measures the correlation between the expected error, and the gradient measure. The second term is exactly the empirical covariance between the target Y and the true gradient component. Since we are taking a fixed concept case, we can simply disregard the truth gradient hence $\mathbb{E}_n[\nabla_k Y]$, for which the term is reduced to $2\left(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y]\right)\left(\mathbb{E}_n[\nabla_k\hat{Y}]\right)$. Under this same assumption applied to the covariance term, we obtain

$$\begin{aligned}\nabla_{k,n}(\mathcal{L}) &= \frac{2}{n}\sum_{i=1}^{n}\left(\hat{Y}_i - Y_i\right)\partial_k\hat{Y}_i \\ &= 2\left(\left(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y]\right)\mathbb{E}_n[\nabla_k\hat{Y}]\right)\end{aligned} \tag{117}$$

Looking into some particular algebraic representation, we can write this simply as

$$\nabla_{k,n}\mathcal{L} = 2\,\mathbb{E}_n\left[(\hat{Y} - Y)\,\nabla_k\hat{Y}\right] \tag{118}$$

$$= 2\Bigg(\underbrace{\left(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y]\right)}_{\left[\mathbb{B}(\hat{Y},Y)\right]^{1/2}}\mathbb{E}_n[\nabla_k\hat{Y}] \tag{119}$$

$$+ \underbrace{\mathbb{E}_n[\hat{Y}\,\nabla_k\hat{Y}] - \mathbb{E}_n[\hat{Y}]\,\mathbb{E}_n[\nabla_k\hat{Y}]}_{\mathrm{Cov}_n(\hat{Y},\nabla_k\hat{Y})} \tag{120}$$

$$- \underbrace{\mathbb{E}_n[Y\,\nabla_k\hat{Y}] - \mathbb{E}_n[Y]\,\mathbb{E}_n[\nabla_k\hat{Y}]}_{\mathrm{Cov}_n(Y,\nabla_k\hat{Y})}\Bigg) \tag{121}$$

The first term can be attributed to self-correct of bias-error gradient correlation, while the two covariance term suggests correlation between model randomness and its gradient, and the last one deal with the truth-gradient correlation. What those gradients mean will have to be tested in some preliminary models. Generally, we should not expect the result to be quite effective. Similar decomposition can be conducted for the other decomposition, as suggested, however, the gradient effect is particularly interesting, and will have to be resolved.

Appendix 3. Details of supervised learning structure

In a typical machine learning setting, we obtain assume the following figure 18. This includes the ground, input space (the setting of which the observations are observed), the concept $c \in \mathcal{C}$, the hypothesis $c \in \mathcal{H}$, and a supervisor ∇ that 'correct' the behaviours of the hypothesis to match with c. This is called a supervised learning setting, in which the learning part is governed by the supervisor and the observations from the concept.

Phase 1.

Begin with the construction and initialization of the feature space $\mathcal{X}^{\infty}$. By the assumption of a probabilistic process, $\mathcal{X}_m$ or simply $\mathcal{X}$ representing the dataset is assumed to be sampled, or appeared from the distribution $\mathcal{D}(p, \cdot)$ for p an arbitrary probabilistic system.

The *ground space*, or in literature generally called **feature space**, is created. This results in the first component of the tuple D, being $\mathcal{X} \subseteq \mathbb{R}^{m\times k}$, where $|\mathcal{X}| = m$, but $size(x) = k$, for $x \in \mathcal{X}$. We assume there exists the random variable x of a given distribution $\mathcal{D}$, such that the set $\mathcal{X} \sim \mathcal{D}$ of all observation is given by an underlying distribution. A well-known assumption in this case is that $x \in \mathcal{X}$ has no dependent components. That is, for example, there does not exist any function $\psi_x : \{x_i\} \to \{x_j\}$, for $i \neq j$. (Thought, this only helps in Bayesian priori analysis). Furthermore, by specifying that $\mathcal{X}$ belongs to a specific distribution, we argue of the assumption that $x \in \mathcal{X}$ is *independently and identically distributed* (i.i.d.), that is, for a given random sampling interpretation $S \subset \mathcal{X} \sim \mathcal{D}$, all data is sampled with the events space governed by the distribution $\mathcal{D}$ for every instance, and the existence of x_i to x_j for $i \neq j$ is none.

The **role** of the distribution configuration is important. If $\mathcal{D}$ is uniform, that is, $D \sim \mathcal{U}(a, b)$, then we can disregard the aspect of $\mathcal{X}$ with random variables. For $[a, b]$ to be $(-\infty, +\infty)$, the problem changed to a *regression problem*. If $\mathcal{D}$ is some other distribution function, then the problem of *representative data* and *population size* becomes apparent, which requires statistical analysis to be taken into consideration.

Phase 2.

For the constructed feature space $\mathcal{X}$, let $EX(c, \mathcal{D})$ be a procedure (or *oracle*) acting on $\mathcal{C}$ and $\mathcal{X}$ to output $\langle x, c(x)\rangle$. The hypothesis then contains a similar procedure $EX_{\mathcal{H}}(h, \mathcal{D})$, such that to output $\langle x, h(x)\rangle$. We call this the *inference phase*.

The feature space $\mathcal{X}$ is provided to h. We assume c is processed of the same process, resulted in $\mathcal{Y}_c$. Then, $h(\mathcal{X})$ outputs the set of hypothesis' target $\mathcal{Y}_h$. Thereby, we are approximating the *process c* itself. We can approach this in two main ways[t]: either *deterministic* or *probabilistic*[u]. We define such as followed. For the constructed feature space $\mathcal{X}$, let $EX(c, \mathcal{D})$ be a procedure (or *orcale*) acting on $\mathcal{C}$ and $\mathcal{X}$ to output $\langle x, c(x)\rangle$. The hypothesis then contains a similar procedure $EX_{\mathcal{H}}(h, \mathcal{D})$, such that to output $\langle x, h(x)\rangle$. We call this the *inference phase*.

Definition Appendix 3.1 (Deterministic - *discriminative modelling*). *A* ***deterministic model*** h_d *assumes its internal space as followed. For any* $h \in \mathcal{H}_d$ *of deterministic model, h is characterized by the mapping* $h_d : \mathcal{X} \to \mathcal{Y}_h$, *such that* $h \in C^n$ *of n-differential space.*

Similarly, if the interpretation of the model is *probabilistic*, we have the following definition of probabilistic-based model.

Definition Appendix 3.2 (Probabilistic - *generative modelling*). *A* ***probabilistic model*** h_p *assumes its internal space as followed. For any* $h \in \mathcal{H}_p$ *of all probabilistic hypothesis, h is characterized by the probability distribution* $\Gamma(p_X(X))$, *where* Γ *is the discrete decision process outside the probabilistic interpretation.* [v]

Notice that the process and the learning setting is probabilistic, however, the interpretation of the hypothesis h can be either deterministic or probabilistic, or anything else. This puts the flexibility into the setting of the model, and permits us to consider various representation of the model structure. Furthermore, the hypothesis for the learning process is evaluated relative to the same probabilistic setting, in which the training takes place, and we allow hypotheses that are only approximation of the target concept Sugiyama 2015; Kearns and Vazirani 1994.

The similarity between both of the two general types is that the model is characterized by their **parameters**, both **model-specific parameters** and **hyperparameters**.

Phase 3.

There exists an evaluator (or *supervisor*) $\nabla(h, c)$ that evaluates specific *loss framework* $\ell\{h(x), c(x)\}$ based on available information, and a *update modifier* $U(\ell, \mathcal{A})$ of certain objective to algorithm $\mathcal{A}$.

In this step of the procedural setting, the *supervisor* includes a loss function, ℓ, which takes discrete arguments, and 'binarily'

t. Note that, in some aspect, this is similar to the usage of mathematical simulation of certain dynamical system, or *any* system that has certain patterns or relations observable.

u. The model itself can be entirely deterministic, while the learning process is not. in fact, one of the very first assumption and axiomatic view of the learning problem is that learning occurs in a *probabilistic setting*.

v. The discrete decision process Γ is very simple to argue. For example, given a Naive Bayes model with $p(C_k \mid D[i])$ of the dataset, the probability distribution is regarded as the argument

$$Y^* = \arg\max_{c_k\in C} P_\theta(c_k \mid x) = \arg\max_{c_k\in C} P_\theta(x \mid c_k)\cdot P(c_k)$$

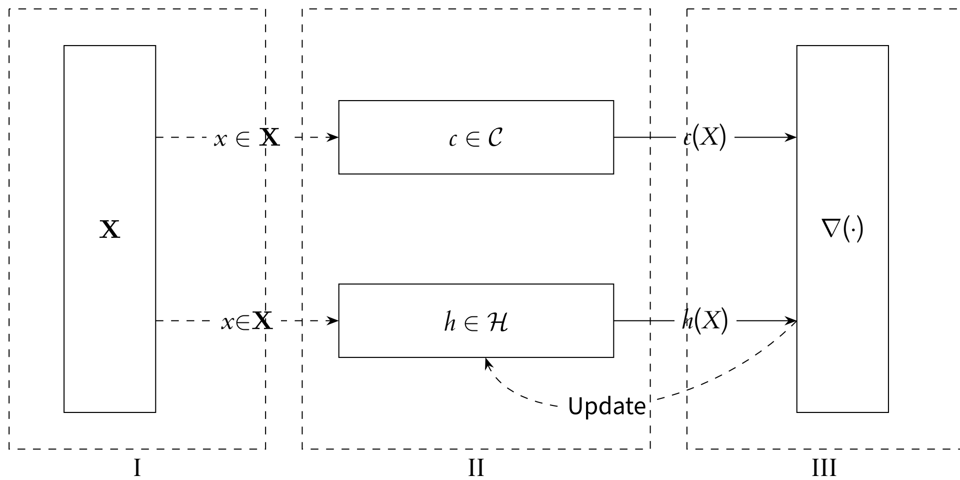


Figure 18. An illustration of the (supervised) statistical process. Phase III contains two parts: First is the evaluation $\nabla(h, c)$ according to the data $\mathcal{D}$, and second is the Update process to re-align c to the actual target.

evaluate the discrete result between h and c. This loss function is supposed to have a global minimum, or at least a certain region of minimal approximation. We have the following definition.

Definition Appendix 3.3 (Loss function). *A loss function is a non-negative function $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, +\infty)$.*

In general, the following is supposed of the loss function:

Conjecture Appendix 3.1 (Loss function convexity). *For any supervisor ∇ of a learner framework $\mathcal{L}(\mathcal{H}, \mathcal{A})$, the loss function ℓ is assumed to be a p-converging function, or as a convex function. That is, for $\ell : \mathbb{R}^n \to \mathbb{R}$, then ℓ is convex on its domain, or is locally convex on $[a, b] \subset \mathbb{R}^n$, if, for $x, y \in \mathbb{R}^n$ of such interval, and for $\lambda \in [0, 1]$, it satisfies*

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \tag{122}$$

Under a single structure, that is, for example, a class of mapping $\mathbb{R}^n \to \mathbb{R}$, there can exist many loss functions to be considered. Take for example the setting of regression analysis. Then, ℓ can be either the absolute error, $|h(x) - c(x)|$, or the L^2 norm, $||h(x), c(x)||_2$. Different loss function provides different path and landscape, though they still share somewhat similar interpolation patterns, or either some might be subjected to, under the consideration of a loss landscape, local minima than others. Furthermore, the form and representation of loss functions is not discrete - but rather based of its interpretation and the supposition of the underlying notion required for the loss function to operate in. For example, if the setting is *generative*, the loss function would take the form of a *divergence measure* between two probability distribution. Notice however, that the action potential provided by the model in their operation dissapears, or rather, is not considered, when using the loss function.

Using such loss function, for the case of no noises or interference uncertainty, if the goal is to minimize the empirical risk on such dataset, then we call this the method of *empirical risk minimization*, or ERM. This is born out of the consideration that in practice, the oracle $EX(c, \mathcal{D})$, and the actual generalization error $R(h)$ is not obtainable.

Appendix D - Bias-variance decomposition derivation in Geman, Bienenstock, and Doursat 1992

Proof. We present the original derivation of bias-variance tradeoff and its analysis from Geman, Bienenstock, and Doursat 1992. Geman's paper is, by his own word, concerned of *parametric models*, i.e. hypothesis class without strong assumption of parameters defined, though the definition of bias-variance tradeoff afterward and its justification is made in the sense of parametric model. We partially clarify[w] this with the following definition as customary.

Definition Appendix 3.4 (Parameterization). *A **parametric model** is one that can be parameterized by a finite number of parameters. In general, for a hypothesis class $\mathcal{H}$of all parametric hypothesis is then expressed as:*

$$\mathcal{H} = \{f(x; \theta) : \theta \in \Theta \subset \mathbb{R}^d\} \tag{123}$$

*where Θ is called the parameter space. A **nonparametric model** is one which cannot be parameterized by a fix number of parameters.*

Suppose of a training dataset $\mathcal{D}$ of N 2-tuples $(\mathbf{x}_i, y_i)$, that is:

$$\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)\} \quad |\mathcal{D}| = N, \mathbf{x} \in \mathbb{R}^d, y \in \mathbb{R} \tag{124}$$

The regression problem is to construct a function $f(\mathbf{x}) \to y$ based on $\mathcal{D}$, which is denoted $f(\mathbf{x}; \mathcal{D})$ to show this dependency.

w. Generally, there is no definite definition on what would be considered non-parametric. Though, an example might be drawn from the (vanilla) neural network and Gaussian Process Regression (GPR)

Then,

$$\begin{aligned}
&\mathbb{E}\left[(y - f(\mathbf{x};\mathcal{D}))^2 \mid \mathbf{x},\mathcal{D}\right] \\
&= \mathbb{E}\left[\left((y - \mathbb{E}[y \mid \mathbf{x}] + (\mathbb{E}[y \mid \mathbf{x}] - f(\mathbf{x};\mathcal{D})))\right)^2 \mid \mathbf{x},\mathcal{D}\right] \\
&= \mathbb{E}\left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x},\mathcal{D}\right] + \left(\mathbb{E}[y \mid \mathbf{x} - f(\mathbf{x};\mathcal{D})]\right)^2 \\
&\quad + 2\mathbb{E}\left[(y - \mathbb{E}[y \mid \mathbf{x}]) \mid \mathbf{x},\mathcal{D}\right] \cdot (\mathbb{E}[y \mid \mathbf{x}] - f(\mathbf{x};\mathcal{D}))^2 \\
&= \mathbb{E}\left[(y - \mathbb{E}[y \mid \mathbf{x}]) \mid \mathbf{x},\mathcal{D}\right] + (\mathbb{E}[y \mid \mathbf{x}] - f(\mathbf{x};\mathcal{D}))^2 \\
&\geq \mathbb{E}\left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x},\mathcal{D}\right]
\end{aligned} \tag{125}$$

Hence, we decompose it to:

$$\begin{aligned}
&\mathbb{E}\left[((y - f(\mathbf{x};\mathcal{D})))^2 \mid \mathbf{x},\mathcal{D}\right] \\
&= \mathbb{E}\left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x},\mathcal{D}\right] + (f(\mathbf{x};\mathcal{D}) - \mathbb{E}[y \mid \mathbf{x}])^2
\end{aligned} \tag{126}$$

Where $\mathbb{E}\left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x},\mathcal{D}\right]$ is regarded to be a relative constant of variance on y given $\mathbf{x}$. Taking the expectation on $\mathcal{D}$, we gain:

$$\begin{aligned}
&\mathbb{E}_{\mathcal{D}}\left\{\mathbb{E}\left[((y - f(\mathbf{x};\mathcal{D})))^2 \mid \mathbf{x},\mathcal{D}\right]\right\} \\
&= \mathbb{E}_{\mathcal{D}}\left\{\mathbb{E}\left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x},\mathcal{D}\right] + (f(\mathbf{x};\mathcal{D}) - \mathbb{E}[y \mid \mathbf{x}])^2\right\} \\
&= \mathbb{E}_{\mathcal{D}}\left\{\mathbb{E}\left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x},\mathcal{D}\right]\right\} + \\
&\qquad \mathbb{E}_{\mathcal{D}}\left\{(f(\mathbf{x};\mathcal{D}) - \mathbb{E}[y \mid \mathbf{x}])^2\right\}
\end{aligned} \tag{127}$$

The second term is of importance, and is further decomposed to:

$$\begin{aligned}
&\mathbb{E}_{\mathcal{D}}\left\{(f(\mathbf{x};\mathcal{D}) - \mathbb{E}[y \mid \mathbf{x}])^2\right\} \\
&= \mathbb{E}_{\mathcal{D}}\left\{(f(\mathbf{x};\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(\mathbf{x};\mathcal{D})] + \mathbb{E}_{\mathcal{D}}[f(\mathbf{x};\mathcal{D})] - \mathbb{E}[y \mid \mathbf{x}])^2\right\} \\
&= \mathbb{E}_{\mathcal{D}}\left[(f(\mathbf{x};\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(\mathbf{x};\mathcal{D})])^2\right] \\
&\quad + \mathbb{E}_{\mathcal{D}}\left[(\mathbb{E}_{\mathcal{D}}[f(\mathbf{x};\mathcal{D})] - \mathbb{E}[y \mid \mathbf{x}])^2\right] \\
&\quad + 2\mathbb{E}_{\mathcal{D}}\left[f(\mathbf{x};\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(\mathbf{x};\mathcal{D})]\right] \\
&\quad \cdot (\mathbb{E}_{\mathcal{D}}[f(\mathbf{x};\mathcal{D})] - \mathbb{E}[y \mid \mathbf{x}]) \\
&= \underbrace{\left\{\mathbb{E}_{\mathcal{D}}[f(x;\mathcal{D})] - \mathbb{E}[y \mid x]\right\}}_{\text{bias term}} \\
&\quad + \underbrace{\mathbb{E}_{\mathcal{D}}\left\{(f(x;\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(x;\mathcal{D})])^2\right\}}_{\text{variance term}}
\end{aligned} \tag{128}$$

which yields the desired bias-variance tradeoff. □

Appendix 4. Practical consideration

While we have been talking in the learning setting pretty loosely on the side of the data itself, it is the data that gives us the problem setting. For example, if the data is a pair (X, Y) of input output representation, then we know that our working space would be $\mathbb{R}^n \times \mathbb{R}^m$ or any of its subset, given the fact that data must be encoded into numerical representations, even in discrete or logical case (in such case, typically, $0 - 1$ pair is enough). In more complex and complicated setting, for example, on the structure of a graph and its embedded representation, it is much more difficult to see the *form of dataset* to be, hence, also the working space.

Another thing with data is the *availability of the data* and the *distribution of the data* also matters. In some cases, our data consists of only skewed or partitioned sections of the actual concept range, at least up to the observable criterion of the concept (some concept does not exist in certain range, at least in a numerical example, and some other reasons). In the other case, the availability of data, meaning the density of data, as well as added *observational effect* also alters the learning setting, simply because learning theory relies on, at least from our glance, simple statistical, observational learning from observation space, without the added structures. An actual, practical concern however, lies in the range of **practical empirical-generalization** setting.

Appendix 4.1 Observational space partitioning

Previously, we established that at least in said formal setting, generalization errors cannot be known beforehand. Thereby, all of our operations, procedures, processes has to be conducted using the observed data, or the sample space $\mathcal{S}$. A solution to utilize that is to consider the generalization region as the region of *new, unseen observations* that while still is of the same concept c, but it is indeed, unseen if the model is never provided of such instances. Hence, it prompts the utilization of limited resources by then to partition the sample space into various subspaces, that would be then fed into the learning process as observational space. This is simply called **sample set partitioning.** Denoted the number of partition (discrete) as k, then for $k = 2$, then this is called the **train-test partition** (or dataset). This contains a dataset $\mathbf{x}_1$ and $\mathbf{x}_2$ of the same original sample set, such that $\mathbf{x}_1/\mathbf{x}_2 \geq 1$ (one is bigger than another, usually). The ratio does indeed affect the overall running procedure, accordingly [x].

Based on the above conclusion, the empirical error and generalized error can be re-calculated to fit the general scheme. For $|\mathcal{S}| = m$ and partition $k + p = m$ for train and test partition

x. If $k > 3$ and above, we call it the nth partitioning. If we further consider the choice of observational spaces being provided, then if $k > 3$, and subspaces are chosen randomly, it is called **cross-validating partition**. The effect of them are fairly well-understood, and we might as well refer to authoritative sources on such issue.

respectively,

$$\hat{R}_k(h) = \frac{1}{k}\sum_{i=1}^{k} \ell\{h(x_i), c(x_i)\}, \tag{129}$$

$$\hat{R}_p(h_{\mathcal{A}}) = \frac{1}{k}\sum_{i=1}^{p} \ell\{h_{\mathcal{A}}(x_i), c(x_i)\} \tag{130}$$

where $h_{\mathcal{A}}$ is the result from the learning algorithm (procedure), for $h^* \neq h_{\mathcal{A}}$. Then, the empirical learning seeks to optimize the first term, that is, for the objective

$$\arg\min_{h\in\mathcal{H}} \hat{R}_k(h) = \arg\min_{h\in\mathcal{H}} \frac{1}{k}\sum_{i=1}^{k} \ell\{h(x_i), c(x_i)\} \tag{131}$$

and the generalization learning as:

$$\begin{aligned}
&\arg\min_{h\in\mathcal{H}} \hat{R}_p(h_{\mathcal{A}}) \\
&= \hat{R}_k(h_{\mathcal{A}}) + \arg\min_{h\in\mathcal{H}} \hat{R}_p(h_{\mathcal{A}}) \\
&= \arg\min_{h\in\mathcal{H}} \left[\frac{1}{k}\sum_{i=1}^{k} \ell\{h_{\mathcal{A}}(x_i), c(x_i)\} + \frac{1}{k}\sum_{i=1}^{p} \ell\{h_{\mathcal{A}}(x_i), c(x_i)\}\right] \\
&= \arg\min_{h\in\mathcal{H}} \left[\epsilon + \frac{1}{k}\sum_{i=1}^{p} \ell\{h_{\mathcal{A}}(x_i), c(x_i)\}\right]
\end{aligned} \tag{132}$$

By the definition order, the generalization error can only be calculated post-$\mathcal{A}$. Hence, we often called the empirical, or ERM-generalizer case to be dynamic, and the solution for ERM is tested statically on generalization set. This partition of order of computation though, brings up the problem of updating factors and others contributing potential that will damage or diffuse the measure.

For cross-validation and other randomize partition, the formulation requires more care in the notion by itself. The structural risk minimization scheme dilemma can be understood from the above consideration naturally. We have effectively defined the two proxies, **heuristic empirical risk** and **heuristic generalization risk**.